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Epigenetic delineation of the earliest cardiac lineage segregation by single-cell multi-omics

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eLife Assessment

This study provides new single-cell multi-omics datasets that may be **useful** in the study of early cardiac lineages. However, the authors' conclusions regarding the mutual regulation of key regulators for cardiac specification and new cardiac lineage trajectories are **inadequately** supported by persuasive analysis and do not align with prior published studies. If revised to address the serious caveats adequately, the findings may be of interest to researchers in the field of cardiac development and congenital heart disease.

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Abstract

Summary

The mammalian heart is formed from multiple mesoderm-derived cell lineages. However, it remains largely unknown when and how the specification of mesoderm towards cardiac lineages is determined. Here, we systematically depict the transcriptional trajectories toward

cardiomyocyte in early mouse embryo, and characterize the epigenetic landscapes underlying the early mesodermal lineage specification by single-cell multi-omics analyses. The analyses also reveal distinct core regulatory networks (CRN) in controlling specification of mesodermal lineages. We further demonstrate the essential role HAND1 and FOXF1 in driving the earliest cardiac progenitors specification. These key transcription factors occupy at distinct enhancers, but function synergistically and hierarchically to regulate the expression of cardiac-specific genes. In addition, HAND1 is required for exiting from the nascent mesoderm program, while FOXF1 is essential for driving cardiac differentiation during juxta-cardiac field (JCF) specification. Our findings establish transcriptional and epigenetic determinants specifying the early cardiac lineage, providing insights for the investigation of congenital heart defects.

Introduction

Heart development requires coordinated specification of multiple lineages, each characterized by serial cell fate determination events. Identifying the developmental trajectories is the key to understanding heart formation [1, 2]. In past decades, a stepwise determination of early cardiac lineage hierarchy has been primarily established [3]. The current model designates cardiac progenitor cells into discrete pools, including the first and second heart field (FHF and SHF) [1], and a newly classified juxta-cardiac field (JCF) [4]. FHF mainly contributes to the left ventricle (LV) and the atria. SHF, located in a dorsal-medial region to FHF, progressively develops into cells in the right ventricle (RV), the outflow tract (OFT) and the atria [5]. Interestingly, JCF contributes to not only epicardium and pericardium, but also cardiomyocytes (CMs) of LV and the atria [4, 6].

Cardiac progenitors are mostly generated from the *Mesp1*-expressing (*Mesp1*$^+$) nascent mesoderm (NM) cells during gastrulation [7, 8]. Pioneering work has revealed that the developmental capacities of each cardiac progenitor pool are highly related to the spatial-temporal constriction during the specification of NM cells [6, 9, 10]. Temporally inducible lineage tracing indicates that E6.5 *Mesp1*$^+$ cells mostly contribute to LV, whereas E7.25 *Mesp1*$^+$ cells give rise to RV, atria, OFT, and inflow tracts (IFT) [9]. It seems that early-streak stage NM cells differentiate into FHF pools, while late-streak stage NM cells relate to SHF progenitors. Interestingly, JCF population is also derived from the *Mesp1*$^+$NM cells in the gastrula [6]. Recent studies on single-cell transcriptomic data of the late headfold stage embryos have revealed that JCF shares a number of molecular markers with FHF, but lacks *Nkx2.5* expression and exhibits specific *Mab21l2* expression [4]. However, unlike FHF cells, JCF cells are largely located at the embryonic-extraembryonic mesodermal interface, as revealed by the *Mab21l2* expression, rostrally to the *Nkx2-5* positive cardiac crescent region [4]. It remains unclear the molecular signaling underlying the early specification of NM cells into JCF and FHF population.

In this study, by bridging the transcriptional landscapes between the gastrula and the headfold stage in early mouse embryos, we systematically depict the transcriptional trajectories leading to CMs during early mouse development, and characterize the epigenetic landscapes that underlie early mesodermal lineage specification. The analyses reveal two dinstinct developmental trajectories towards CMs, namely the JCF trajectory (the *Hand1*-expressing early extraembryonic mesoderm – JCF and FHF – CM) and the SHF trajectory (the pharyngeal mesoderm cells – SHF – CM). Through single-cell multi-omics analysis, a predicted core regulatory network (CRN) in JCF is identified, consisting of transcription factors (TFs) GATA4, TEAD4, HAND1 and FOXF1. Further functional analysis indicates that HAND1 and FOXF1 are activated sequentially, and both required for mesodermal specification and the expression of the JCF specific genes. Taken together, our study unveils the transcriptional and epigenetic dynamics during early cardiac specification, demonstrates the crucial roles of HAND1 and FOXF1 in driving early cardiac specification, and provides insights for the investigation of congenital heart defects.

Results

Transcriptional dynamics during the specification of early cardiac progenitors

To delineate the origin of cardiac progenitors, we constructed the E6.5-8.5 developmental trajectories using the published mouse gastrulation cell atlas by performing the Waddington-Optimal-Transport (WOT) analysis to infer ancestor-descendant fates of cells [11, 12]. WOT models the temporal dynamics of cell state transitions by computing probabilistic couplings between adjacent timepoints, effectively inferring ancestor-descendant relationships based on transcriptomic similarity and temporal continuity. Here, CMs were used as trajectory endpoints and traced back to the E6.5 epiblast. Clear trajectory separation was observed within E7.5-7.75 (**Figure 1A**). Besides the pharyngeal mesoderm (PM) cell cluster at E7.75, a subset of E7.5 early extraembryonic mesoderm (EEM) cells [6] was specifically identified in a distinct developmental trajectory. PM population was marked by the expression of *Isl1*, *Sfrp1*, *Tcf21*, *Tbx1* and *Irx3*, suggesting its relationship with the SHF progenitors; the EEM cells exhibited highly expressing *Hand1*, *Pmp22*, *Foxf1* and *Spin2c* (Figure S1A). In addition, cells along the EEM trajectory also expressed higher levels of the FHF signature genes, including *Tbx5* and *Hcn4*, suggesting their contribution to the FHF [6]. Compared with the PM trajectory, the EEM trajectory was mainly composed of cells in the later stages of cardiac development (**Figure 1B** and S1B). By the heart looping stage (E8.5), the two trajectories indeed exhibited distinct contributions to cardiac structures LV and OFT, respectively (**Figure 1C**).

Thus, we here refer to these two transcriptional trajectories as JCF and SHF (**Figure 1A**). The spatial correlation of these two trajectories with heart fields was confirmed using the dataset from manually micro-dissected mesodermal cells in cardiac regions of E7.75-8.25 mouse embryos (Figure S1C) [4]. The JCF and SHF trajectories contained overlapping but temporally distinct cell types (**Figure 1B** and Figure S1D). We performed analysis of fate divergence between two trajectories, which suggests, before E7.0, mesodermal cells have similar probabilities to choose either trajectory (Figure S1E).The separation of the two trajectories can be observed as early as E7.0 (**Figure 1B** and Figure S1E). NM cells, marked by *Mesp1* and *Lefty2*, at E7.0 were more likely to be the multipotent progenitors of the SHF trajectory; whereas mixed mesoderm (MM) cells, marked by *Hand1* and *Msx2*, in later developmental stage and EEM cells tended to contribute to the JCF trajectory (**Figures 1D**, 1E and Figure S1F). To illucidate the contribution of JCF and SHF lineages to the cardiac crescent (CC), we used early CC progenitor population (E7.75 *Nkx2- 5+*; *Mab21l2-* CMs) as the starting point and performed WOT lineage inference (Figure S2A). Results suggest that both JCF and SHF progenitors contribute to the CC, consistent with live imaging-based single cell tracing by Dominguez et al [13] and lineage tracing results by Zhang et al [6]. We also analyzed the expression levels of CC marker genes (*Tbx5*, *Hcn4*) and observed their activation along both trajectories (Figure S2B).

We then systematically characterized the stage specific marker genes in the JCF and SHF trajectories. These two trajectories exhibited discrete and dynamic gene expression profiles during development (**Figure 1F**). The JCF marker gene, *Mab21l2*, showed transient expression [4], in contrast to the late but continued expression of SHF markers (*Isl1*, *Tbx1*), further supporting the asynchronized fate commitment by the two lineages. Notably, along the pseudotime trajectory, the NM marker *Mesp1* displays a transient and early expression in the JCF lineage, whereas it is persistently expressed during the mid-to-late stages in the SHF lineage, which suggests distinct mechanisms of mesodermal specification underlying the two lineages. We also observed relatively similar inhibitory Wnt and Nodal, as well as active Fgf and Notch, signaling activities along the two trajectories (**Figures 1G** and S3A). Interestingly, the early stage of the JCF trajectory seems show higher Bmp and Yap signaling activities (**Figure 1G**). Temporal expression profiles of the

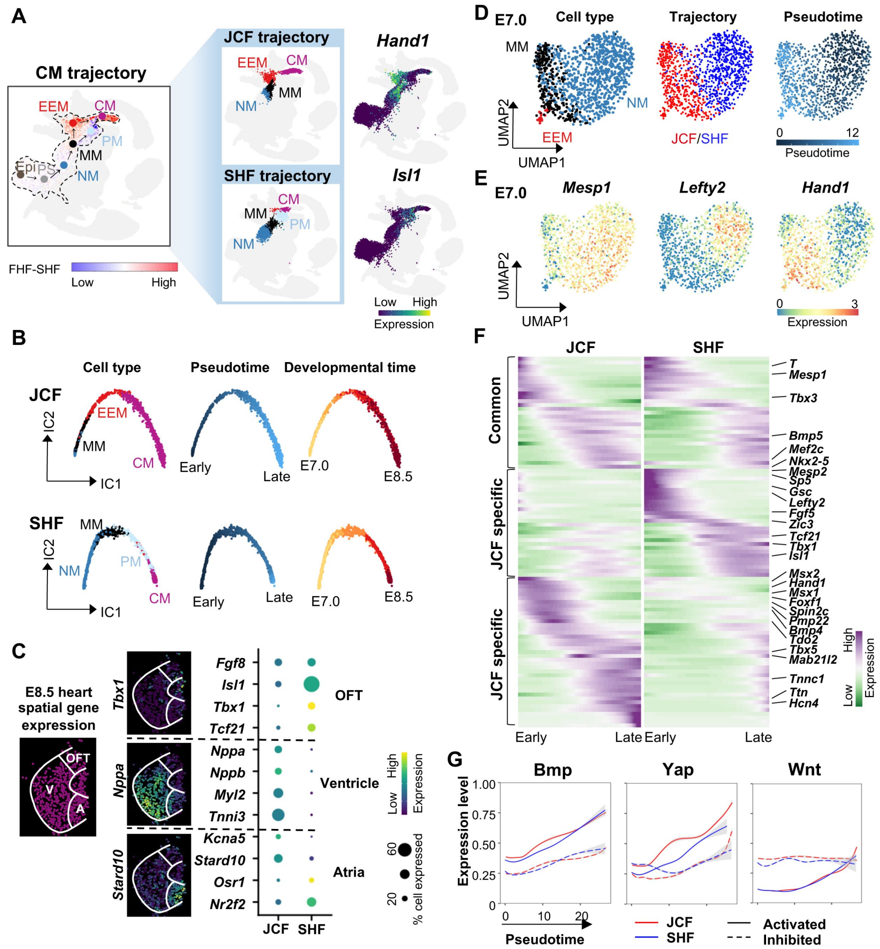

Figure 1

Inferred trajectories reflect two distinct development routes of CMs.

(A) Two distinct trajectories, the JCF trajectory and the SHF trajectory, were inferred during CM differentiation. UMAP layout from Pijuan-Sala et al. [12] is highlighted by cells belonging to the WOT predicted developmental trajectories for CM (left), EEM (upper middle) and PM (lower middle), respectively. UMAP layout for the CM trajectory with cells is colored by the difference between FHF- and SHF-gene signature scores. UMAP layout for the JCF or SHF trajectory is colored by cell types. Gene expression levels of *Hand1* (upper right) and *Isl1* (lower right) of cells along the CM trajectory is overlaid onto the above-mentioned UMAP layout. Epi, epiblast. PS, primitive streak. NM, nascent mesoderm. MM, mixed mesoderm. EEM, early extraembryonic mesoderm. PM, pharyngeal mesoderm. CM, cardiomyocyte.
(B) Independent component (IC) layout showing pseudotemporal trajectories for JCF trajectory (upper) and SHF trajectory (lower) cells, colored by cell type (left), pseudotime (middle) and developmental time (right).
(C) The JCF and SHF trajectories showing distinct contributions to cardiac structures. Spatial plot showing spots in cardiac subregions from E8.5 embryos (left) [29]. White lines denoting the region including outflow tract (OFT), ventricle (V) and Atria (A). 'Virtual' in situ hybridization (vISH) confirming spatial specificity of marker genes corresponding to cardiac subregions (middle). Dot plot showing expression difference in subregion specific genes in the JCF and SHF trajectories (right).
(D) Mesodermal lineage segregation at E7.0. UMAP layout for E7.0 CM trajectory cells is colored by cell type (left), trajectory (middle) and pseudotime (right).
(E) UMAP layout (same as d) showing the expression of the E7.0 marker genes of NM (*Mesp1*), JCF (*Hand1*) and SHF (*Lefty2*).
(F) Heatmap showing the expression of pseudotime-dependent genes for the JCF trajectory (right) and the SHF trajectory (left). Rows and columns represent genes and cell bins, respectively. Genes used for the heatmap are listed in Supplementary Table 1.
(G) Smoothened fitting curves showing expression levels of activated (solid line) and inhibited (dotted line) signaling markers in the JCF (red) and SHF (blue) trajectories.

Bmp genes indicated that *Bmp4*/*5*/*7* were dynamically expressed during cardiac specification, with *Bmp4* demonstrating higher JCF specificity and at least 0.5 days earlier activation (Figure S3B). Geo-seq data analysis indicated that *Bmp4* was highly specific to mesoderm, and enriched at the proximal mesodermal ends (layer 11 at E7.0, layer 9-10 at E7.5) with distinct anterior-posterior preference at E7.5 (Figure S3C). For the target genes of Bmp signaling, several genes (*Hand1*, *Car4*, *Arl4c* and *Pmp22*) showed JCF specific activation-to-repression dynamics, similar to *Bmp4* (**Figures 1F** and S3D).

Epigenetic signatures of the early JCF and SHF progenitors

To investigate the epigenetic regulation during cardiac cell fate decisions, we performed multi-omic analysis by combining single-nucleus RNA-sequencing (snRNA-seq) and scATAC-seq to generate paired, cell type specific transcriptome and chromatin accessibility profiles of 13,226 cells in E7.0 mouse embryos. The single-cell transcriptomic data were integrated with the published E7.0 mouse embryo cell atlas data, followed by label transfer and gene expression-based cell type identification (Figures S4A and S4B). For the scATAC-seq data, we scored the genome-wide ATAC activities with bin sizes of 10 kb prior to UMAP analysis, which yielded cell clusters similar to transcriptome-based analysis (Figure S4A).

Nine clusters of cells were identified through clustering analysis at both transcriptional and epigenetic levels, which are NM cells (Clusters 0, 1 and 2), primordia germ cells (PGC, Cluster 4), hematoendothelial progenitors (Haem, Clusters 5 and 6), and EEM cells (Clusters 3, 7 and 8) (**Figures 2A**, 2B and S4C). RNA velocity also supported the four possible trajectories mentioned above for the earliest mesodermal cell specification (**Figure 2C**). WOT analysis revealed that Clusters 3, 7, and 8 showed intermediate to high probabilities of belonging to the JCF trajectory; pseudotime analysis indicated that Cluster 8 represented the late differentiated EEM populations (Figure S5A). Although Cluster 2 represented the relatively late stage of NM cell population by pseudotime analysis, Cluster 1, 0 and 2 demonstrated similar probabilities of belonging to the SHF trajectory (Figure S5A). Thus, the analyses further indicated that EEM cells were clearly separated from the nascent mesoderm, while the SHF trajectory related cells still remained at the early NM stage at E7.0.

We further analyzed the chromatin accessibilities of these cell clusters. Total 90,661 chromatin accessible elements (CAEs) were detected, 7,206 of which were differentially accessible elements (DAEs) across the 9 clusters (**Figure 2D**). The DAEs were annotated to their target genes by enhancer-promoter (EP) pairing analysis. Consistent with the clustering analysis based on gene expression (Figure S4C), DAVID functional term analysis revealed that the DAE target genes in Clusters 3, 7 and 8 , such as *Hand1*, *Foxf1*, *Bmp4* and *Msx1*, were mainly associated with heart morphogenesis, that those in Clusters 5 and 6, like *Tal1*, *Lmo2* and *Fli1*, were related to angiogenesis and vasculature development, and that those in Clusters 0, 1, 2, for examples, *T*, *Zic2*, *Lhx1* and *Gata6*, were enriched in gastrulation and mesoderm development (**Figures 2D** and **2E**).

To characterize the spatiotemporal chromatin dynamics of the DAEs in JCF and SHF, we quantified the occupancies of the enhancer marks H3K4me1 and H3K27ac, as well as the promoter mark H3K4me3 at these DAEs across the E6.5-7.5 developmental stages[14]. JCF/Cluster 8 and SHF/Cluster 2 specific DAEs could potentially function as enhancers and become activated at anterior regions of E7.5 embryos, as the DAEs were generally marked by H3K27ac and H3K4me1, but not H3K4me3 (**Figure 2F**). The enrichment of H3K4me1 at E7.0 and even earlier at E6.5 stage, along with the higher levels of H3K27ac at these DAEs at E7.5 stage, suggested that many of these DAEs could be dormant or inactive enhancers during earlier stages like E6.5, but primed for later activation during lineage specification (**Figure 2F**). Indeed, the integrated analysis on chromantin accessibility of the DAEs, shown by ATAC-seq, and their target gene expression levels supported that a large portion of the JCF/Cluster 8 DAEs were primed before the full activation of their target genes, like *Bmp4*, *Hand1* and *Foxf1* (**Figure 2G**). For example, seven DAEs associated

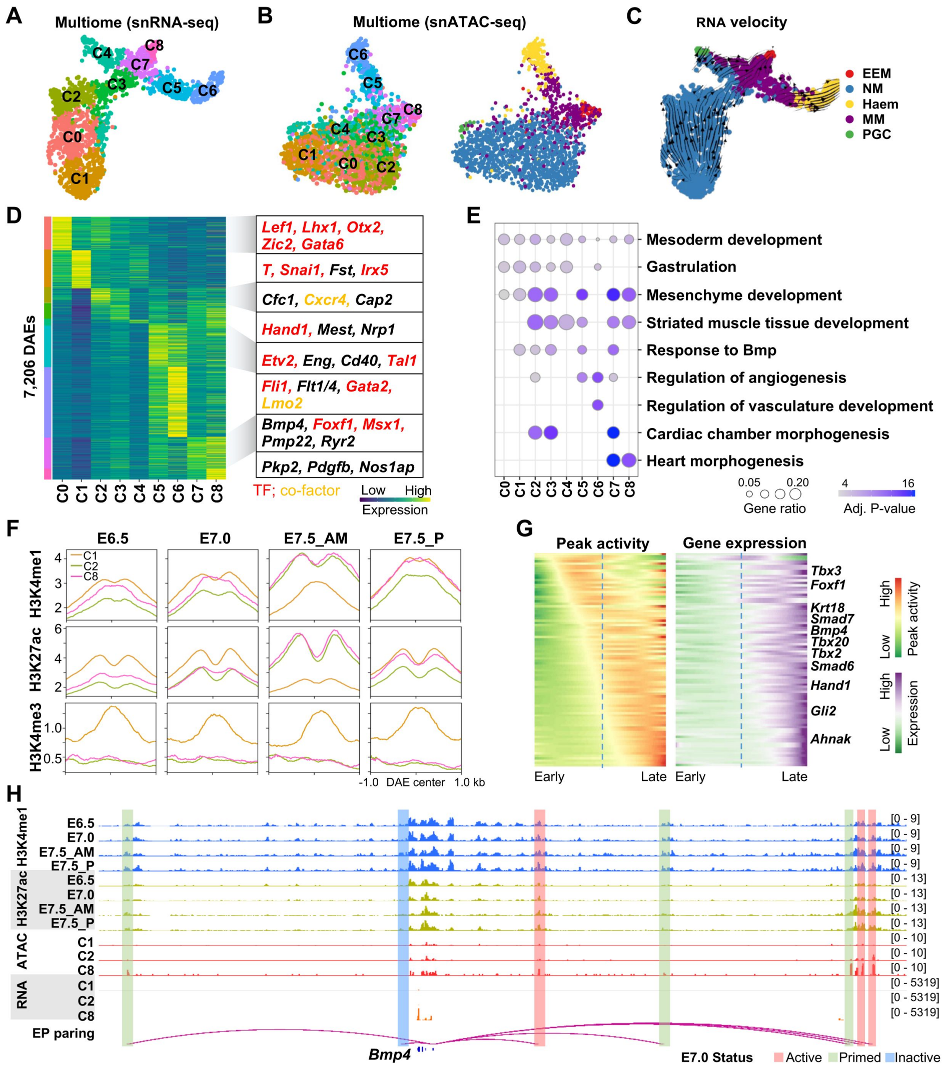

Figure 2

Multi-omics analysis reveals epigenetic signatures of the early JCF and SHF progenitors.

(A-C) Clustering analysis of E7.0 single-cell multi-omics data and comparison between modalities of transcriptome (snRNA-seq) and chromatin accessibility (snATAC-seq). UMAP layout, using only snRNA-seq (A,C) or snATAC-seq (B) data, is colored by cluster identities (left) or cell types (right). For both (A) and (B), cluster identities are determined by snRNA-seq data. For (C), developmental directions, shown as arrows, are indicated by RNA-velocity analysis.

(D) Heatmap showing the activity of 7,206 differentially accessible elements (DAEs) between clusters. For the DAE motif analysis, we firstly inferred the motif and TF families, then tested which specific TFs are expressed in the corresponding cell cluster. Rows and columns represent DAEs and clusters. Colors indicate levels of accessibility averaged among cells from each cluster. The length of the corresponding color bar on the left represents the number of DEAs. Representative DAE-associated genes are shown on the right side.

(E) Dot plot indicating functional enrichment of DAE-associated genes of each cluster.

(F) Epigenetic status of DAEs during mesodermal specification of E6.5-7.5 mice embryo. H3K4me1, H3K27ac and H3K4me3 profiles were averaged among DAEs of cluster C1 (NM), C2 (SHF-direction frontier) and C8 (JCF-direction frontier). Published H3K4me1, H3K27ac and H3K4me3 ChIP-seq data were collected from E6.5 epiblast, E7.0/7.5 posterior epiblast (P) and E7.5 anterior mesoderm (AM) [14].

(G) Smoothened heatmap showing dynamic gene expression (right) and enhancer accessibility (left) along E7.0 mesoderm pseudotime trajectories for gene-enhancer pairs. Dashed lines indicating the pseudo-temporal midpoint.

(H) Representative genome browser snapshots of ATAC/RNA-seq (aggregated gene expression and chromatin accessibility for each cluster), H3K4me1 and H3K27ac at the *Bmp4* locus. Putative enhancers status at E7.0 are highlighted by colors.

with the *Bmp4* gene were identifited, three of which were primed at E7.0 as marked by low levels of H3K27ac but high levels of H3K4me1 (**Figure 2H**). Taken together, the combined transcriptome and chromatin accessability analysis further supported the early lineage segregation of JCF and the epigenetic priming at gastrulation stage for early cardiac genes.

Identification of lineage specific key TFs

An integrated analysis of motif enrichment at the DAEs and TF expression data allowed us to identify potential lineage specific key TFs. The SHF/Cluster 2 specific DAEs showed motif enrichment similar to the recognition sequences of known NM specific TFs, including GATA4, ZIC3, EOMES, OTX2, and LHX1 (Figure S5B). Binding sites for hematoendothelium-related TFs, such as GATA2, FLI1, JUNB, and SOX7, were enriched in the hematoendothelial progenitors/Cluster 6 DAEs. In contrast, the binding motifs of GATA4, HAND1, FOXF1 and TEAD4 were highly over-represented in the JCF/Cluster 8 specific DAEs (**Figure 3A**). Among those JCF-related TFs, GATA4 and TEAD4 showed similar expression and motif activities at Clusters 2, 7, 8 of both JCF and SHF lineages. HAND1 and FOXF1 demonstrated both strong motif activities and specific expression at Cluster 7 and Cluster 8. Interestingly, the expression of HAND1 and FOXF1 seemed relatively transient at NM and EEM cells, and then became downregulated at CM of E7.75 (Figure S5C).

Based on E7.0 single-cell multi-omics data analysis, we predicted a core regulatory network (CRN) centering on the four TFs (**Figure 3B**). Functional enrichment analyses indicated that this CRN could control key aspects of JCF specification, including Wnt signaling, epithelium cell migration, cell number maintenance, mesenchyme development, and cardiac development. In the CRN, GATA4 and TEAD4 controlled larger gene sets related to the transition from epiblast to mesodermal status, necessary for both JCF and SHF. HAND1 and FOXF1 co-regulated functionally more specific gene sets critical for differentiation to EEM status (**Figure 3B**). Consistently, most HAND1 and FOXF1 target genes were specifically expressed in EEM, in contrast to GATA4 and TEAD4 target genes (**Figure 3B**).

We also performed Chromatin immunoprecipitation sequencing (ChIP-Seq) analysis to profile the chromatin occupancies of HAND1 and FOXF1 in mesoderm (MES) and cardiac progenitor (CP) cells derived from the step-wise directed cardiomyocyte differentiation of mouse embryonic stem cells[15]. We also collected published GATA4 ChIP-seq data[16] of E12.5 mouse embryonic heart, FLI1, ZIC2, ZIC3 and MESP1 ChIP-seq data[17] of 2.5 days EB differentiation. Direct comparison of ChIP-seq occupancy profiles with DAEs confirmed the specific enrichment of GATA4, HAND1, and FOXF1 at clusters 7 and 8 of JCF lineages, Fli1 at Clusters 5 and 6, while MESP1 is specifically enrichment at mesoderm cell clusters (**Figure 3C**). We also noticed the enrichment of HAND1 at early Cluster 3-specific DAEs, while FOXF1 tends to show more speicifc enrichment at cardiac specific enhancers at later CP cells.

HAND1 and FOXF1 regulate the JCF specific genes

In order to further investigate the molecular roles of HAND1 and FOXF1 in JCF specification, we generated *Hand1* and *Foxf1* KO mouse embryonic stem cell (mESC) lines (Figures S6A-D), followed by *in vitro* mesoderm differentiation and RNA-seq analyses. 2,331 down-regulated and 1,714 up-regulated genes in *Hand1* KO mesoderm (MES) cells, and 870 down-regulated and 970 up-regulated genes in *Foxf1* KO MES cells with fold-change (FC) > 1.5 and P-value < 1e-5 were identified (Supplementary Table 2). To explore whether HAND1 and FOXF1 are required for the proper expression of the JCF related genes, we examined the expression of the signature genes of the 9 cell clusters in control , *Hand1* KO and *Foxf1* KO MES cells. First of all, over 90% of the cluster specific genes were detected in the control MES transcriptome, indicating that the *in vitro* differentiation model could be a reliable tool to study the regulation of these cluster specific signature genes (**Figure 4A**). Indeed, whole transcriptome cosine similarity analysis revealed that the *in vitro* differentiated MES cells were more close to MM and EEM cell state transcriptome-wide (**Figure 4B**). *Hand1* KO and *Foxf1* KO led to down-regulated expression of a large portion of

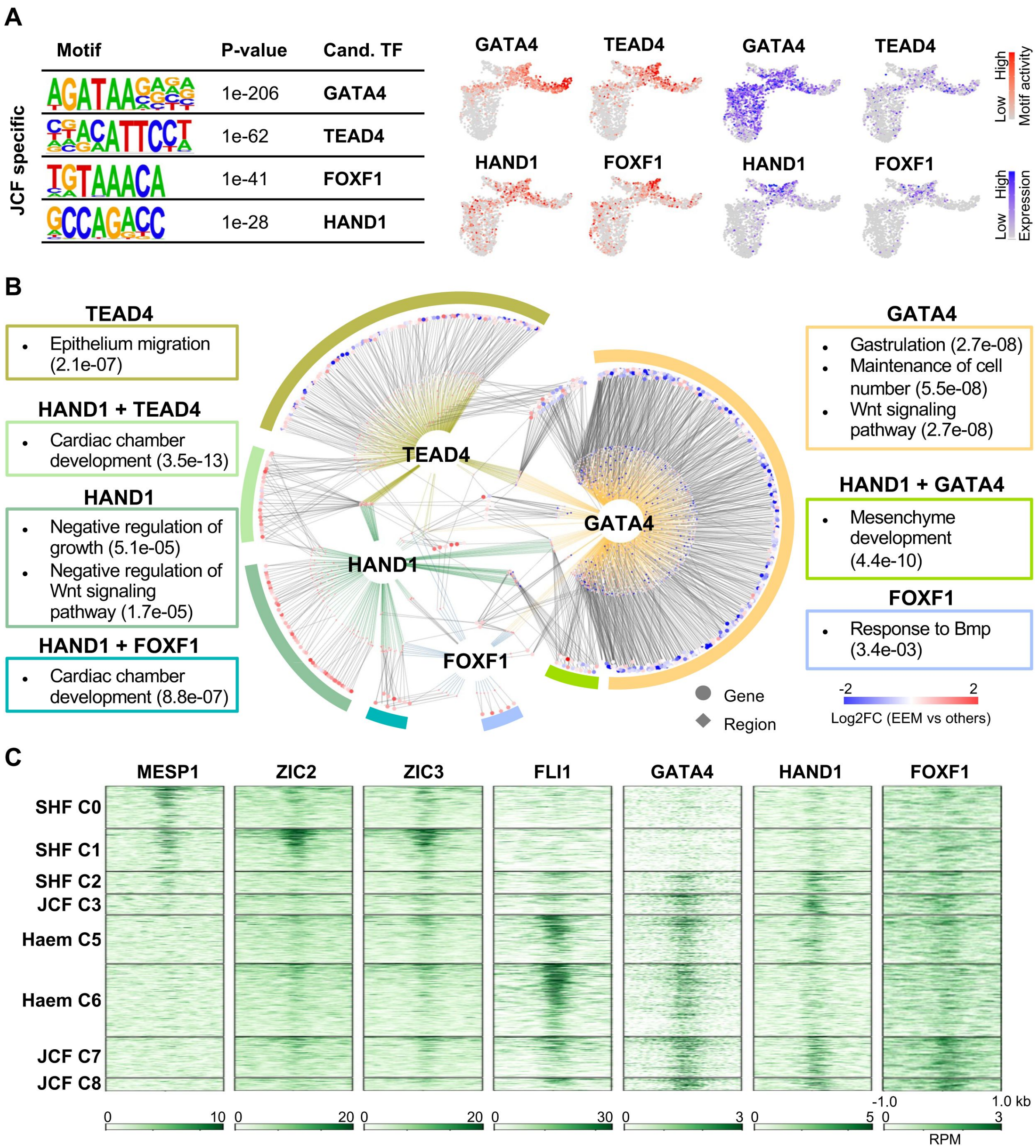

Figure 3

The CRN is identified centering on GATA4, HAND1, FOXF1 and TEAD4 in driving JCF specification.

(A) Identification of top JCF specific DNA-binding motifs and corresponding candidate TFs. The JCF specific DAEs were defined by comparing C8 with C1 snATAC-seq data using SnapATAC [30] 'findDAR' function. Motif calling was performed by the HOMER [31] 'findMotifsGenome.pl' function. Motif activity (colored in red) and TF expression (colored in blue) levels of trajectory specific candidate TFs, are overlaid on the UMAP layout from **Figure 2A**.

(B) Core regulatory network (CRN) of E7.0 EEM cells. The network is composed of TFs (GATA4, HAND1, FOXF1 and TEAD4), TF binding regions (diamond shapes), target genes (dot shapes), colored lines (TFs to regions) and grey lines (regions to genes). Colors of genes and regions representing the log2 FCs of E7.0 EEM cells over other mesodermal cells, measured by scRNA-seq and snATAC-seq data, respectively. Functional enrichment terms of target genes are shown in boxes with subtitles indicating corresponding TFs. *Q*-values, using hyper-geometric tests, are shown in parenthesis.

(C) Heatmaps representing the enrichment of MESP1, ZIC2, ZIC3, FLI1, GATA4, HAND1 and FOXF1 across the DAEs of each cluster.

the Clusters 3, 7 and 8 JCF marker genes, but up-regulated expression of many of the Clusters 0 and 1 SHF marker genes in MES cells (**Figure 4A**). Consistently, the cosine similarity suggested that *Hand1* and *Foxf1* depletion could lead to EEM differentiation defects (**Figure 4B**). In addition, many of the down-regulated JCF specific genes and up-regulated SHF specific genes were directly bound by HAND1 and FOXF1 (**Figures 4A** and S6E). The analysis indicated that HAND1 and FOXF1 were able to directly activate the JCF specific genes.

Mutual regulation between HAND1 and FOXF1 in driving cardiac specific gene expression

We found that around 50% of the dysregulated genes in *Foxf1* KO MES cells were also dysregulated in *Hand1* KO MES cells, suggesting their synergistic function in transcriptional regulation (**Figure 4C**). For example, the key JCF specific genes, including *Hand1*, *Foxf1*, *Bmp4*, *Tbx20* and *Pmp22* were significantly down-regulated, while the epiblast genes *Dnmt3b*, *Sema6a* and *Fst*, and the NM specific genes *Otx2* and *Zic2* were substantially up-regulated in both KO cells (**Figures 4C** and S6E). Depletion of *Hand1* blocked the activation of *Foxf1* during cardiac progenitor differentiation, and vice versa, while *Hand1* overexpression was able to activate *Foxf1* and the other JCF specific genes, and vice verse (Figures S7A-D).

The functional relevance between HAND1 and FOXF1 in target gene regulation could be attribuited to the enrichment of HAND1 at the *Foxf1* enhancers, and vice versa (**Figure 4D**). We then used CRISPR-Cas9 technology to delete the putative enhancers of *Hand1* and *Foxf1* (Figures S7E-F). Indeed, deletion of the HAND1-bound *Foxf1* enhancer (*Foxf1*-eH) abolished the activation of *Foxf1*; deletion of the FOXF1-bound *Hand1* enhancer (*Hand1*-eF) also significantly reduced the expression of *Hand1* during cardiac differentiation (**Figures 4E** and **4F**). Importantly, the induction of the JCF specific genes *Bmp4*, *Pmp22* and *Spin2c* was severely impaired after deletion of either *Foxf1*-eH or *Hand1*-eF. Overexpression of HAND1 in the *Hand1*-eF KO cells and FOXF1 in the *Foxf1*-eH KO cells were able to rescue the levels of these JCF specific genes during cardiac differentiation. To further investigate the role of *Foxf1*-eH in *Foxf1* expression, we performed CRISPR activation (CRISPRa) assay to activate *Foxf1*-eH (**Figures 4G**). CRISPRa of *Foxf1*-eH led to a specific increase the expression of *Foxf1* and its downstream target genes, but not its neighboring genes (**Figure 4H**). Together, our in vitro experimental data indicated that mutual regulation between HAND1 and FOXF1 could play a key role in activation of JCF specific genes.

Hand1 KO leads to MES overproliferation but cell death after exiting from the MES status

Meta analysis demonstrated that HAND1 occupied the early Cluster 3 specific DAEs, while FOXF1 tended to show more specific enrichment at the late cardiac specific Clusters 7 and 8 enhancers (**Figures 3C** and **5A**). We also noticed that the *Hand1* KO MES colonies were evidently much larger than those of WT control and *Foxf1* KO, though the same number of EB cells were seeded at equal density for MES differentiation (**Figure 5B**). While *Foxf1* KO barely affected cell proliferation rate, the count of the *Hand1* KO cells was substantially increased when the cells were differentiated from EB towards MES state (**Figure 5C**). The *Hand1* KO cells gradually lost viability upon *in vitro* cardiac lineage induction. Consistent with the increased proliferation rate observed, gene ontology (GO) analysis revealed that the genes involved in negative regulation of proliferation and positive regulation of cell migration were specifically down-regulated in *Hand1* KO, but not *Foxf1* KO, MES cells (**Figures 5D** and S7G).

The *Foxf1* KO MES colonies derived from MES differentiation appeared phenotypically normal, but were not able to further differentiate into beating CMs (**Figure 5B** and Supplementary Video 1). The expression of the mature CM markers *Tnnt2*, *Myh6* and *Myh7* were also substantially lower after *Foxf1* KO (Figure S7H). To further examine the function of FOXF1 in cardiac progenitor

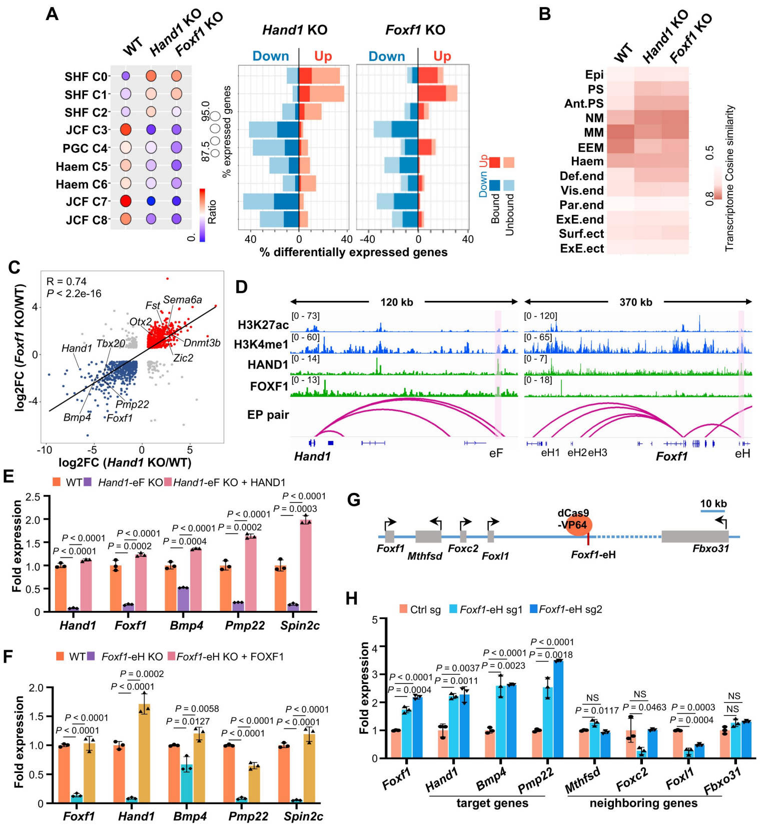

Figure 4

HAND1 and FOXF1 are mutually regulated and required for the expression of the JCF specific genes.

(A) Comparison of the genes affected by *Hand1* KO or *Foxf1* KO with cluster specific genes of E7.0 mesodermal cells. Dot plot showing that absolute (dot size) and relative (dot color) ratio of the cluster specific genes in WT, *Hand1* KO or *Foxf1* KO MES cells. >90% cluster specific genes were expressed in MES. Higher enrichment in C0-2 and lower enrichment in other clusters can be observed in *Hand1* KO or *Foxf1* KO MES cells. Bar plots showing the up- or down-regulated genes in *Hand1* KO or *Foxf1* KO MES cells. Lengths of bars representing the percentage of cluster specific genes which were up- or down-regulated. Dark colors indicating direct targets of *Hand1* (left) and *Foxf1* (right).
(B) Heatmap showing the transcriptomic similarity of MES cells and cell types of E7.0 mouse embryo. MES transcriptome data were generated using bulk RNA-seq. Cell-type specific transcriptomes of E7.0 mice were determined as the average of single-cell transcriptomes from each cell type. The gene set for comparison was defined as the collection of top 50 marker genes of each E7.0 cell type. Cosine similarity metric was used. Ant.PS, anterior primitive streak. Haem, haematoendothelial progenitors. Def.end, def.endoderm. Vis.end, visceral endoderm. Par.end, parietal endoderm. ExE.end, ExE endoderm. Surf.ect, surface ectoderm. ExE.ect, ExE ectoderm.
(C) Scatter plot showing gene expression FCs after *Hand1* KO or *Foxf1* KO. Dots representing genes with FC > 1.5 and adjusted P (Padj)-value < 1e-5 (Wald tests). Red and blue colors indicating genes co-activated/inhibited in *Hand1* KO and *Foxf1* KO cells. Correlation coefficient of FCs between *Hand1* KO and *Foxf1* KO is 0.74, P-value < 2.2e-16, t-test.
(D) Representative genome browser snapshots showing the localization of H3K27ac, H3K4me1, HAND1 and FOXF1 at the *Hand1* and *Foxf1* loci. The FOXF1-binding *Hand1* enhancer (*Hand1*- eF) and the HAND1-binding *Foxf1* enhancer (*Foxf1*-eH), for enhancer KO experiments, are highlighted.
(E-F) RT-qPCR showing that the reduction in RNA levels of the EEM marker genes after *Hand1*- eF KO can be rescued by HAND1 OE (e), and after *Foxf1*-eH KO can be rescued by FOXF1 OE.
(G) Cartoon illustrating target *Foxf1*-eH of by dCas9-VP64 and the genes in the vicinity of *Foxf1*- eH.
(H) RT-qPCR showing that the substantial increase in RNA levels of *Foxf1* and the FOXF1 target genes, but not the *Foxf1*-eH neighbouring genes in *Foxf1*-eH CRISPRa cells. Data are the mean ± standard error of the mean from three independent experiments. Two-tailed unpaired Student's t-test was performed.

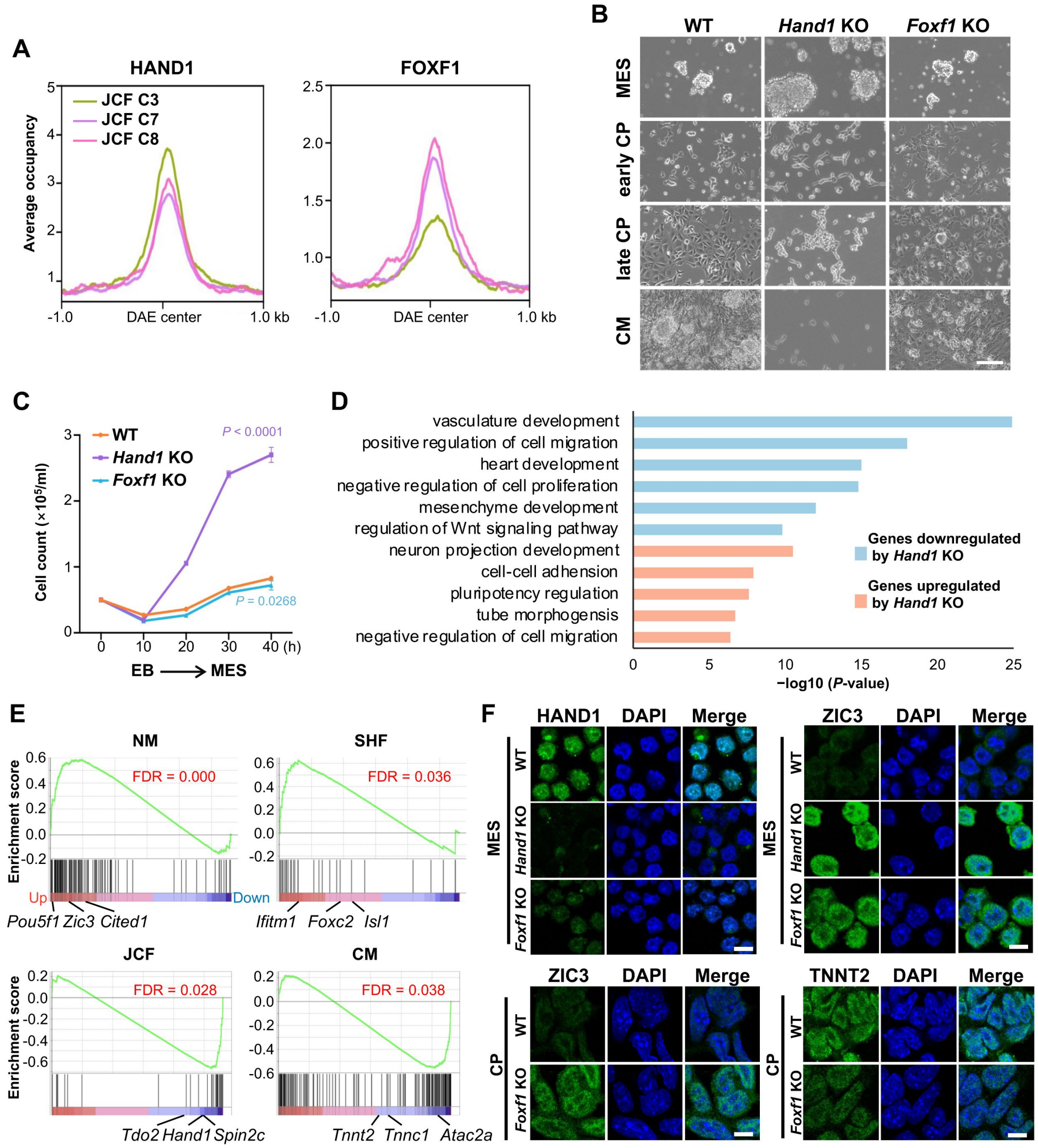

Figure 5

HAND1 and FOXF1 are required for the cardiac lineage specification.

(A) Meta plot analysis showing the occupancies of HAND1 (left) and FOXF1 (right) at the C3/7/8 JCF specific DAEs from **Figure 2D**.

(B) Bright field microscopic images of WT, *Hand1* KO and *Foxf1* KO cells at MES, early CP, late CP and CM stages. Scale bar: 100 μm.

(C) Growth curve showing numbers of WT, *Hand1* KO and *Foxf1* KO cells counted at different time intervals during EB to MES differentiation. Data are the mean ± standard error of the mean from three independent experiments. Two-tailed unpaired Student's *t*-test was performed.

(D) Enriched GO terms of the genes up- or down-regulated after *Hand1* KO. One-sided Fisher's Exact test with Benjamini-Hochberg multiple testing correction was performed.

(E) GSEA showing the distribution of the marker genes within the ranking of *Foxf1* KO affected genes of *in vitro* differentiated CP cells (FDR calculated by permutation tests). *Foxf1* KO affected genes were ranked from up- (red) to down-regulation (blue).

(F) Immunofluorescence staining of HAND1, ZIC3, and TNNT2 in WT and *Hand1/Foxf1* KO lines at MES/CP stage. Scale bar: 10 μm.

specification, we performed RNA-seq analysis in control and the CP cells derived *Foxf1* KO mESCs. Gene Set Enrichment Analysis (GSEA) revealed the expression levels of the NM specific genes (like *Zic3*, *Pou5f1* and *Cited1*) and the SHF specific genes (like *Isl1*, *Ifitm1* and *Foxc2*) were remarkably up-regulated, while the expression levels of the JCF specific genes (like *Hand1*, *Spin2c* and *Tdo2*) and the early CM specific genes (like *Acta2*, *Tnnc1* and *Tnnt2*) were significantly reduced (**Figure 5E**). Thus, our data further supported the specific and synergistic roles of HAND1 and FOXF1 in JCF cardiac progenitor specification. In addition, we performed IF staining of mesodermal (ZIC3), JCF (HAND1) and cardiac markers (TNNT2), followed by cell quantification (**Figure 5F**). Results indicate that *Hand1* and *Foxf1* knockout leads to reduced commitment to the JCF lineage, evidenced by the loss of *Hand1* expression, accumulation of undifferentiated ZIC3+ mesoderm, and impaired cardiomyocyte formation (TNNT2+), consistent with the up-regulation of JCF lineage specific genes and the down-regulation of SHF lineage specific genes. These results suggest that HAND1 and FOXF1 may cooperatively regulate early cardiac lineage specification by promoting JCF-associated gene expression and suppressing alternative mesodermal programs.

Genetic loss of *Hand1* blocks the specification of mesoderm along JCF

To further invesigate the roles of HAND1 in JCF specification *in vivo*, we generated floxed allele of *Hand1* by inserting loxP sites flanking the exon 1 of the *Hand1* gene. Genetic crosses of the *Hand1*$^{fl/fl}$ mice with *Mesp1-Cre* mice[7] allowed specific deletion of *Hand1* in mesodermal cells (Figure S8A). Consistently, *Foxf1* is also co-expressed in HAND1-positive EEM cells and its expression was also drastically down-regulated in MESP1-CRE driven *Hand1* conditional KO (*Hand1* CKO) embryos (**Figure 6A**). Mesodermal deletion of *Hand1* generally led to smaller embryos at E7.0 and also later stages (Figure S8B). Data suggest that by E8.5 when heart looping initiate in control group (14/17), the hearts of *Hand1* CKO embryos (3/3) still demonstrate a linear tube morphology (Figure S8C). By E9.5 when atrium and ventricle become distinct in WT embryos, heart looping of *Hand1* CKO embryos is abnormal (**Figure 6B**). The *Hand1* CKO embryos appeared to die by E9.5 due to embryonic turning failure and heart looping abnormality, mirroring the previously reported phenotype of *Hand1* KO mice [18, 19].

To assess how *Hand1* loss affects early mesoderm development *in vivo*, we analyzed the single- cell transcriptomics of control and *Hand1* CKO embryos at E7.0. Integrated analysis indicated the loss of EEM cells, but the abnormal accumulation of primitive streak (PS), NM and MM cells in *Hand1* CKO embryos (**Figures 6C** and **6D**), which suggested that specific deletion of *Hand1* in mesodermal cells strongly affected early mesoderm differentiation. Detailed analysis regarding the percentage of each cell type revealed the specific reduction of cell numbers from EEM, hematoendothelial progenitors, and ExE ectoderm cell clusters in *Hand1* CKO embryos (**Figure 6D**). To further support this finding, we performed immunofluorescence staining of the EEM marker *Vim*, which revealed a marked reduction of *Vim*+ cells in *Hand1* CKO embryos compared to controls, indicating impaired progression toward the JCF lineage (Figure S8D).We then compared the gene expression pattern for each cell cluster and calculated the significantly affected genes between Control and *Hand1* CKO embryos (**Figure 6E**). Consistently, EEM cell cluster was the most affected clusters upon *Hand1* loss (**Figures 6E** and S8E). Although the percentages of hematoendothelial progenitors and ExE ectoderm cells were reduced, the expression of both cell type specific marker genes did not seem affected drastically. To further illustrate the developmental progression of the mesodermal lineage, we performed paired URD lineage inference analysis[20], further confirming the specific development block of NM specification towards EEM in JCF (**Figure 6F**). We also performed label transfer analysis to identify JCF and SHF progenitor cells from the E7.0 scRNA-seq data (Figure S9A). Our analysis showed that the fraction of JCF progenitors increased by over 2-fold, whereas the fraction of SHF progenitors remained unchanged (Figures S9B and S9C), supporting that *Hand1* deletion driven by MESP1- CRE led to accumulated JCF progenitor cells and blocked the JCF direction of mesoderm differentiation.

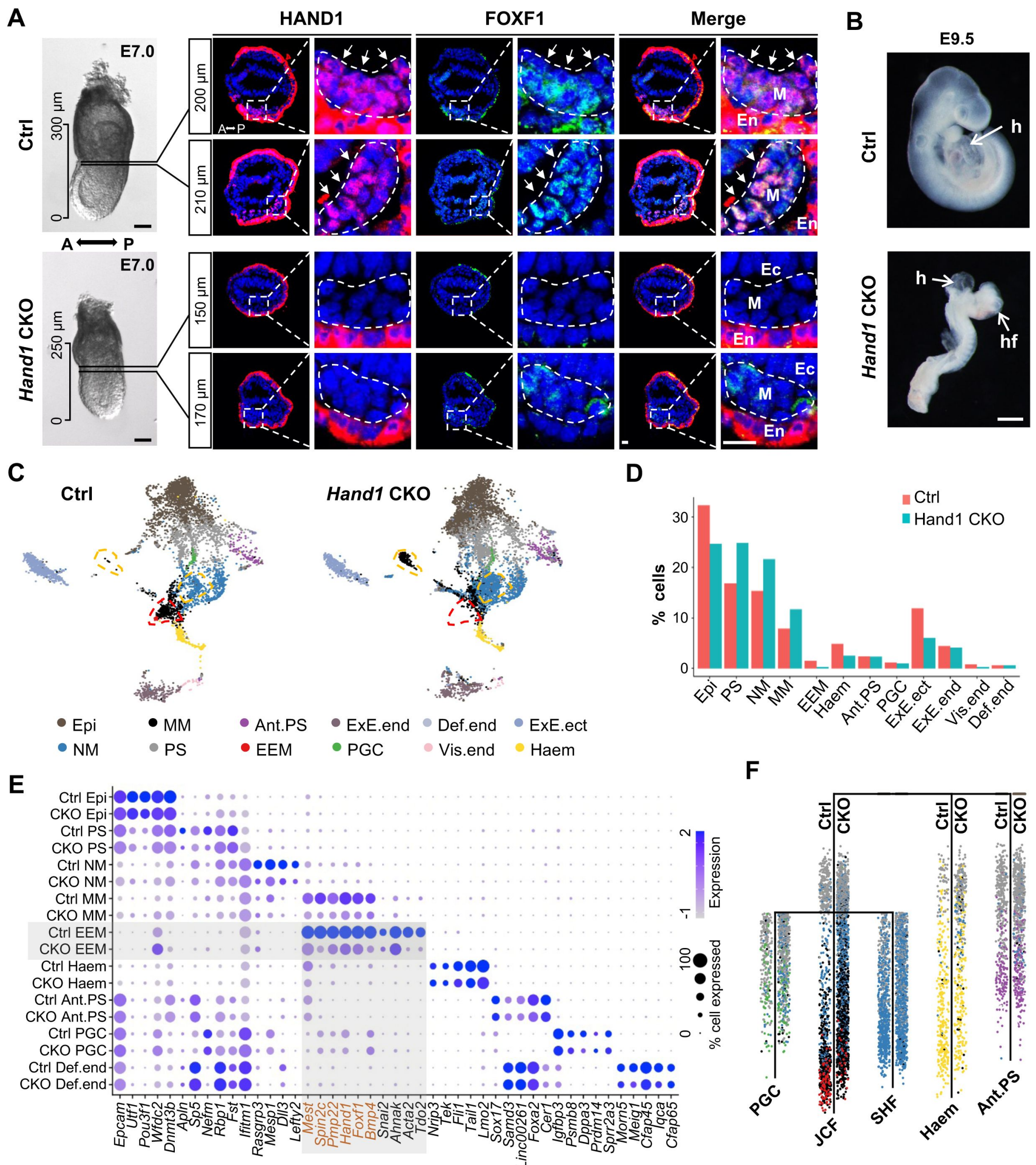

Figure 6

Hand1 KO in mesodermal cells blocks JCF specification in mouse embryos.

(A) Immunofluorescence staining showing a substantial reduction in protein levels of HAND1 and FOXF1 at the embryonic-extraembryonic boundary transverse sections (indicated by dashed lines and arrows) of E7.0 *Hand1* CKO embryos. Bright filed images (lateral view) of corresponding embryos are shown on the left side. The distance for each section to the distal tip of the embryo is labelled at the upper right corner for each image. Scale bars: 100 μm (bright field images, left) and 20 μm (transverse sections, right).
(B) The bright field images of E9.5 Control (Ctrl) and *Hand1* CKO mouse embryos. The arrows indicating the embryonic heart (h) and head folds (hf). Scale bar: 500 μm.
(C) UMAP layout of integrated E7.0 mouse embryo scRNA-seq data. Integration was performed using the scRNA-seq data of Ctrl and *Hand1* CKO mice. Colors indicating cell types. Dashed lines emphasizing cells with increased (red) or decreased (yellow) numbers in *Hand1* CKO mice.
(D) Bar plot showing the percentage of each cell type of Ctrl and *Hand1* CKO mice.
(E) Dot plot showing marker gene expression in Ctrl and *Hand1* CKO mice. Red highlighting markers of MM and EEM cell type.
(F) URD inferred trajectory tree revealing the developmental hierarchy of E7.0 mesoderm cells (Ctrl and *Hand1* CKO snRNA-seq), colored by cell types. Ctrl and *Hand1* CKO cells are distributed on both sides of the URD tree trunk, with Ctrl cells on the left and *Hand1* CKO cells on the right.

The abnormal accumulation of PS, NM and MM cells in *Hand1* CKO embryos appeared to be consistent with the phenotypes observed in *in vitro* mesoderm differentiation of *Hand1* KO mESCs (**Figures 4B** and **5B**). Notably, the genes involved in negative regulation of proliferation and down- regulated in *Hand1* KO MESs were also reduced in the JCF, but not SHF, trjectory of E7.0 *Hand1* CKO embryos (**Figure 7A**). In addition, cell migration related genes were also affected in *Hand1*- depleted MES and embryos. To further validate this phenotype, we performed sequential DAPI staining on cryo-sectioned E7.0 control and *Hand1* CKO embryos, followed by cellular segmentation and cell density measurement (**Figures 7B** and **7C**). The analysis revealed that the mesoderm cells near the extraembryonic region in *Hand1* CKO embryos were more compacted (**Figures 7B**'', 7C'', and 7D), while the distal region of the *Hand1* CKO embryos showed no obvious difference from the control embryos (**Figures 7B**''' and 7C'''). In addition, reduced exocoelomic cavity (EC) size and increased number of mesodermal cells in extraembryonic region were also observed in *Hand1* CKO embryos (**Figures 7B**' and 7C'). Hematoxylin and eosin (H&E) staining of E7.5 embryos also supported reduced embryo size and accumulated cells at mesodermal regions upon loss of *Hand1* (**Figure 7E**). These data together establish HAND1 as a factor in promoting the specification of mesodermal cells toward JCF.

Discussion

In this study, we described the transcriptional trajectories and epigenetic landscapes of early cardiac specification event in mouse embryos, and identified a predicted CRN underlying the early mesodermal lineage specification. Mechanistically, we demonstrated that this earliest cardiac JCF specification event was tightly regulated by HAND1 and FOXF1. HAND1 and FOXF1 were mutually regulated, but also performed their respective functions in JCF cell fate determination. *In vitro* differentiation and *in vivo* mouse model analyses indicated that HAND1 was essential for exiting from NM program during cardiac specification. Depletion of FOXF1 impaired the capacity of mesodermal cells in cardiac specification (**Figure 7F**). In sum, our findings provided new insights into the transcriptional determinants that specify the early cardiac progenitors, paving the way for the identification of potential therapeutic targets for treating congenital heart defects.

Recent studies using scRNA-seq have reported the roadmaps of mammalian embryonic lineage development[12, 21]. Together with studies which focus on cardiac progenitors[4, 6, 22], several models have been proposed to explain the multi-lineage process of heart formation. In this study, we observed a clear divergence between the JCF and SHF trajectories around E7.5-E7.75, and performed forward-backward tracing to reconstruct their developmental paths. Although cells were assigned to distinct trajectories based on dominant fate probabilities, fate divergence analysis revealed that JCF and SHF likely originate from a common progenitor pool prior to E7.0. The two trajectories are consistent with the previous clonal analysis and also the HAND1+ cells lineage tracing by Zhang et al. [23]. Importantly, we here identified specific trajectory-related gene cohorts throughout the whole process, shedding light on the molecular mechanisms by which HAND1 is crucial for promoting the exit of NM program during the early JCF specification process.

MESP1 serves as a master regulator in the establishment of the cardiac lineage. An analysis of previously published MESP1-regulated genes revealed that GATA4, HAND1 and FOXF1 were directly controlled by MESP1. GATA4 was rapidly induced by MESP1 within 12 hours, while HAND1 and FOXF1 were activated after 24 hours of MESP1 induction. On a pseudotime scale, we noticed a sequential-temporal expression pattern of GATA4, HAND1 and FOXF1 in JCF. Our transcriptional and epigenomic analyses suggested a sequential activation model of MESP1-GATA4-HAND1-FOXF1. However, it should be noted that MESP1 and GATA4 were activated in both JCF and SHF. Thus, additional regulation must exist and instructs the process of JCF-SHF lineage segregation. It has been reported that BMP signaling activated the expression of *Hand1* during heart formation [24]. Since Bmp4 displayed higher specificity for JCF, it might explain the

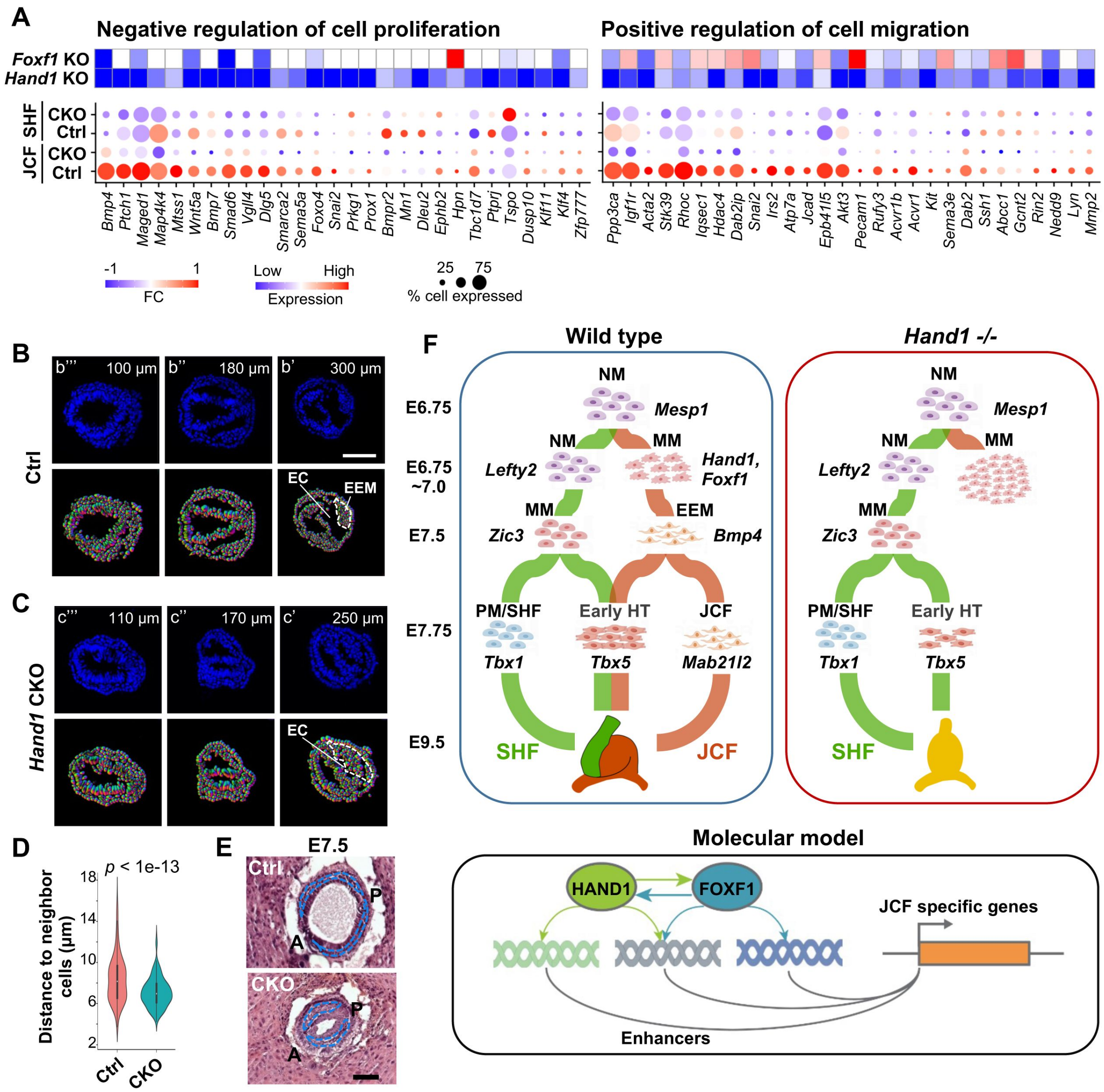

Figure 7

HAND1 is required for exiting from the NM program.

(A) Co-regulated changes in the genes related to negative regulation of cell proliferation (left) and positive regulation of cell migration (right) *in vivo* and *in vitro*. Heatmap showing the fold changes (FC) of those co-regulated genes after *Hand1* KO and *Foxf1* KO in MES cells *in vitro* (top). Dot plot showing expression levels of those co-regulated genes after *Hand1* KO in the JCF and SHF trajectories *in vivo* (bottom).

(B-C) DAPI staining in the transverse sections of E7.0 Ctrl (b) and *Hand1* CKO (c) embryos. The distance for each section to the distal tip of the embryo was labelled at the upper right corner for each image. Dashed white lines denoting locations of the exocoelomic cavity (EC) and NM.

(D) Violin plot showing the distance between neighboring mesoderm cells in the transverse sections of E7.0 Ctrl and *Hand1* CKO embryos. *P*-value was calculated using one-sided Mann- Whitney U test.

(E) H&E staining showing reduced embryo size and accumulation of mesodermal cells of E7.5 *Hand1* CKO mouse embryos. Scale bars: 100 μm.

(F) Working model of early cardiac lineage differentiation. Cardiac fate specification initiates around E6.75-7.0 by segregation of the JCF and SHF trajectories. At E7.0, multipotent progenitor cells in JCF lineage present *Hand1*+ mixed mesodermal (MM) status, in contrast to SHF cells showing *Lefty2*+ nascent mesodermal (NM) status. JCF and SHF specifically contribute to JCF and SHF, respectively, while they both contribute to the formation of early heart tube (HT). HAND1 and FOXF1 bind to enhancers, regulate the JCF specific genes, and drive mesodermal differentiation toward JCF direction. *Hand1* deletion led to blocked JCF specification and accumulation of early mesodermal cells.

activation of *Hand1*, and subsequent *Foxf1* activation in JCF, but not the SHF. Thus, HAND1 and FOXF1 might be subject to feed-forward activation by MESP1 and GATA4 in concert with BMP signalling, thereby promoting JCF cardiac specification. Future studies are worthwhile to further investigate how TFs work together with signaling pathways to promote cardiac lineage segregation.

We further demonstrated the synergistic roles of HAND1 and FOXF1 in early cardiac specification. *Foxf1*-/- mice were not able to survive beyond E10.0, with incomplete separation of splanchnic and somatic mesoderm, leading to abnormal coelom formation. Interestingly, FOXF1 marked the mesothelium lining (a monolayer epithelial lining of the exocoelomic cavity), the anterior distal part of which extends to the presumptive cardiac mesoderm [25]. Such an expression pattern is coincident with that of HAND1 in JCF [4]. The role of FoxF is also essential for the early cardiac development in the ascidian Ciona intestinalis [26]. During the early heart development, HAND1 demonstrates multi-facet functions and contribute to the formation of FHF [6], heart loop [19], and outflow tract [27]. A recent study suggests that *Hand1* knockout biases progenitor cells toward SHF lineage, affecting the expression of cardiomyocyte markers and delaying differentiation onset in human pluripotent stem cells [28]. Our studies showed that FOXF1 was indispensable for the specification of mesoderm toward early cardiac progenitors, through binding the JCF progenitor specific enhancers for their later activation. Together, the combined analyses on the developmental route of CMs and their transcriptional determinants will further enhance our understanding of the etiology behind congenital heart defects, ultimately providing insights into potential regenerative strategies for heart disease treatment.

STAR Methods

Mice

All mouse experiments were approved by the Animal Care and Use Committee at Southeast University, and performed in accordance with institutional guidelines. Mice were housed in cages under SPF conditions and had free access to water and food. *Hand1*$^{fl/fl}$ mice were produced by Cyagen with loxP sites inserted into the regions surrounding exon 1 in the *Hand1* locus. *Hand1*$^{fl/+}$; *Mesp1-Cre* mice were generated by crossing *Hand1*$^{fl/fl}$ mice with *Mesp1-Cre* mice [32].

The *Hand1*$^{fl/fl}$ or *Hand1*$^{fl/+}$ female mice caged with male *Hand1*$^{fl/+}$; *Mesp1-Cre* mice. Females were screened for vaginal plugs following morning (E0.5). To obtain post-implantation embryos, female mice at 7.0 or 9.5 days post-coitum (d.p.c.) were executed and the uteri were dissected and transferred to a petri dish with PBS. For 7.0 d.p.c embryos, each decidua was carefully freed from the uterine muscle layers using properly sharpened forceps. Then the embryos were carefully separated from decidua. Reichart's membrane and the ectoplacental cone were also removed from the 7.0 d.p.c embryos. The uteri surrounding the 9.5 d.p.c embryos were cut open with a small incision, the embryos were genetly squeezed out, and then the amniotic membranes were removed. Images of the embryo were acquired on a stereo microscope (Mshot, MZ62; Olympus, SC180).

Genomic DNA for mouse genotyping was obtained from mouse tail biopsies. Genotyping of mouse embryos was performed with genomic DNA, which was obtained by collecting extra- embryonic regions subsequently digestion using Mouse Direct PCR Kit (Bimake, B40013). PCR reactions were used to detect the *Cre* transgene and the *Hand1* loxP site. Thermal cycle reactions were as follows: 3 min at 94 °C, 35 cycles of 30 s at 94 °C, 35 s at 60 °C, 45 s at 72 °C and a final 5 min extension at 72 °C. Primer sequences used in this study are listed in Supplementary Table 3.

Antibodies

Antibody against HAND1 (sc-390376) (WB: 1:1000, IF for embryos: 1:100, IF for cells: 1:200) was purchased from Santa Cruz. Antibody against FOXF1 (Abclonal, A13017) (WB: 1:2000, IF for embryos: 1:100, IF for cells: 1:500) was purchased from Abclonal. Antibody against HA (ChIP: 3 μg) was generated in house. Antibody against TUBULIN (66031-1-Ig) (WB: 1:50, 000) was purchased from Proteintech. Goat anti-mouse IgG Alexa Fluor 488(A-11001) and goat anti-rabbit IgG Alexa Fluor 546 (A-11035) were purchased from Thermo Fisher Scientific.

Mouse ESC culture

Mouse E14 ESCs were maintained in DMEM (Hyclone, SH30243.01) supplemented with 15% FBS, 1 × nonessential amino acids (STEMCELL, 07600), 1 × GlutaMAX (Gibco, 35050-061), 1 × penicillin streptomycin solution (Sangon Biotech, E607011-0500), b-mercaptoethanol (ALDRICH), 0.1 μg/ml leukemia inhibitory factor (Novoprotein, C690), 3 μM CHIR99021 and 1 μM PD0325901 in gelatin-coated plates at 37 °C under 5% CO2.

CRISPR-Cas9 guided KO

SgRNAs targeting *Hand1*, *Foxf1* and their enhancer sites were cloned into lentiCRISPR v2. These constructs were transfected into 70% confluent 293T cells together with 6 μg of psPAX2 packaging plasmids and 2 μg of pMD2.G envelope plasmids using Highgene (Abclonal, RM09014). The media was half-replaced with fresh DMEM supplemented with 10% FBS 6 h after transfection. The lentiviral supernatants were harvested at 24 h, 48 h, and 72 h post-transfection, filtered through 0.45 μm filters, and concentrated at 50,000 × g for 0.5 h. Mouse E14 ESCs were infected with concentrated lentiviral particles with polybrene (Sigma) at the concentration of 8 μg/ml. 2 μg/ml puromycin was used for selection for 48 h and individual colonies were picked and expanded in 48-well plates. The clones were screened with genomic PCR, and confirmed by TA cloning and Sanger sequencing. Oligo sequences used in this study are listed in Supplementary Table 3.

In vitro cardiac differentiation

Mouse E14 ESCs were differentiated into embryoid bodies (EBs) at a density of 75,000 cells/ml in 6 cm dishes for 48 h in serum-free media (DMEM (Hyclone, SH30243.01), DMEM/F12 (1:1) (Gibco, 11320-033), 0.05% BSA (Sigma, A1933), 1 × GlutaMax (Gibco, 35050-061), B27 supplement (Gibco, 12587010), N2 supplement (Gibco, 17502048), supplemented with 50 mg/ml ascorbic acid (Alfa Aesar, 50-81-7) and 4.5×10^{-4} M monothioglycerol). EBs were dissociated into single cells and re-aggregated as MES cells for 40 h at 50,000 cells/ml in the presence of 5 ng/mL human VEGF (Novoprotein, C083) and 10 ng/ml human Activin A (Peprotech, 96-120-14-10) and 0.3 ng/ml human BMP4 (R&D, 5020-BP-010). MES cells were dissociated and plated as a monolayer in gelatin-coated 12-well plate at 50,000 cells/ml in StemPro-34 (Gibco, 10639011) supplemented with 5 ng/mL VEGF (Novoprotein, C083), 10 ng/mL human basic FGF (Gibco, PMG0035) and 25 ng/mL FGF10 (R&D, 345-FG-250). CP cells were harvested after differentiation for 32 h. Contracting CMs can be observed after another 5 days. Differentiation stages were confirmed by the expression of marker genes.

Immunofluorescence

Mouse Embryos: Embryos were fixed in 4% PFA, then rinsed in PBS and incubated in 30% sucrose at 4 °C until sinking. Embryos were then embedded in OCT medium and snap-frozen in liquid nitrogen and stored at −80 °C. Sections were cut at a thickness of 10 μm. To perform IF, embryo sections were washed three times for 5 min each, then antigen repair was performed using citrate. The samples were permeated for 40 min with 0.3% TritonX-100 followed by washing three times with PBS for 5 min each. The samples were then blocked with ReadyProbes 2.5% Normal Goat Serum (Thermo) for 1 h at room temperature. The samples were incubated overnight at 4 °C with primary antibodies followed by washing three times with PBS for 10 min each. The samples were incubated with secondary antibodies for 1 h at room temperature, then washed three times with PBS for 10 min each. The samples were then incubated with DAPI at room temperature for 10 min followed by washing one time with PBS for 5 min. After the final wash, the samples were mounted with mounting buffer. Images were captured using a Zeiss 700 laser confocal microscope.

MES and CP cells were dissociated into single cells and fixed on 12-well plate cell slide with 4% PFA at room temperature for 15 min. The fixed cells were washed 3 times with PBS and permeabilized with 0.2% Triton X-100 for 5 min at room temperature. The cells were then blocked with PBS containing 2% BSA and 0.3% TritonX-100 for 1 h at room temperature. Appropriate dilution of primary antibody was added and cells were incubated overnight at 4 °C. After washing with PBS for 3 times, the cells were stained with secondary antibodies and DAPI for 1 h at room temperature followed by mounting on slides. Images were captured using a Zeiss 700 laser confocal microscope.

H&E staining

Whole E7.5 embryos were fixed overnight with 4% PFA, embedded vertically in clean paraffin, then sliced to obtain 7 μm paraffin sections. Standard hematoxylin and eosin (H&E) staining methods were used to stain the sections. The images of H&E staining were obtained on a microscope (Olympus, IX73).

Western blotting

Cells were washed twice in ice-cold PBS and incubated in 1 × SDS lysis buffer for 15 min at 95 ℃. Lysates were pre-cleared by maximum speed centrifugation for 3 min, then separated by SDS-PAGE and transferred to polyvinylidene fluoride (PVDF) membrane. 5% non-fat dry milk was used for blocking and primary antibody was added. The membrane was incubated overnight at 4 °C, then washed, incubated with secondary antibody for 1 h at room temperature. ECL substrate was used for imaging by autoradiography.

Quantitative RT-PCR and bulk RNA-seq

Total RNA was extracted from cells using RNA isolater Total RNA Extraction Reagent (Vazyme). 500 ng of total RNA was used to synthesize cDNA using ABScript II RT Mix (Abclonal, RK20403). Resultant cDNA was diluted in water and 12.5 ng cDNA was used in each qRT-PCR reaction. Reactions were run on CFX96 (Bio-Rad) using 2 × Universal SYBR Green Fast qPCR Mix (Abclonal, RK21203). The relative expression levels of genes of interest were normalized to the housekeeping gene *Actin*. Each experiment contains at least three biological replicates. Primer sequences used in this study are listed in Supplementary Table 3. For RNA-seq, 1 × 10^6 MES or CP cells were harvested and RNA extraction was performed using Rneasy mini plus kit (Qiagen). 1 μg of total RNA was used for the construction of sequencing libraries and sequencing.

snRNA-seq

To prepare single cells for snRNA-seq, embryos at E7.0 were dissected in cold sterile 1 × PBS without Ca^{2+}, Mg^{2+} under a stereo microscope. Embryos were staged based on their morphology. The Reichert's membrane and ectoplacental cone were removed, and genotypic identification of embryos was carried out. Embryos were placed into a 1.5 ml microfuge tube and digested into single cells using TrypLETM Express (Thermo) at 37 °C. Single-cell RNA sequencing was performed on single-cell suspensions using 10 × Genomics.

10 × Multiome library preparation and high throughput sequencing

Six embryos staged at E7.0 were collected and washed with cooled 0.5% BSA/PBS for twice. Sufficient embryos were pooled and then subjected to 200 μL TrypLE for cell dissociation for 10 min at 37 °C with frequent gentle mixture. Single-cell suspension of embryos were then quenched and washed with 0.5% BSA/PBS, and finally filtered using 40 μm Flowmi cell strainer. The acquired single cell suspension were then subjected to nuclei isolation, library preparation by following the manufacturer's instruction. The library was sequenced on Novaseq 6000 platform with recommended sequencing depths and read lengths.

ChIP, ChIP-seq library preparation

$5 \times e^6$ cells were crosslinked with 1% formaldehyde for 10 min and quenched with 0.125 M glycine for 5 min at room temperature. Fixed cells were incubated in 0.5 ml lysis buffer (50 mM Tris-HCl [pH 8.0], 1 % SDS, 5 mM EDTA) for 10 mins on ice. After adding 1 ml dilution buffer (20 mM Tris-HCl [pH 8.0], 150 mM NaCl, 2 mM EDTA, 1% Triton X-100), chromatins were sonicated into 200-800 bp fragments using Bioruptor (Diagenode) and immunoprecipitated with protein A agarose beads and specific antibody at 4 °C for 12 h. Immunoprecipitates were washed with Wash Buffer I (20 mM Tris-HCl [pH 8.0], 150 mM NaCl, 2 mM EDTA, 1% Triton X-100, 0.1% SDS), Wash Buffer II (20 mM Tris-HCl [pH 8.0], 500 mM NaCl, 2 mM EDTA, 1% Triton X-100, 0.1% SDS), Wash Buffer III (10 mM Tris-HCl [pH 8.0], 0.25 M LiCl, 1 mM EDTA, 1% deoxycholate, 1% NP-40) and TE, respectively. After the final wash, DNA was eluted and reverse-crosslinked at 65 °C for at least 6 h. DNA was then purified and used for PCR amplification or ChIP-seq library preparation. ChIP-seq libraries were prepared with VAHTS Universal DNA Library Prep Kit for sequencing.

Data processing and quality control

Single cell multi-omics data: sequence alignment

We used Cell Ranger ARC software suite for the analysis of single cell multi-omics (ATAC & Gene Expression) data (*https://support.10xgenomics.com/single-cell-multiome-atac-gex/software/pipelines/latest/what-is-cell-ranger-arc*). We applied the "cellranger-arc count" function for barcode counting, adapter/primer removal and sequence alignment. The processed reads were aligned to mm10 using the BWA-MEM algorithm. For ATAC sequencing data, BAM files were generated for downstream analysis. For Gene Expression (GEX) data, "cellranger-arc count" generated gene-barcode matrices were used for further analysis. Version of Cell Ranger ARC genomic sequence and gene annotation: refdata-cellranger-arc-mm10-2020-A-2.0.0.

snRNA-seq data

snRNA-seq datasets of the current study include: (1) E6.5-8.5 whole mouse embryos; (2) micro-dissected anterior cardiac regions of mouse embryos at early crescent to linear heart tube (~E7.75- 8.25) stages. (3) GEX part of the single cell multi-omics data of E7.0 mouse embryos. (4) Control and *Hand1* CKO E7.0 mouse embryos.

We used R package 'Seurat' for QC and normalization purposes [33]. Cells with abnormal sequencing depth (nFeature_RNA < 2000 or nCount_RNA > 1e5) or with high mitochondrial ratio (percent.mt > 5) were excluded. We used the 'SCTransform' to perform normalization, variance stabilization, and regression of cell cycle scores (using the 'CellCycleScoring' function). For dataset (1), we selected E7.0 samples, removed cells labelled as 'doublet' or 'stripped' and regressed out sequencing batches. Doublet removal was performed using R package 'DoubletFinder' [34].

snATAC-seq data

Bam files of snATAC-seq from multiome data were processed using Snaptools [30] function 'snap- pre' to remove low-quality fragments (MAPQ < 30), over-sized fragments (length > 1,000 bp), secondary alignments and PCR duplicates. This function also generates cell-by-bin matrices of variable resolution (bin sizes: 1kb, 5kb, 10kb) for downstream analysis. The range of fragment coverage for bin selection was set between 500 and 20,000.

ChIP-seq data analysis

Adapter sequences were removed from Fastq files using trim_galore (v0.6.7) with default parameters. After trimming, ChIP-seq reads were aligned to the mm10 mouse genome using Bowtie2 (v2.3.5.1) [35] with '--no-mixed' and '--no-discordant' parameters. The aligned files were sorted and converted to BAM format using SAMtools (v1.9). BigWig files were subsequently generated using deepTools bamCoverage (v3.5.0) [36], employing CPM normalization and ignoring duplicates. Peak calling was performed with MACS3 (3.0.0a6) [37] for HAND1 (*Q*-value < 1e-5) and FOXF1 (*Q*-value < 1e-10) ChIP-seq data.

Tracing the CM, JCF and SHF trajectories

The trajectories of each in **Figure 1** were inferred using the Waddington-OT Python package (v1.0.8) [11] with a predefined starting cell set (E8.5 CM cells for tracing the CM trajectory, E7.5 EEM cells in the CM trajectory for tracing JCF trajectory, and E7.75 pharyngeal mesoderm cells in the CM trajectory for tracing the SHF trajectory), and cells of each trajectory are selected if WOT score > 0.0001. To specifically distinguish between JCF and SHF, the difference in the WOT score of the two trajectories was calculated, and if the difference is greater than 0, it belongs to the JCF trajectory and vice versa to the SHF trajectory. WOT is designed for time-series scRNAseq data where the time/stage each single cell is given. At any adjacent time points ti and ti+1, WOT estimates the transition probability of all cells at ti to all cells at ti +1. One can select a cell set of interest at any time point ti infer their ancestors at ti -1 or their descendants at ti +1 by sums of the transition probabilities. The WOT package was used with default parameters as in the Waddington-OT online tutorial (*https://broadinstitute.github.io/wot/tutorial/*).

Signaling pathway enrichment analysis

Potential signaling-activated/inhibited genes of Bmp, Yap, Wnt, Nodal, Notch and Fgf were collected from Peng et al. 2019 [38]. Enrichment of signaling target gene sets was determined by the average expression levels of activated/inhibited genes. To assess dynamic changes along pseudotime, we calculated the smoothed expression trends of these gene sets using LOESS regression. Notably, an increase in the expression of inhibited genes over pseudotime may reflect a gradual reduction in signaling activity or potential compensation by alternative pathways. These interpretations are consistent with the dynamic and context-dependent nature of signaling regulation during early development.

Spatial mapping of single cells

Locations of mouse E7.0/7.5 mesodermal cells were inferred by comparison with the GEO-seq data, where each sample represents 5–40 cells with defined spatial locations. Transfer component analysis (TCA) [39] was performed to achieve shared representation of scRNA-seq and GEO-seq samples. Single cells were mapped to GEO-seq locations with the highest correlation coefficients.

Identification of pseudotime-dependent genes

Single-cell pseudotime trajectory analysis was performed using R package Monocle 2 (v2.22.0) [40] according to the online tutorials (*http://cole-trapnell-lab.github.io/monocle-release/*). Monocle object was directly constructed using Monocle implemented new Cell Data Set function from Seurat (v4.1.1) object, and Monocle implemented differentialGeneTest function was used to find highly variable genes for ordering. Based on these, we selected key and specific genes in JCF and SHF trajectory, respectively, and further visualized them in the heatmap using cell bin-by-gene matrix.

snRNA-seq data integration and label transfer

This research included data integration for the following datasets (dataset numbers in 'Data processing and quality control: snRNA-seq data'): (1) and (2) for comparison of the predicted trajectories and JCF cells; (1) and (3) for cell-type annotation of multiome single cells; (1) and (4) for cell-type annotation of Ctrl/*Hand1* CKO single cells. For multiome dataset, integration was performed by two steps: 1) whole embryo integration; 2) according to the predicted cell types of (1), mesoderm specific integration was done by selecting relevant cell types (NM, MM, EEM, Haem, PGC).

The integration and label transfer process follows the pipeline provided by Seurat [33]: *https://satijalab.org/seurat/articles/integration_introduction.html*. 4,000 genes were selected for integration based on variable gene sets of individual datasets. We selected "SCT" as the normalization method for identification of integration or transfer anchors.

Clustering analysis of snRNA-seq data

We performed clustering analysis for cells of the integrated mesodermal lineage. Seurat [33] function "FindNeighbors" and "FindClusters" were applied to the Top 20 principle compoments with the following parameter setting: annoy.metric = "cosine", resolution = 0.95. This analysis resulted in nine clusters (C0-8), which is also used for snATAC-seq analysis.

FHF and SHF gene signature scores

For each gene set, we defined the average z-score normalized expression levels as the signature score per cell. The FHF and SHF gene sets were collected from Soysa et al. 2019 [22] Supplementary Table 1 (FHF: "FHF" genes of tab "Figure1de_n = 21,366 cells"; SHF: "MP" genes of tab "Fig 2ab_n = 2,103 cells").

snATAC-seq data analysis

snATAC-seq bam files were combined for each cluster, C0-8, for identification of accessible elements (AE). AEs were then collected as a basic set of peak annotation. We used Snaptools [30] function 'add_pmat' to generate a cell-by-peak matrix (pmat), where the value of each element represent the accessibility per cell per peak. Diffusion map followed by UMAP analysis was performed to generate the snATAC-seq version of 2D data visualization.

Cluster specific AEs were identified using SnapATAC function 'findDAR'. For analysis of C2-8 specific AEs, we used C1, which represents the least differentiated cell group, as the background cluster. For C0-1, we used cells from all other clusters as background. Cluster specific AEs were provided to HOMER [31] function 'findMotifsGenome.pl' for motif analysis, and to deepTools [36] for plotting heatmaps of ChIP-seq data.

Definition of enhancer regulated target genes

Enhancer-promoter (EP) pairs were predicted by SnapATAC function ‘predictGenePeakPair’ [30]. This method performs logistic regression using peak accessibility (snATAC-seq) and expression (snRNA-seq) of neighboring genes to identify EP links. Candicate EP pairs need to be less than 50kb apart. Cluster specific target genes were defined as EP-associated genes of cluster specific AEs. Functional enrichment analysis was performed using R package clusterProfiler [41] (‘compareCluster’ function, ontology was selected as ‘BP’).

Analysis of TF activity

We analyzed TF activity, defined as the relative accessibility of AEs containing the motif sequence[42] of a TF, using R package chromVAR (*https://greenleaflab.github.io/chromVAR/articles/Introduction.html*). SnapATAC-generated cell- by-peak matrices were converted to ‘SummarizedExperiment’ format as the input of chromVAR. The TF activity is calculated by the ‘computeDeviations’ function by comparing the accessibility of motif-containing AEs with background peaks with similar GC content. Genomic coordinates of motifs were acquired by the ‘getMatrixSet’ function and R package ‘JASPAR2020’. Synergy scores between TF pairs were computed by the ‘getAnnotationSynergy’ function.

Classification of activation modes of gene-enhancer pairs

We performed pseudotime trajectory analysis for E7.0 JCF trajectory (C0, C3, C7 and C8) in multiome using the abovementioned method. To further explore the underlying mesoderm developmental mechanism of enhancer clusters in regulating gene expression, we first built gene-enhancer pairs for preselected clusters based on snATAC-seq and snRNA-seq datasets. Secondly, we partitioned cells from E7.0 mesoderm lineage into pseudotime bins and calculated the cell bin- by-gene matrix of snRNA-seq and the cell bin-by-peak matrix of snATAC-seq, respectively. Finally, we divided activation modes of gene-enhancer pairs into three groups, fast, sync., and slow, according to the order of bins corresponding to their activation time.

Inference for URD lineage tree

To reconstruct branching developmental trajectory trees for E7.0 mesoendodermal cells (Ctrl and *Hand1* CKO), we used URD (v1.1.1) [20]. We used all cells assigned as epiblast as the root of the tree, and cells of clusters that contained the most differentiated cell-types at the latest pseudotime as the tips, where the CM branch was divided into JCF and SHF trajectories. By the movement of coordinates, the cells of Ctrl and *Hand1* KO are located on either side of the branch, respectively.

CRN construction and in silico KO

The CRNs based on multi-omics (scRNA + snATAC) data was constructed by SCENIC+ (v1.0.1) (ref) (details available via *https://scenicplus.readthedocs.io/en/latest/pbmc_multiome_tutorial.html*). CRN-based in-silico KO was conducted by SCENIC + module "scenicplus.simulation". In this step the expression level of a TF was set to zero and SCENIC + propagates the effect through CRN in an iterative manner to obtain the perturbed expression matrix. Cell-state transition vectors were generated and projected to the tSNE map using "plot_perturbation_effect_in_embedding" function.

Cell nucleus segmentation

We employed the cellpose (V2.1.1) [43]algorithm for segmentation of DAPI staining for embyo transverse sections. The pre-trained CP model (--pretrained_model CP) was used to obtain the masks of images. We used option “--diameter 8” to resize the image to conform to the input parameters of the pre-trained model. We utilized the MorphoLibJ plugin of Fiji to obtain morphological metrics of the segmented nuclei. Finally, we utilized the VAA3D (2.938) [44] software to perform partitioning of the embryo, to identify cells belonging to the endoderm, mesoderm or epiblast.

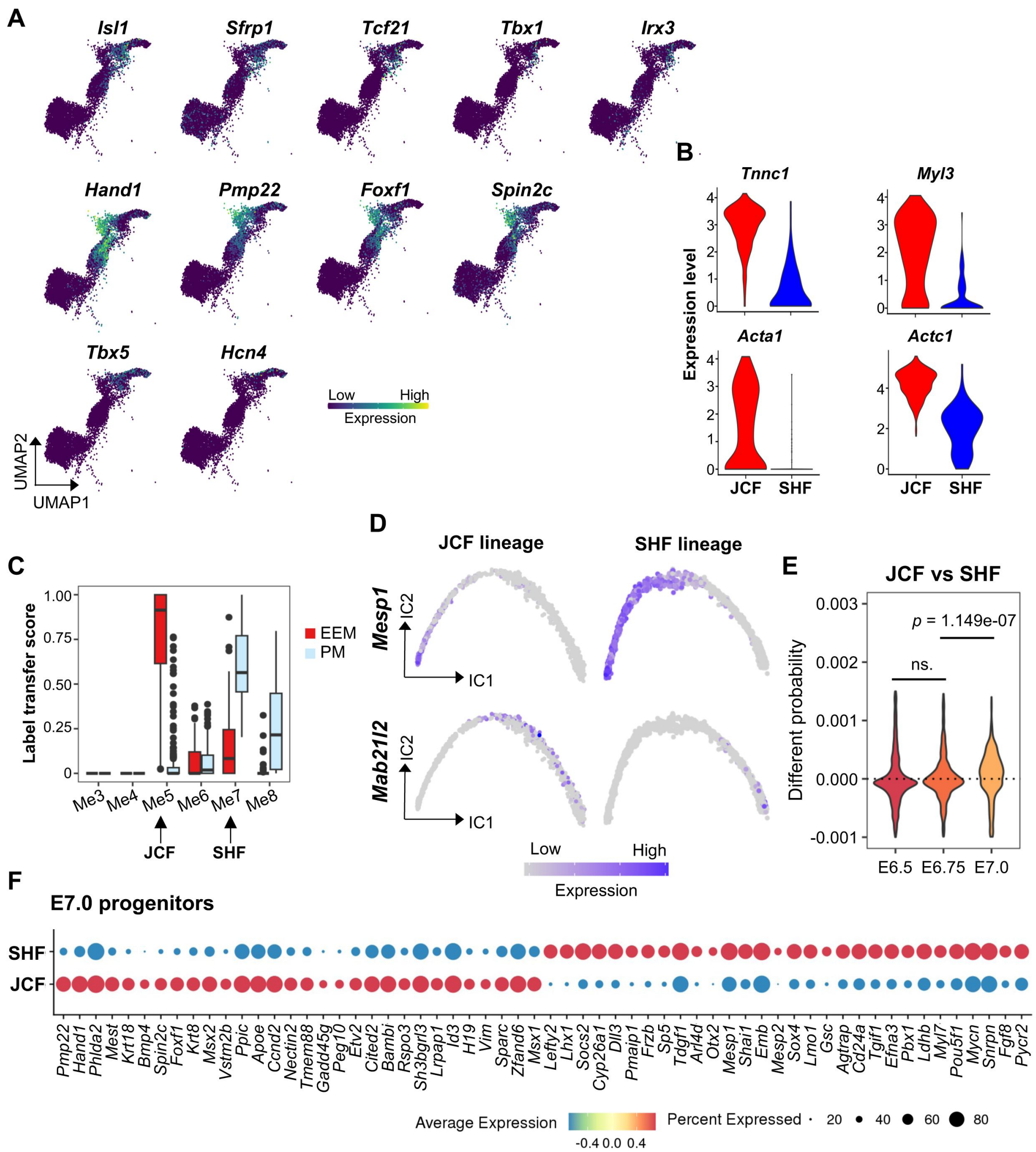

Figure S1

Data integration showing the developmental trajectories of cardiac progenitors

(A) Gene expression levels of the SHF (upper), JCF (middle) and FHF markers (lower) overlaid on the UMAP layout of the CM trajectory. Only cells belonging to the CM trajectory are shown.

(B) Violin plots showing the distribution of mature myocardium markers expression levels in E8.5 CM cells of the JCF and SHF trajectories.

(C) Fraction of labels transferred from Tyser et al. [4] for E7.5 EEM cells and E7.75 PM cells.

(D) Independent component (IC) layout showing pseudotemporal trajectories for JCF trajectory (left) and SHF trajectory (right) cells, colored by *Mesp1* and *Mab21l2* expression.

(E) Inference for fate divergence between two trajectories during E6.5-7.0. The violin plots show the differential WOT scores of JCF and SHF lineages. Two-tailed unpaired Student's *t*-test was performed.

(F) Marker gene analyses based on JCF/SHF lineages for E7.0 progenitors.

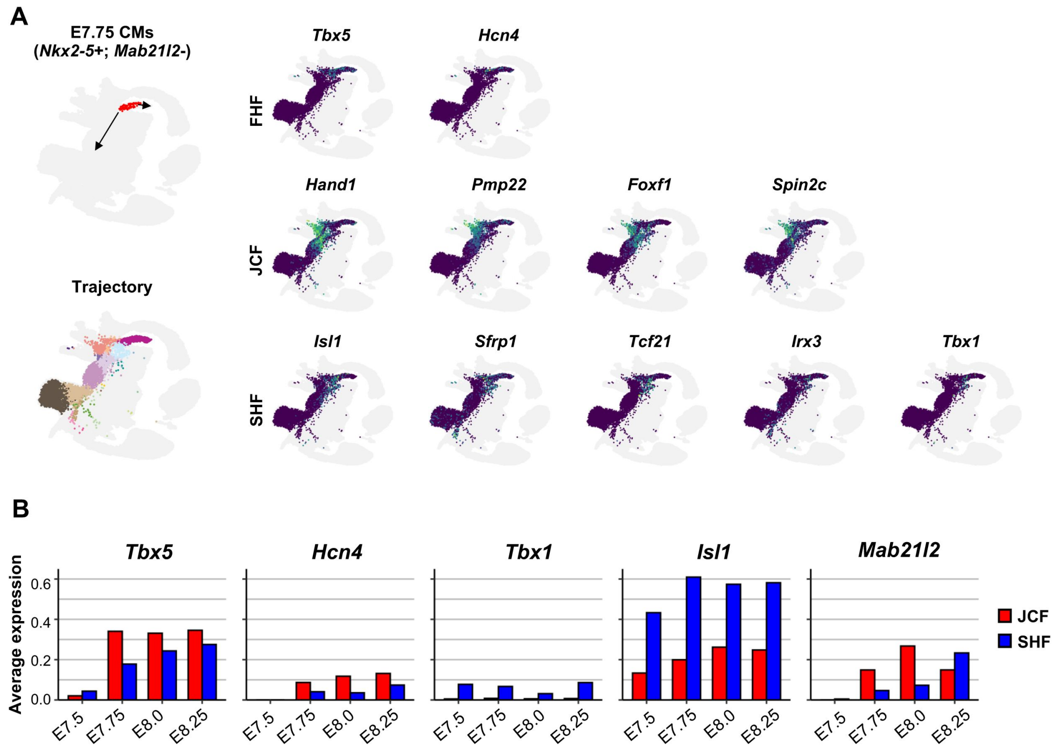

Figure S2

JCF and SHF progenitors contribute to the FHF

(A) WOT analysis for early FHF progenitor population. UMAP layout from Pijuan-Sala et al. [12] is highlighted by cells belonging to the WOT predicted developmental trajectories for CM (E7.75 *Nkx2-5+*; *Mab21l2-* CM cells). Expression levels of representative marker genes for FHF, SHF JCF and SHF, projected onto the same UMAP.

(B) Barplots indicates variable contribution of JCF and SHF to FHF. Bar hights represent the expression levels of FHF (*Tbx5, Hcn4*), SHF (*Tbx1, Isl1*) and JCF (*Mab21l2*) marker genes, averaged among single cells of each lineage.

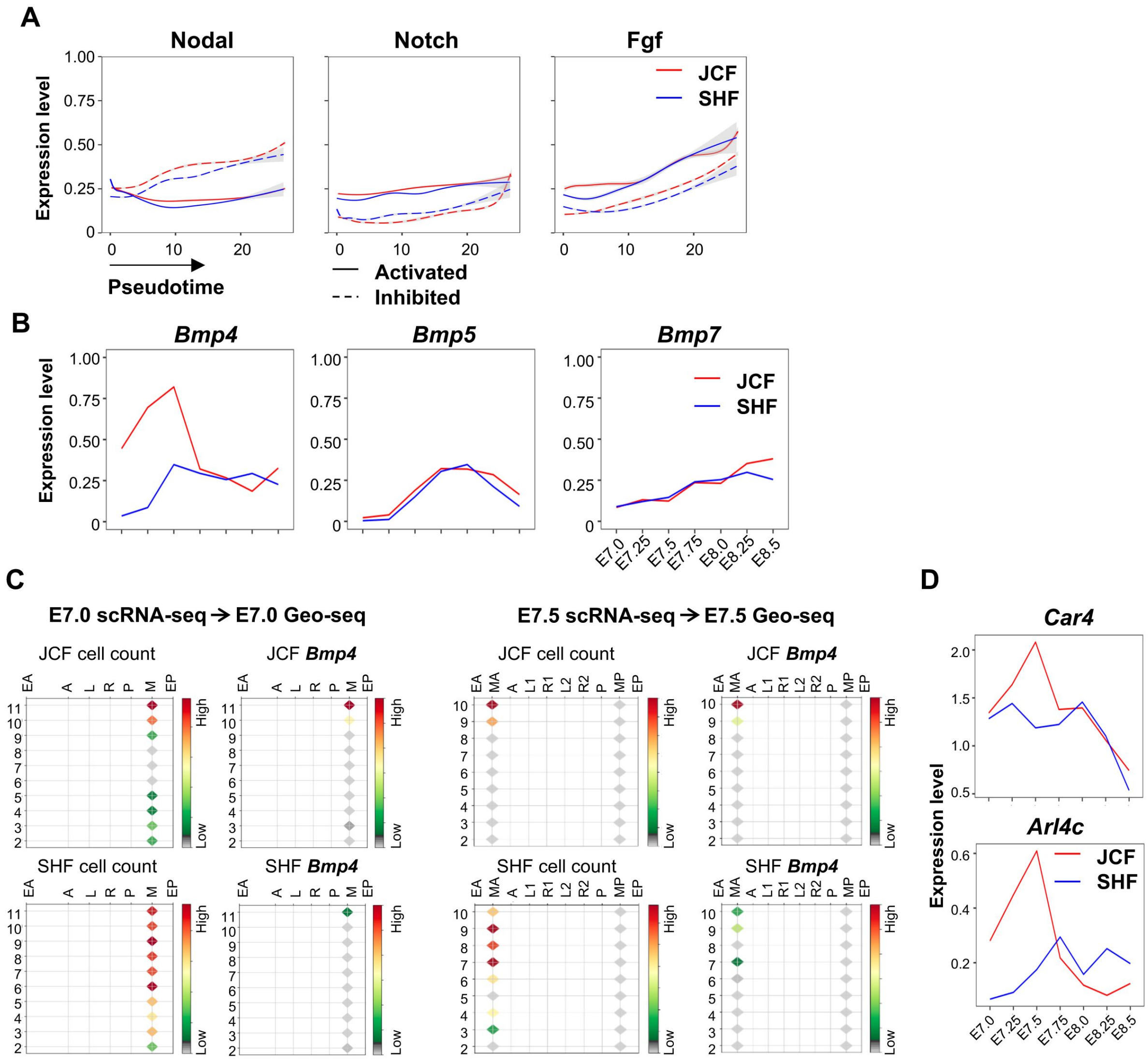

Figure S3

Differential gene expression and signaling activaity in JCF and SHF

(A) Smoothened fitting curves showing expression levels of activated (solid line) and inhibited (dotted line) signaling target genes in JCF (red) and SHF (blue).
(B) Dynamic expression of *Bmp4*/*5*/*7* in JCF and SHF between E7.0-E8.5 stages.
(C) Corn plots showing spatial JCF (upper) and SHF single cell mapping at the E7.0 (left) and E7.5 (right) stages. Columns representing micro-dissected locations in germ layers and rows representing distal (slice 2) to proximal ends (slices 10/11) of the mouse embryos. Colors indicating the number of single cells mapped to each location, or the aggregated expression level of *Bmp4*. EA, anterior endoderm; EP, posterior endoderm; M, whole mesoderm; MA, anterior mesoderm; MP, posterior mesoderm; A, anterior ectoderm; P, posterior ectoderm; L, left lateral ectoderm; R, right lateral ectoderm.
(D) Dynamic expression of the Bmp signaling target genes *Car4* and *Arl4c* in JCF and SHF between E7.0-E8.5 stages.

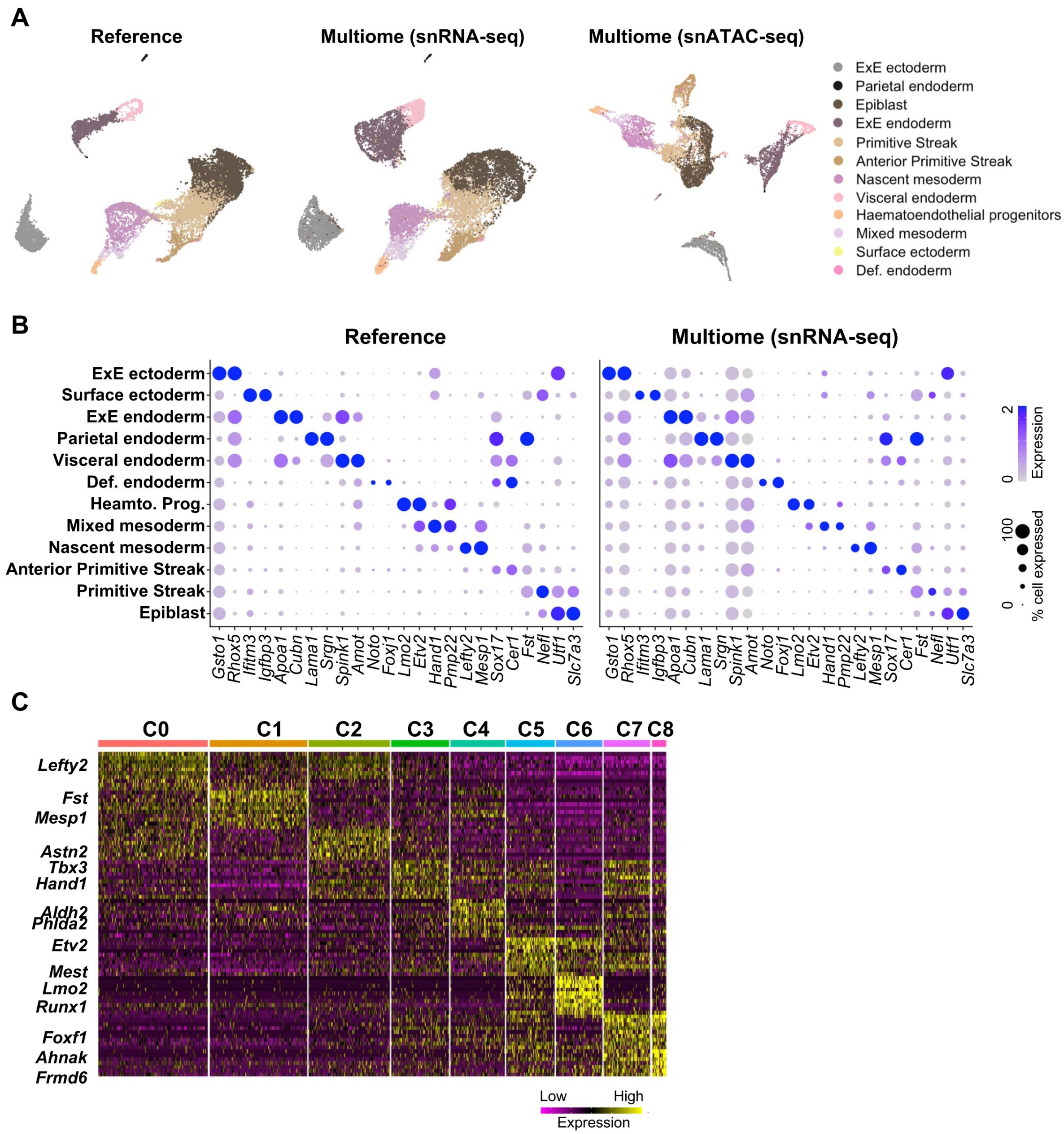

Figure S4

Data integration of E7.0 multi-omics snRNA-seq with reference scRNA-seq

(A) UMAP layout of E7.0 snRNA-seq or snATAC-seq data. UMAPs of snRNA-seq was generated by integrating published [12] and multi-omics data of this study. Cells are colored by cell types.

(B) Dot plot showing marker gene expression in reference and multi-omics dataset.

(C) Heatmap showing expression levels of marker genes across mesodermal clusters of E7.0 mouse embryos.

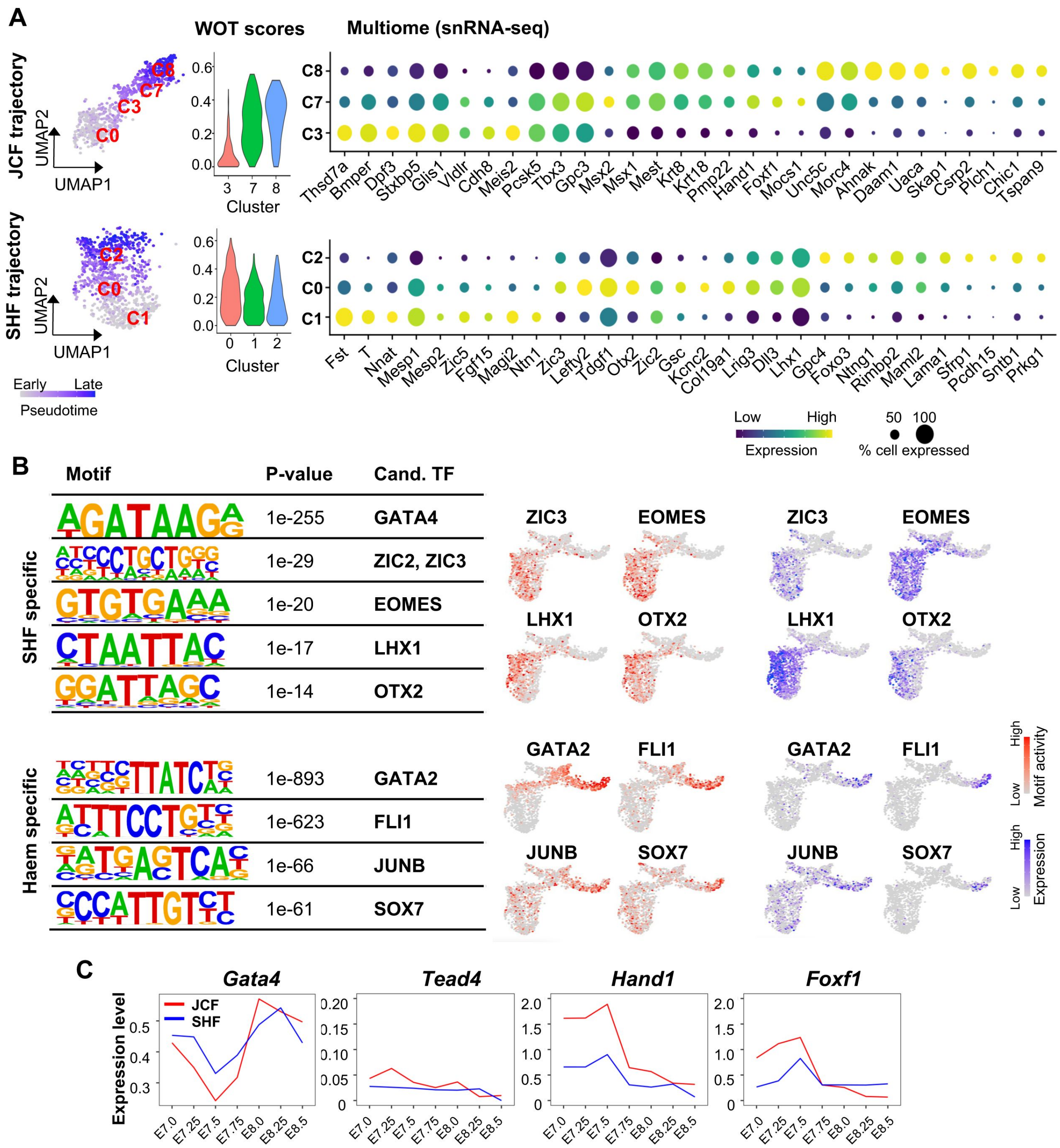

Figure S5

Identification of the key lineage specific TFs

(A) UMAP layout (left) of the JCF (C0/3/7/8) and SHF (C1/0/2) developmental trajectories. Color shades indicating pseudotime for developmental stages. Violin plots (middle) showing the WOT score of C3/7/8 and C0/1/2 cells belong to JCF and SHF lineages, respectively. Dot plots (right) showing the marker gene expression for JCF specific clusters (C3/7/8) and SHF specific clusters (C1/0/2) in snRNA-seq.

(B) Identification of top SHF (upper)/Haem (lower) specific DNA-binding motifs and corresponding candidate TFs. The SHF/Haem specific DAEs were defined by comparing C2/C6 with C1 snATAC-seq data using SnapATAC [30] 'findDAR' function. Motif calling was performed by the HOMER [31] 'findMotifsGenome.pl' function. Motif activity (colored in red) and TF expression (colored in blue) levels of trajectory specific candidate TFs, are overlaid on the UMAP layout from **Figure 2A**.

(C) Dynamic expression of *Gata4*, *Tead4*, *Hand1* and *Foxf1* in JCF and SHF between E7.0-E8.5 developmental stages.

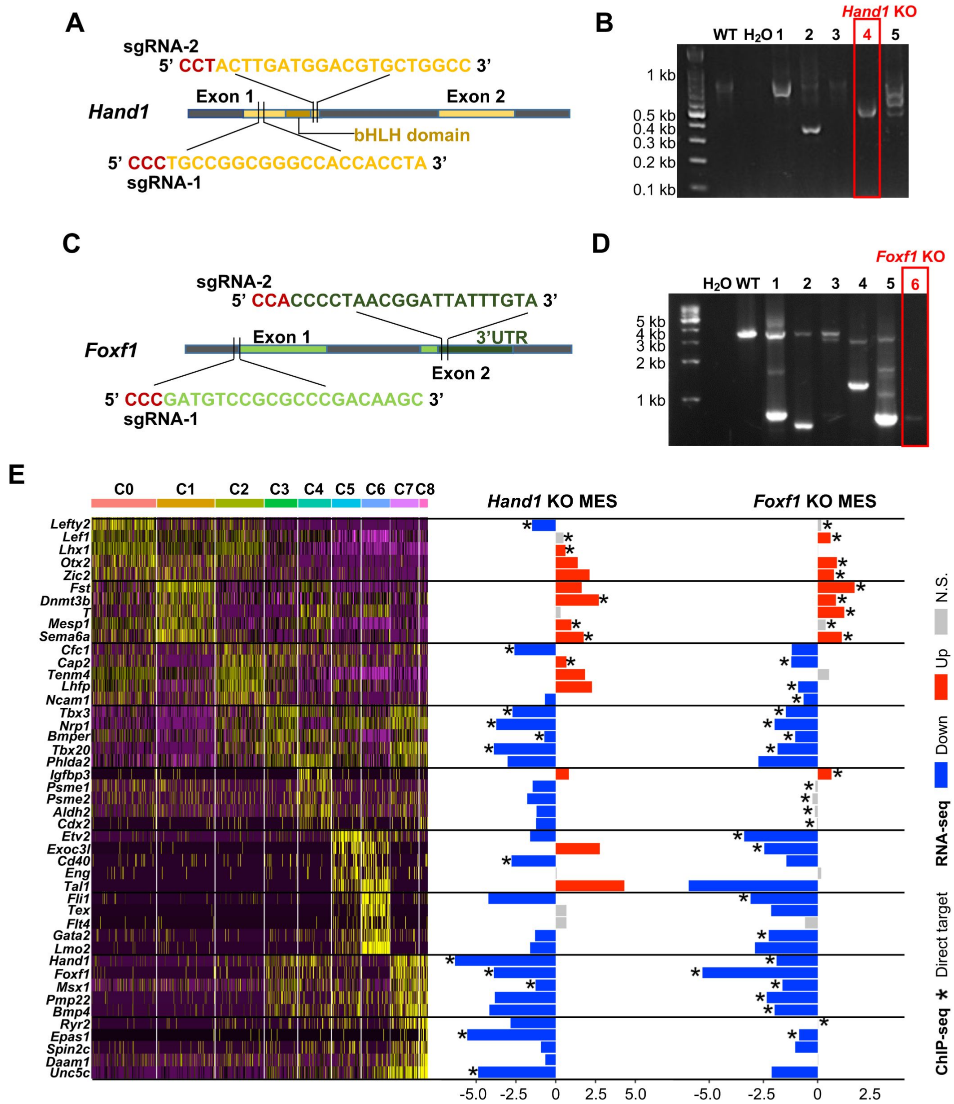

Figure S6

Generation of the Hand1 and Foxf1 KO ESC lines

(A,C) Schematic diagram of the positions of sgRNAs targeting the CDS of *Hand1* (A) and *Foxf1* (C).
(B,D) Genomic PCR analyses using the primer sets flanking the cleavage sites verifying the genomic DNA deletion of *Hand1* (B) and *Foxf1* (D).
(E) Heatmap showing expression levels of the marker genes across mesodermal clusters of E7.0 mouse embryos (left). Bar plots showing expression FC upon *Hand1* and *Foxf1* KO (right). Asterisks indicating direct binding by the corresponding TFs.

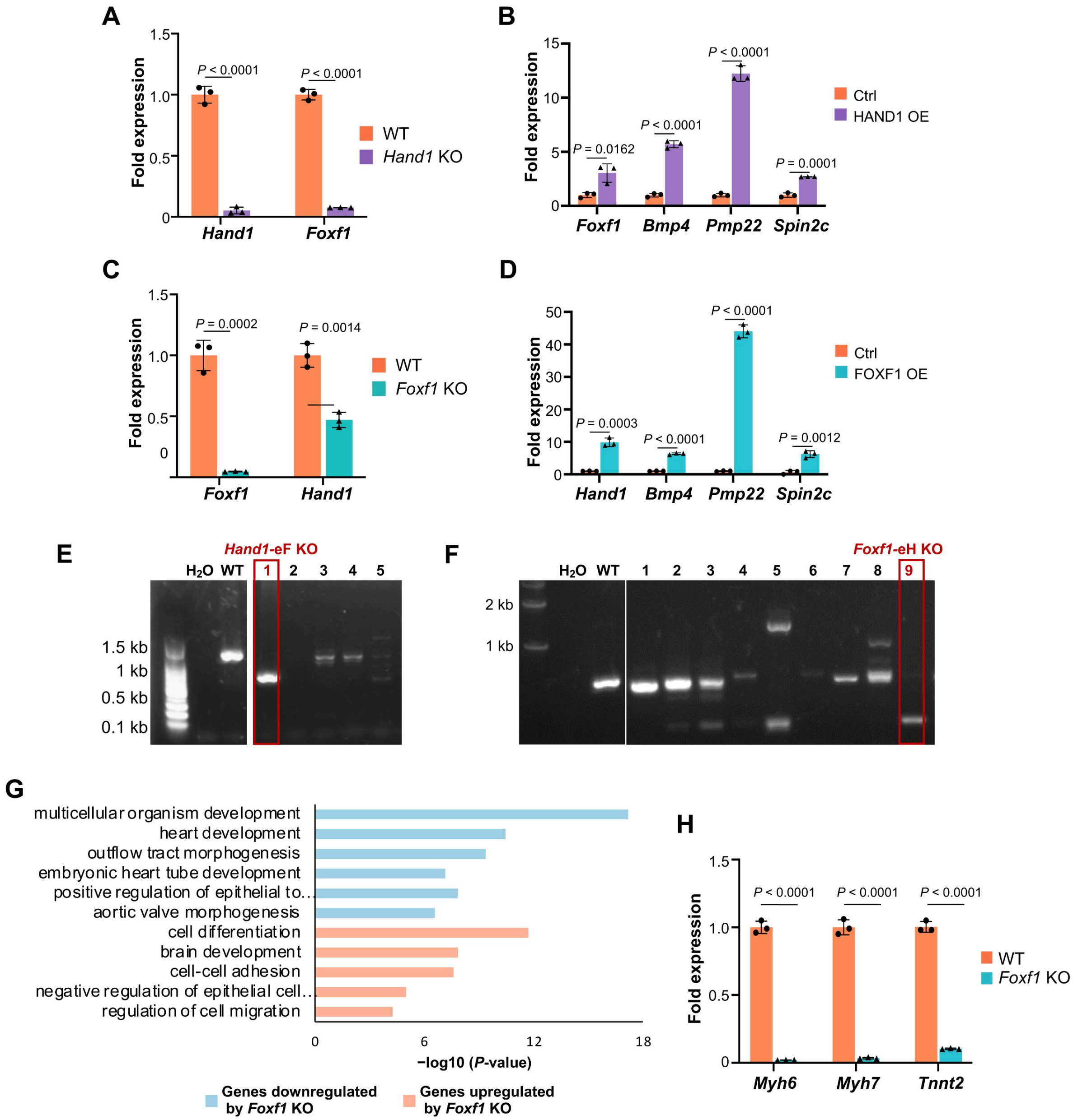

Figure S7

Mutual regulation between HAND1 and FOXF1

(A-D) RT-qPCR showing the levels of HAND1 and FOXF1 in *Hand1* and *Foxf1* KO, the EEM marker genes (*Foxf1, Bmp4, Pmp22* and *Spin2c*) after HAND1 and FOXF1 OE.
(E-F) Genomic PCR analyses using the primer sets flanking the cleavage sites verifying the genomic DNA deletion of the *Hand1* enhancer (a) and the *Foxf1* enhancer (c).
(G) Enriched GO terms of the genes up- or down-regulated after *Foxf1* KO. One-sided Fisher's Exact test with Benjamini-Hochberg multiple testing correction was performed.
(H) RT-qPCR showing that the expression levels of *Myh6*, *Myh7* and *Tnnt2* at CM stage are impaired by *Foxf1* KO. Data are the mean ± standard error of the mean from three independent experiments. Two-tailed unpaired Student's *t*-test was performed.

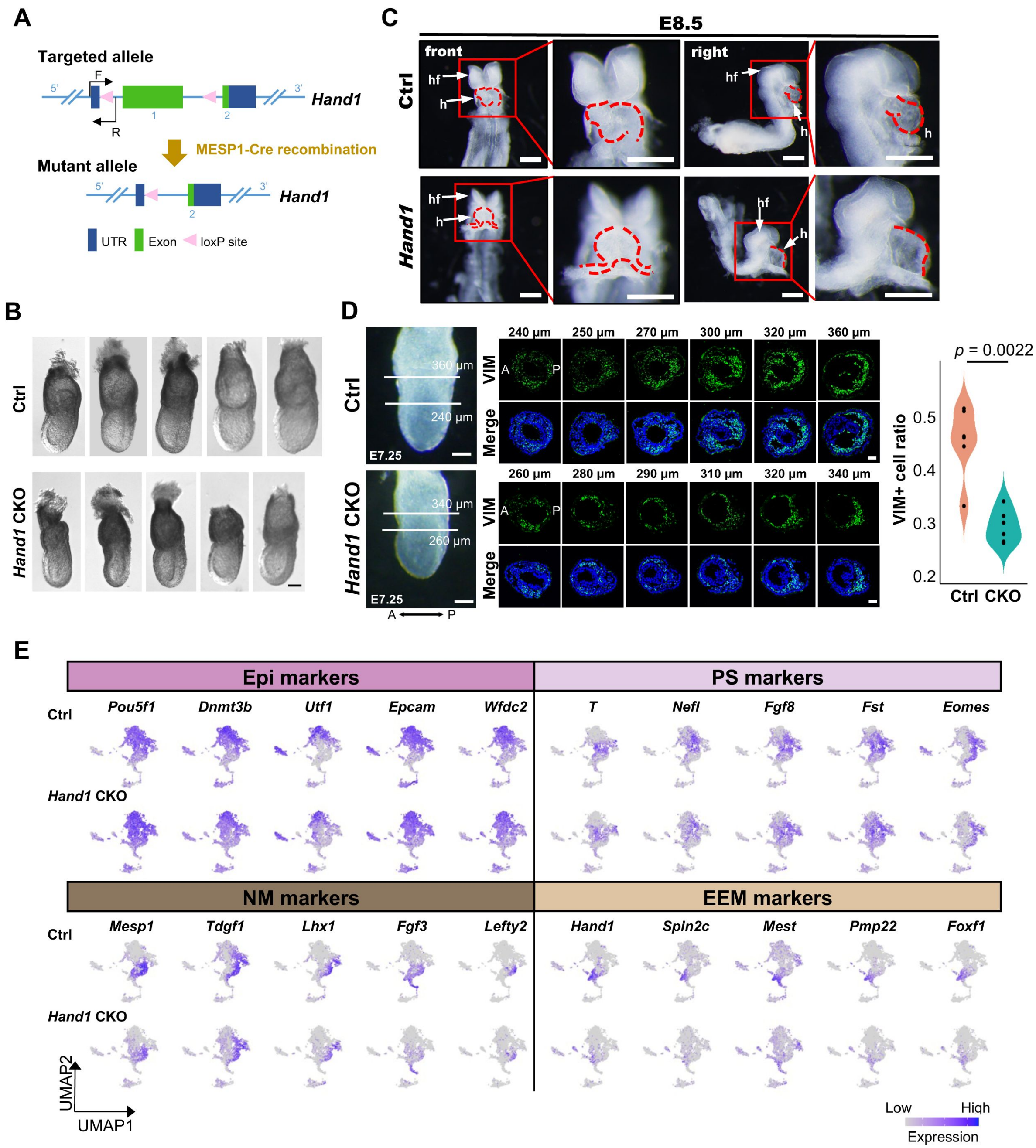

Figure S8

HAND1 is required for the expression of the EEM specific genes in vivo

(A) Schematic diagram showing strategy for the mesodermal cell specific *Hand1* KO embryo generation.
(B) Bright field images of representative E7.0 Ctrl and *Hand1* CKO embryos. Scale bar: 0.1 mm.
(C) The bright field images of E8.5 Ctrl and *Hand1* CKO mouse embryos. The arrows indicating the embryonic heart (h) and head folds (hf). Scale bar: 500 µm.
(D) The bright field image of E7.25 Ctrl and *Hand1* CKO mouse embryos (left), scale bar: 100 µm. Immunofluorescence staining of VIM (EEM marker) on serial embryo sections (middle), scale bar: 50 µm. Quantification of the proportion of VIM$^+$ EEM cells, *P*-value was calculated using one-sided Mann-Whitney U test.
(E) UMAP layout as in **Figure 6C**. Colors indicating expression levels of the marker genes of Epi, PS, NM and EEM.

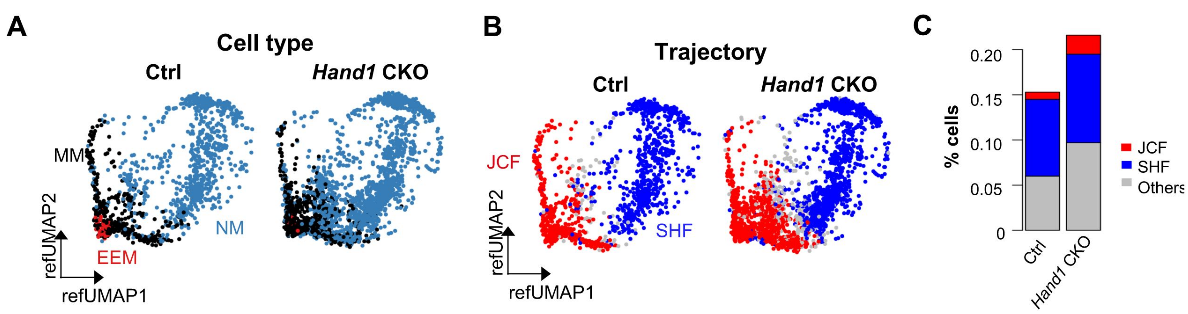

Figure S9

Integration of E7.0 progenitors and E7.0 Ctrl/Hand1 KO scRNA-seq

(A-B) Projection of E7.0 scRNA-seq onto the reference UMAP structure from **Figure 1D**. Reference UMAP layout for E7.0 CM trajectory cells colored by cell type (A) and trajectory (B).
(C) Bar plot showing the percentage of the JCF and SHF trajectories in NM of Ctrl and *Hand1* CKO mice.

Supplemental Tables

Gene	Cluster
Mixl1	Common
Tdgf1	Common
T	Common
Eomes	Common
Mesp1	Common
Pou5f1	Common
Car2	Common
Ccnd1	Common
Fgf3	Common
Sgk3	Common
Tbx3	Common
Phlda2	Common
Dkk1	Common
Hoxb2	Common
Meg3	Common
Vim	Common
Nrp1	Common
H19	Common
Hand2	Common
Rem1	Common
Bmp5	Common
Hapln1	Common
Cd24a	Common
Ephb1	Common
Sfrp5	Common
Gata5	Common
Mef2c	Common
Sox4	Common
Nkx2-5	Common
Mesp2	SHF specific
Sp5	SHF specific
Frzb	SHF specific
Arl4d	SHF specific
Gsc	SHF specific

Cyp26a1	SHF specific
Lefty2	SHF specific
Lhx1	SHF specific
Dll3	SHF specific
Pmaip1	SHF specific
Fam212a	SHF specific
Fgf5	SHF specific
Lhfp	SHF specific
Zic3	SHF specific
Cpa2	SHF specific
Sfrp1	SHF specific
Crabp1	SHF specific
Tbx1	SHF specific
Tcf21	SHF specific
Apoe	SHF specific
Hoxb3os	SHF specific
Pard6g	SHF specific
Isl1	SHF specific
Meis1	SHF specific
Aldh1a2	SHF specific
Fam213a	SHF specific
Tspan7	SHF specific
Hoxb1	SHF specific
Hoxa1	SHF specific
Dbn1	SHF specific
Vstm2b	JCF specific
Msx2	JCF specific
Hand1	JCF specific
Ppic	JCF specific
Msx1	JCF specific
Foxf1	JCF specific
Spin2c	JCF specific
Pmp22	JCF specific
Bmp4	JCF specific
Arl4c	JCF specific
Hoxd1	JCF specific
Tdo2	JCF specific
Krt18	JCF specific
Krt8	JCF specific

Mest	JCF specific
Hsd11b2	JCF specific
Dlk1	JCF specific
Tmem108	JCF specific
Csrp1	JCF specific
Tbx5	JCF specific
Mab21l2	JCF specific
Crip2	JCF specific
Tgfb1i1	JCF specific
Rbm38	JCF specific
Myocd	JCF specific
Wnt2	JCF specific
Tnnc1	JCF specific
Tnnt2	JCF specific
Acta2	JCF specific
Actc1	JCF specific
Myl7	JCF specific
Ttn	JCF specific
Hcn4	JCF specific
Myl9	JCF specific
Alcam	JCF specific
Hspb1	JCF specific
Nppb	JCF specific
Tnni3	JCF specific
Nppa	JCF specific

Table S1.1

Pseudotime-dependent genes for JCF/SHF trajectoyies. Related to

Figure 1F

Gene	Peak
Bmper	chr9:23205101-23205836
Hs3st3b1	chr11:63793400-63793901
Daam1	chr12:71813806-71814148
Pik3ap1	chr19:41388870-41389628
Rbm24	chr13:46522878-46523765
Dnah2	chr11:69437127-69437715
Masp1	chr16:23308079-23308586
Nrp1	chr8:128362384-128363365
Gfpt2	chr11:49781554-49782328
Nfatc2	chr2:168553368-168553873
Nfatc2	chr2:168633039-168633743
Tbx3	chr5:119588186-119589099
Nectin2	chr7:19695082-19695480
Rbm20	chr19:53670532-53671133
P4ha2	chr11:54064566-54064817
Kdr	chr5:76008228-76008806
Foxf1	chr8:120859927-120860544
Runx1	chr16:92685575-92686199
Slc6a6	chr6:91546003-91546701
Itga9	chr9:118774313-118774658
Tmem88	chr11:69510823-69511366
Vldlr	chr19:27183395-27183815
Cxcr4	chr1:128551753-128552340
Frmd6	chr12:70766970-70767397
Uaca	chr9:60787496-60788106
Zfhx3	chr8:108606479-108607224
Cgnl1	chr9:71749980-71750457
Kdm6b	chr11:69474662-69475132
Krt18	chr15:102013681-102013965
Rbfox2	chr15:77313482-77314091
Cgnl1	chr9:71709031-71709501
Smad7	chr18:75295165-75296030
Gm15511	chr9:69234348-69234819
Bmp4	chr14:46582969-46583613
Tbx20	chr9:24848198-24848669

Bmp2	chr2:133429498-133430016
Ryr2	chr13:12030416-12030844
Frem1	chr4:83095438-83095824
Tbx2	chr11:85803990-85804510
Klf6	chr13:5824290-5824668
Fpgs	chr2:32655607-32656216
St8sia1	chr6:142831256-142831930
Smad6	chr9:63919240-63919570
Kcnma1	chr14:23902557-23903307
Ackr3	chr1:90240337-90240673
Gm15222	chr14:46582969-46583613
Nrp1	chr8:128251495-128252057
Arfgef1	chr1:10285267-10285743
Dok4	chr8:94900700-94901649
Kif26b	chr1:178645037-178645334
Alpk3	chr7:81057414-81058331
Tpm1	chr9:67163861-67164376
Tbx20	chr9:24888287-24888799
Capn5	chr7:98149520-98150564
Rubie	chr14:46578136-46578796
Hand1	chr11:57929446-57929965
Bmp5	chr9:76007435-76007997
Podxl	chr6:31587404-31587997
Ano1	chr7:144786403-144786986
Mest	chr6:30703300-30704166
Gli2	chr1:118904950-118905383
Afdn	chr17:13709118-13709846
Ksr1	chr11:79123505-79123959
Rubie	chr14:46573109-46573707
St3gal6	chr16:58538961-58539278
Gm15511	chr9:69270897-69271224
Map3k5	chr10:20001907-20002282
Dok4	chr8:94892725-94893264
1700017B05Rik	chr9:57354263-57355026
Peg10	chr6:4634494-4634975
Rubie	chr14:46612475-46612905
Lypd6	chr2:50175810-50176459
Nckap5	chr1:126719707-126720255
Kif26b	chr1:178661078-178661894

Ahnak	chr19:9050275-9050663
Frem1	chr4:82959967-82960424
Hapln1	chr13:89401161-89401629
Col1a1	chr11:95004154-95004501
Cmtm7	chr9:114541247-114541587
Notch2	chr3:97889960-97890632
Hspa12b	chr2:131140351-131140939
Morc4	chrX:139875251-139875622
Pdgfb	chr15:79950260-79951054
Pdgfb	chr15:80000585-80000947
Map1b	chr13:99679373-99679831
Mvb12b	chr2:33877943-33878692
Adgra2	chr8:26887106-26887393
Wt1	chr2:105153091-105153686
Nfatc2	chr2:168589929-168591368
Adgra2	chr8:27088697-27089310
Tmem108	chr9:103805424-103805892
Ppp1r13b	chr12:111896858-111897524

Table S1.2

Gene-enhancer pairs in E7.0 mesoderm pseudotime trajectories. Related to Figure 2G

Table S2. ***Hand1*- and *Foxf1*-KO affected genes in MES cells. Attached as an excel file.**

	Gene	Forward	Reverse
RT-qPCR primer	*Actin*	CTGTCGAGTCGCGTCCACC	CGCAGCGATATCGTCATCCA
	Myh6	GCCCAGTACCTCCGAAAGTC	GCCTTAACATACTCCTCCTTGTC
	Myh7	ACTGTCAACACTAAGAGGGTCA	TTGGATGATTTGATCTTCCAGGG
	Tnnt2	CAGAGGAGGCCAACGTAGAAG	CTCCATCGGGGATCTTGGGT
	Hand1	ACGCACATCATCACCATCAT	CTACTGCGGTGGTAGGTGGT
	Foxf1	ACGCCGTTTACTCCAGCTC	CGTTGTGACTGTTTTGGTGAAG
	Bmp4	TTCCTGGTAACCGAATGCTGA	CCTGAATCTCGGCGACTTTTT
	Pmp22	CCGCAGCACAGCTGTCTTT	AGCAGATTAGCCTCAGGCACAA
	Spin2c	CACCACATTGGCTCTACAACC	CACGATGCTCCTACGGGAT
	Mthfsd	TGAGAAAGGCTGGCGAATTGG	GGGGATGTCTACGACCTGG
	Foxc2	AACCCAACAGCAAACTTTCCC	GCGTAGCTCGATAGGGCAG
	Foxl1	GAGCAGAGGGTCACACTGAAC	CTTCCTGCGCCGATAATTGC
	Fbxo31	CATGCGGTTCAAGCCACTG	GTCTGGTTACACTTGGTGGAG
sgRNA for genomic deletion	*Hand1* sgRNA1	TGCCGGCGGGCCACCACCTA	
	Hand1 sgRNA2	ACTTGATGGACGTGCTGGCC	
	Foxf1 sgRNA1	GATGTCCGCGCCCGACAAGC	
	Foxf1 sgRNA2	CCCCTAACGGATTATTTGTA	
	Hand1-eF sgRNA1	CGCGTCATAACCTATAGGGC	
	Hand1-eF sgRNA2	AAGTGTCGGCCGGTGGCGAT	
	Foxf1-eH sgRNA1	TACGTCCTGGACCTGCTAAC	
	Foxf1-eH sgRNA2	AGGCTGGCTGCGTGCACGTA	
sgRNA for CRISPRa	Ctrl sgRNA	ATGCGTCAGTCGACTGATGC	
	Foxf1-eH sgRNA1	TACGTCCTGGACCTGCTAAC	
	Foxf1-eH sgRNA2	AGTTAGCAGGTCCAGGACGT	
mice genotyping primer	*Hand1*-CKO-5'	GTTGCCTACAGAAACCTTCAAGAGG	ATGGTGATGATGTGCGTAGCTG
	Hand1-CKO-3'	TTTTGGGTCTTCTCCAGTTTGAGT	GACCCCGAGGGTCAGATAAAAG
	Mesp1-Cre	TTCTGGAAGGGGCCCGCTTCA	GCCTGTTTTGCACGTTCACCG

Table S3

Primers/oligos used in the current study

Data availability

Sequencing (sc/snRNA-seq, snATAC-seq, ChIP-seq, and RNA-seq) data that support the findings of this study have been deposited in the Gene Expression Omnibus under accession numbers GSE245713. Previously published ChIP-seq data that were re-analyzed here are available under accession codes GSE165107 (MESP1, ZIC2 and ZIC3, 2.5-day EB), GSE47085 (HAND1, FLK+ MES cells), GSE52123 (GATA4, E12.5 mouse heart) and GSE69099 (FLI1, ES derived hemogenic endothelium). Previously published scRNA–seq data that were re-analyzed here are available under accession codes E-MTAB-6967 (ArrayExpress, E6.5-8.5 mouse embryos) and E- MTAB-7403 (ArrayExpress, micro-dissected heart-related samples of E7.75-8.5 mouse embryos). Previously published GEO-seq data that were re-analyzed here are available under accession codes GSE171588. All other data supporting the findings of this study are available from the corresponding author on reasonable request.

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Additional information

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C.L., Z.L., P.X., and N.J. designed the research. X.J., Q.P., X.Y., and Y.Z. performed most the experiments and analyzed the data. P.X., J.H., C.W., W.Z., S.S., and Y.Y. analyzed transcriptomic and epigenomic data. Z.C., K.F., and H.F. assisted with cell line generation. W.F., K.G., and S.Z. performed the breeding of mouse lines. T.Z., Z.Y., K.W., W.X, Z.L., and C.L. provided the resources. Z.L., C.L., and P.X. wrote the manuscript. C.L. and Z.L. supervised the project.

Statistics and reproducibility

Numbers of biological replicates, statistical tests and *P*-values are reported in the figure legends. If not mentioned otherwise in the figure legend, statistical significance was determined using two-tailed Student's *t*-test, provided by GraphPad Prism9 statistical software.

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Reviewer #1 (Public review):

Summary:

In this study, the authors identified and described the transcriptional trajectories leading to CMs during early mouse development, and characterized the epigenetic landscapes that underlie early mesodermal lineage specification.

The authors identified two transcriptomic trajectories from a mesodermal population to cardiomyocytes, the MJH and PSH trajectories. These trajectories are relevant to the current model for the First Heart Field (FHF) and the Second Heart Field (SHF) differentiation. Then, the authors characterized both gene expression and enhancer activity of the MJH and PSH trajectories, using a multiomics analysis. They highlighted the role of Gata4, Hand1, Foxf1, and Tead4 in the specification of the MJH trajectory. Finally, they performed a focused analysis of the role of Hand1 and Foxf1 in the MJH trajectory, showing their mutual regulation and their requirement for cardiac lineage specification.

Strengths:

The authors performed an extensive transcriptional and epigenetic analysis of early cardiac lineage specification and differentiation which will be of interest to investigators in the field of cardiac development and congenital heart disease. The authors considered the impact of the loss of Hand1 and Foxf1 in-vitro and Hand1 in-vivo.

Weaknesses:

The authors used previously published scRNA-seq data to generate two described transcriptomic trajectories.

(1) Details of the re-analysis step should be added, including a careful characterization of the different clusters and maker genes, more details on the WOT analysis, and details on the time stamp distribution along the different pseudotimes. These details would be important to allow readers to gain confidence that the two major trajectories identified are realistic interpretations of the input data.

The authors have also renamed the cardiac trajectories/lineages, departing from the convention applied in hundreds of papers, making the interpretation of their results challenging.

(2) The concept of "reverse reasoning" applied to the Waddington-OT package for directional mass transfer is not adequately explained. While the authors correctly acknowledged Waddington-OT's ability to model cell transitions from ancestors to descendants (using optimal transport theory), the justification for using a "reverse reasoning" approach is missing. Clarifying the rationale behind this strategy would be beneficial.

(3) As the authors used the EEM cell cluster as a starting point to build the MJH trajectory, it's unclear whether this trajectory truly represents the cardiac differentiation trajectory of the FHF progenitors:
- This strategy infers that the FHF progenitors are mixed in the same cluster as the extra-embryonic mesoderm, but no specific characterization of potential different cell populations included in this cluster was performed to confirm this.

- The authors identified the EEM cluster as a Juxta-cardiac field, without showing the expression of the principal marker Mab21l2 per cluster and/or on UMAPs.

- As the FHF progenitors arise earlier than the Juxta-cardiac field cells, it must be possible to identify an early FHF progenitor population (Nkx2-5+; Mab21l2-) using the time stamp. It would be more accurate to use this FHF cluster as a starting point than the EEM cluster to infer the FHF cardiac differentiation trajectory.

These concerns call into question the overall veracity of the trajectory analysis, and in fact, the discrepancies with prior published heart field trajectories are noted but the authors fail to validate their new interpretation. Because their trajectories are followed for the remainder of the paper, many of the interpretations and claims in the paper may be misleading. For example, these trajectories are used subsequently for annotation of the multiomic data, but any errors in the initial trajectories could result in errors in multiomic annotation, etc, etc.

(4) As mentioned in the discussion, the authors identified the MJH and PSH trajectories as non-overlapping. But, the authors did not discuss major previously published data showing that both FHF and SHF arise from a common transcriptomic progenitor state in the primitive streak (DOI: 10.1126/science.aao4174; DOI: 10.1007/s11886-022-01681-w). The authors should consider and discuss the specifics of why they obtained two completely separate trajectories from the beginning, how these observations conflict with prior published work, and what efforts they have made at validation.

(5) Figures 1D and E are confusing, as it's unclear why the authors selected only cells at E7.0. Also, panels 1D 'Trajectory' and 'Pseudotime' suggest that the CM trajectory moves from the PSH cells to the MJH. This result is confusing, and the authors should explain this observation.

(6) Regarding the PSH trajectory, it's unclear how the authors can obtain a full cardiac differentiation trajectory from the SHF progenitors as the SHF-derived cardiomyocytes are just starting to invade the heart tube at E8.5 (DOI: 10.7554/eLife.30668).

The above notes some of the discrepancies between the author's trajectory analysis and the historical cardiac development literature. Overall, the discrepancies between the author's trajectory analysis and the historical cardiac development literature are glossed over and not adequately validated.

(7) The authors mention analyzing "activated/inhibited genes" from Peng et al. 2019 but didn't specify when Peng's data was collected. Is it temporally relevant to the current study? How can "later stage" pathway enrichment be interpreted in the context of early-stage gene expression?

(8) Motif enrichment: cluster-specific DAEs were analyzed for motifs, but the authors list specific TFs rather than TF families, which is all that motif enrichment can provide. The authors should either list TF families or state clearly that the specific TFs they list were not validated beyond motifs.

(9) The core regulatory network is purely predictive. The authors again should refrain from language implying that the TFs in the CRN have any validated role.

Regarding the in vivo analysis of Hand1 CKO embryos, Figures 6 and 7:

(10) How can the authors explain the presence of a heart tube in the E9.5 Hand1 CKO embryos (Figure 6B) if, following the authors' model, the FHF/Juxta-cardiac field trajectory is disrupted by Hand1 CKO? A more detailed analysis of the cardiac phenotype of Hand1 CKO embryos would help to assess this question.

(11) The cell proportion differences observed between Ctrl and Hand1 CKO in Figure 6D need to be replicated and an appropriate statistical analysis must be performed to definitely conclude the impact of Hand1 CKO on cell proportions.

(12) The in-vitro cell differentiations are unlikely to recapitulate the complexity of the heart fields in-vivo, but they are analyzed and interpreted as if they do.

(13) The schematic summary of Figure 7F is confusing and should be adjusted based on the following considerations:
(a) the 'Wild-type' side presents 3 main trajectories (SHF, Early HT and JCF), but uses a 2-color code and the authors described only two trajectories everywhere else in the article (aka MJH and PSH). It's unclear how the SHF trajectory (blue line) can contribute to the Early HT, when the Early HT is supposed to be FHF-associated only (DOI: 10.7554/eLife.30668). As mentioned

previously in Major comment 3., this model suggests a distinction between FHF and JCF trajectories, which is not investigated in the article.
(b) the color code suggests that the MJH (FHF-related) trajectory will give rise to the right ventricle and outflow tract (green line), which is contrary to current knowledge.

Minor comments:

(1) How genes were selected to generate Figure 1F? Is this a list of top differentially expressed genes over each pseudotime and/or between pseudotimes?

(2) Regarding Figure 1G, it's unclear how inhibited signaling can have an increased expression of underlying genes over pseudotimes. Can the authors give more details about this analysis and results?

(3) How do the authors explain the visible Hand1 expression in Hand1 CKO in Figure S7C 'EEM markers'? Is this an expected expression in terms of RNA which is not converted into proteins?

(4) The authors do not address the potential presence of doublets (merged cells) within their newly generated dataset. While they mention using "SCTransform" for normalization and artifact removal, it's unclear if doublet removal was explicitly performed.

Comments on revised version:

Summary:

The authors have not addressed the major philosophical problems with the initial submission. They interpret their data without care to conform to years of prior publications in the field. This causes the authors to draw fanciful conclusions that are highly likely to be inaccurate (at best).

Q1R1: The authors gave more details about the characterization of cell types and the two identified trajectories.

a) It remains unclear how the authors generated this list. Are they manually selected genes based on relevant literature or an unbiased marker gene identification analysis? Either references should be added, or the bioinformatics explanation should be included in the method section.
b) Revised text satisfies the comment.
c) Revised text satisfies the comment.

Other comments:

Figure 1F: left annotation needs to be corrected (two "JCF specific").

Q2R1: Revised text satisfies the comment.

Q3R1 (1): Revised text satisfies the comment.

Q3R1 (2): a) The explanation of how the authors built the JCF trajectory makes sense and the renaming from "MJH" to "JCF" is correct and better represents the identification that was made using time points from E7.5 to E8.5. However, the explanation given does not answer our original question. Our original comment asked about the FHF differentiation trajectory. The authors built the "MJH" trajectory as the combined "FHF/JCF" trajectory, however, it is not directly established whether the FHF and JCF progenitor differentiation trajectories are the same. The authors did not directly try to identify the FHF and JCF trajectories separately using appropriate real time windows but only assumed that they were the same. Every link between JCF and FHF trajectories assuming that they are shared without prior identification of the FHF progenitor differentiation trajectory should be removed from the manuscript (e.g.

page 4: "namely the JCF trajectory (the Hand1-expressing early extraembryonic mesoderm - JCF and FHF - CM)").

b) Adding the Mab21l2 ICA plot satisfies the comment.

c) The explanation given by the authors regarding the FHF trajectory analysis is missing important details. The authors started the reverse trajectory analysis from E7.75 cardiomyocytes as being the FHF.

- The authors should be mindful with the distinction between FHF progenitors and FHF-derived cardiomyocytes.
- It is unclear whether cells called after the starting point (E7.75 CMs) in the reverse FHF trajectory, were collected prior E7.75. Can the authors add more details, and a real time point distribution along the FHF pseudotime to their analysis? Also, what cells belong to the FHF trajectory after the E7.75 CMs in the reverse direction? These cells should be shown as in Figure 1A and 1B for the JCF and SHF trajectories.
- As the FHF arises first and differentiates into the cardiac crescent prior to or at the same time the JCF and SHF emerge, it is impossible for late progenitors (JCF and SHF) to contribute to the early FHF progenitor pool. Therefore, the observation that "both JCF and SHF lineages contribute to the early FHF progenitor population" can not be correct. It is also not what Dominguez et al showed. This misinterpretation goes against the current literature (e.g. DOI: 10.1038/ncb3024) and will leads to confusion.

Q4R1: Revised text and figure satisfy the comment.

Q5R1: The answer satisfies the comment.

Q6R1: a) The authors did not address the question and did not change their language in the manuscript. As SHF-derived cardiomyocytes are missing (because they are generated after E8.5), the part of the SHF trajectory going from SHF progenitors to the E8.5 heart tube must be inaccurate.

b) The authors correctly mentioned, both JCF and SHF will contribute to the four-chamber heart. However, as the dataset used by the authors spans only to E8.5 (which is days before the completion of the four-chamber heart), and all SHF and the vast majority of JCF contributions don't reach the heart until after E8.5, any claims about trajectories from JCF/SHF progenitor pools to cardiomyocytes should be removed because they do not correspond to prior published and accepted work.

Q7R1: Especially because gene expression levels change over time, the authors might have considered genes as specific and restricted to a pathway based on their expression at a given time (e.g. later time), but at another time (e.g. earlier time), the same genes could have another expression pattern and not be pathway-specific anymore.

Q8R1: Revised text satisfies the comment.

Q9R1: Revised text satisfies the comment.

Q10R1: Thank you for analyzing deeper the cardiac phenotype of the Hand1 cKO embryos.

Regarding the presence of a heart tube, while, following the authors' model the FHF/JCF trajectory is disrupted:

- Renaming the "MSH" to "JCF" is more accurate to the data shown by the authors as mainly the EEM is altered after Hand1 cKO.
- The presence of the heart tube suggests that even if the JCF is altered, the FHF can still produce a cardiac crescent and a heart tube (as observed in Hand1-null embryos DOI: 10.1038/ng0398-266). The schematic Figure 7F suggests that only the SHF contribution will

allow the formation of the heart tube. This unorthodox idea would need to be assessed by an alternate approach. More likely is that the model simply ignores the FHF contribution (the most important up to E8.5). The schematic is therefore incomplete and inaccurate and should be removed or edited to correspond to the prior literature.

Q11R1: It is unclear what "replicates" mean in the authors' answer, as if they have been pooled without replicate-specific barcodes they are no longer replicates and should be considered as a single sample. This should be explicitly written in the method section. Thank you for your IF staining/quantification. If DAPI was used, it should be written in the figure caption.

Q12R1: Revised text satisfies the comment.

Q13R1: The answer given by the authors did not satisfy the comment because of the following:

- The authors investigated two differentiation trajectories (JCF and SHF) in the article but Figure 7F presents three trajectories (JCF, SHF, and Early HT). The "Early HT" is neither mentioned, nor discussed in the manuscript.
- Figure 7F suggests that the "Early HT" trajectory corresponds to a combination of the SHF and JCF trajectories but does not mention the early FHF trajectory. This is going against the current literature. This relates to the comments of Q10R1.
- As the authors rightly point out, the SHF will be contributing to the heart tube, but through a cell invasion of the already differentiated heart tube (10.1016/j.devcel.2023.01.010). Our prior comments did not question the implication of the SHF to the looping and ballooning process but mentioned that the heart tube arises before the invasion from SHF and is FHF-derived. Figure 7F in the context of Hand1-null suggest that the heart tube will form from the SHF lineage, which is confusing as the SHF is known to contribute by invasion of the (already-formed) FHF-derived heart tube. The FHF lineage is missing from the authors' model.
- In the revised manuscript, the FHF trajectory analysis is still unclear and suggests that the JCF and SHF progenitors contribute to the FHF progenitor which is going against current literature. This relates to the comments of Q3R1 (2).

Overall, the schematic Figure 7F is very confusing as it does not follow already published data without being fully validated and therefore is inaccurate and misleading.

Minor comments:

The answers satisfy the minor comments.

https://doi.org/10.7554/eLife.98293.2.sa3

Reviewer #2 (Public review):

Summary of goals:

The aims of the study were to identify new lineage trajectories for the cardiac lineages of the heart, and to use computational and cell and animal studies to identify and validate new gene regulatory mechanisms involved in these trajectories.

Strengths:

Overall: the study addresses the long standing yet still not fully answered questions of what drives the earliest specification mechanisms of the heart lineages. The introduction demonstrates a good understanding of the relevant lineage trajectories that have been previously established, and the significance of the work is well described. The study takes advantage of several recently published data sets and attempts t use these in combination to uncover any new mechanisms underlying early mesoderm/cardiac specification mechanisms. A strength of the study is the use of an in vitro model system (mESCs) to assess the functional relevance of the key players identified in the computational analysis, including innovative technology such as CRISPR-guided enhancer modulations. Lastly, the study generates mesoderm-specific Hand1 LOF embryos and assesses the differentiation trajectories in these animals, which represents a strong complementary approach to the in vitro and computational analysis earlier in the paper. The manuscript is clearly written and the methods section is detailed and comprehensive.

Comments and Weaknesses:

I unfortunately still have the same concerns I had for the original submission. There are many strong claims about lineage trajectories and population relationships that are based purely on the analysis of a number of datasets, some published and a few new datasets.

The methods used involve significant input bias, and some of the less user-biased approaches, such as the new RNA velocity analysis on the JCF/SHF trajectories, are included only in the response to reviewers but not in the manuscript (R1R2), as far as I can tell. This analysis does not seem to suggest that CMs are generated from both trajectories, but it is difficult to know as they provide so little information on what exactly they did.
The conclusions are particularly concerning not only because they are largely based on computational analysis, but also because they contradict well-described concepts (which are supported by in vivo lineage tracing).
I want to give them credit for having done some additional experiments. That said, the new data added for the validation of some of their concepts (mESC Fig 5F and embryos Fig S8C) do not strengthen their conclusions in my opinion. The mESC data were not quantified, and the embryo data looks like quantifications were done in different planes of a single embryo, but it's hard to tell as little information is provided. Even with accurate quantification, I believe the IF analysis for VIM in Hand1 cKO embryos is not sufficient to back up their claims on the role of Hand1 in driving the JCF lineage.

https://doi.org/10.7554/eLife.98293.2.sa2

Reviewer #3 (Public review):

In this manuscript, the Xie et al. delineate two cardiac lineage trajectories using pseudo-time and epigenetic analyses, tracing development from E6.5 to E8.5, culminating in cardiomyocytes (CMs). The authors propose that mutual regulation between the transcription factors Hand1 and Foxf1 plays a role in specifying a first cardiac lineage.

Following the first round of revision, the authors have renamed their EEM-JCF/FHF (MJH) and PM-SHF (PSH) trajectories JCF and SHF. However, their use of this terminology is confusing. The so-called JCF trajectory appears to represent a mixture of JCF and FHF, as Hand1-expressing early extraembryonic mesoderm contributes to FHF-derived cardiomyocytes (e.g., HCN4+, Tbx5+). The authors then argue that JCF arises from Hand1+ cells and is therefore distinct from FHF, yet elsewhere suggest that both JCF and SHF contribute to FHF. This introduces conceptual inconsistencies.

Furthermore, the expression of Hand1, Foxf1, and Bmp4 in the lateral plate mesoderm complicates the assertion that JCF is distinct from FHF (Development 2015; 142: 3307-3320; Nat Rev Mol Cell Biol, https://www.nature.com/articles/nrm2618; Circ Res 2021, https://doi.org/10.1161/CIRCRESAHA.121.318943). Mab21l2 expression also overlaps with the cardiac crescent. The designation of Tbx20 as a "key JCF-specific gene" is problematic, why should it not equally be considered an FHF-specific marker (https://pmc.ncbi.nlm.nih.gov/articles

/PMC10629681)? Perhaps the JCF trajectory represent a subset of FHF. A designation such as "JCF/FHF" may therefore be more appropriate.

In Figure 1A, the decision to define a single CM state as the endpoint of both trajectories is also problematic. FHF and SHF are known to give rise to distinct CM subtypes, yet in the authors' reconstruction both lineages converge on one CM population. This was the point raised in Question 1 of my initial review. If both trajectories converge on the same CM state, are they truly independent lineages? This interpretation remains unclear and potentially misleading.

https://doi.org/10.7554/eLife.98293.2.sa1

Author response:

The following is the authors' response to the original reviews

> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *In this study, the authors identified and described the transcriptional trajectories leading to CMs during early mouse development, and characterized the epigenetic landscapes that underlie early mesodermal lineage specification.*
>
> *The authors identified two transcriptomic trajectories from a mesodermal population to cardiomyocytes, the MJH and PSH trajectories. These trajectories are relevant to the current model for the First Heart Field (FHF) and the Second Heart Field (SHF) differentiation. Then, the authors characterized both gene expression and enhancer activity of the MJH and PSH trajectories, using a multiomics analysis. They highlighted the role of Gata4, Hand1, Foxf1, and Tead4 in the specification of the MJH trajectory. Finally, they performed a focused analysis of the role of Hand1 and Foxf1 in the MJH trajectory, showing their mutual regulation and their requirement for cardiac lineage specification.*
>
> *Strengths:*
>
> *The authors performed an extensive transcriptional and epigenetic analysis of early cardiac lineage specification and differentiation which will be of interest to investigators in the field of cardiac development and congenital heart disease. The authors considered the impact of the loss of Hand1 and Foxf1 in-vitro and Hand1 in-vivo.*
>
> *Weaknesses:*
>
> *The authors used previously published scRNA-seq data to generate two described transcriptomic trajectories.*

We agree that a two-route cardiac development model has been described, which is consistent with our analyses. However, the developmental origins and key events by early lineage specification is unclear. Our study provided new insights from the following aspects:

a) Computational analyses inferred the earliest cardiac fate segregation by E6.75-7.0.

b) Provided the new-generated E7.0 multi-omics data which revealed the transcriptomic and chromatin accessibility landscape.

c) Utilized multi-omics and ChIP-seq data to construct a core regulatory network underlying the JCF lineage specification.

d) Applied in vitro and in vivo analyses, which elucidated the synergistic and different roles of key transcription factors, HAND1 and FOXF1.

> *Q1R1: Details of the re-analysis step should be added, including a careful characterization of the different clusters and maker genes, more details on the WOT analysis, and details on the time stamp distribution along the different pseudotimes. These details would be important to allow readers to gain confidence that the two major trajectories identified are realistic interpretations of the input data.*

R1R1: Thank you for the valuable suggestion. In the last version, we characterized the two major trajectories by identifying their common or specific gene sets, and by profiling the expression dynamics along pseudotime (Figure 1F). But we realized a careful description was not provided. In the revised manuscript, we have made the following improvements:

a) Provided marker gene analyses based on cell types as well as developmental lineages to support the E7.0 progenitor clusters (Figure S1F).

b) For Figure 1F: revised the text and introduced characteristic genes for the two trajectories.

c) For WOT analysis: provided more details in the first paragraph of the 'Results' section.

> *R2R1: The authors have also renamed the cardiac trajectories/lineages, departing from the convention applied in hundreds of papers, making the interpretation of their results challenging.*

R2R1: Agreed. We have changed the MJH as JCF lineage and PSH as SHF lineage.

> *Q3R1: The concept of "reverse reasoning" applied to the Waddington-OT package for directional mass transfer is not adequately explained. While the authors correctly acknowledged Waddington-OT's ability to model cell transitions from ancestors to descendants (using optimal transport theory), the justification for using a "reverse reasoning" approach is missing. Clarifying the rationale behind this strategy would be beneficial.*

R3R1: Thank you for pointing out the unclear explanation. As mentioned in R1R1, we have clarified the rationale in the revised manuscript.

We would like to provide some additional details: WOT is designed for time-series scRNA-seq data where the time/stage each single cell is given. At any adjacent time points t_i and t_{i+1}, WOT estimates the transition probability of all cells at t_i to all cells at t_{i+1}. One can select a cell set of interest at any time point t_i to infer their ancestors at t_{i-1} or their descendants at t_{i+1} by sums of the transition probabilities. As introduced in the original paper, WOT allows for both 'forward' and 'reverse' inference (DOI: 10.1016/j.cell.2019.01.006).

> *Q3R1: As the authors used the EEM cell cluster as a starting point to build the MJH trajectory, it's unclear whether this trajectory truly represents the cardiac differentiation trajectory of the FHF progenitors:*
>
> *- This strategy infers that the FHF progenitors are mixed in the same cluster as the extra-embryonic mesoderm, but no specific characterization of potential different cell populations included in this cluster was performed to confirm this.*

To build the MJH trajectory, we performed a two-step analysis:

(1) Firstly, we used E8.5 CM cells as a starting point to perform WOT computational reverse lineage tracing and identify CM progenitors at each time point.

(2) Secondly, we selected EEM cells from the E7.5 CM progenitor pool, as a starting point to perform WOT analysis. Cells along this trajectory consist of the JCF lineage (Figure 1B).

The reason why we chose to use this subset of E7.5 EEM cells was due to its purity. It is distinct from the SHF lineage as suggested by their separation in the UMAP. It is also different from FHF cells as no FHF/CM markers were detected by E7.5.

It is admitted that it is infeasible to achieve 100% purity in this single cell omics analysis, but we believe the current strategy of defining the JCF lineage is reasonable. The distinct gene expression dynamics (Figure 1F) and spatial mapping results (Figure 1C), between JCF and SHF lineages, also supported our conclusion.

> *- The authors identified the EEM cluster as a Juxta-cardiac field, without showing the expression of the principal marker Mab21l2 per cluster and/or on UMAPs.*

Thank you for your suggestion. We have added Mab21l2 expression plots in the ICA layout (new Figure S1D), showing its transient expression dynamics, consistent with Tyser et al (DOI: 10.1126/science.abb2986).

> *- As the FHF progenitors arise earlier than the Juxta-cardiac field cells, it must be possible to identify an early FHF progenitor population (Nkx2-5+; Mab21l2-) using the time stamp. It would be more accurate to use this FHF cluster as a starting point than the EEM cluster to infer the FHF cardiac differentiation trajectory.*

We appreciate your insights. We used the early FHF progenitor population (E7.75 Nkx2-5+; Mab21l2- CM cells) as the starting point and identified its progenitor cells by E7.0 (Figure S2A). Results suggest both JCF and SHF lineages contribute to the early FHF progenitor population, consistent with live imaging-based single cell tracing by Dominguez et al (DOI: 10.1016/j.cell.2023.01.001).

> *These concerns call into question the overall veracity of the trajectory analysis, and in fact, the discrepancies with prior published heart field trajectories are noted but the authors fail to validate their new interpretation. Because their trajectories are followed for the remainder of the paper, many of the interpretations and claims in the paper may be misleading. For example, these trajectories are used subsequently for annotation of the multiomic data, but any errors in the initial trajectories could result in errors in multiomic annotation, etc, etc.*

Thank you for your valuable comments. In the revised manuscript, we have added details about the trajectory analysis including the procedure of WOT lineage inference, marker gene expression and early FHF lineage tracing. We also renamed the two trajectories to avoid confusion with prior published heart field trajectories. Generally, our trajectories are consistent with the published evidence about two major lineages contributing to the linear heart tube:

a) Clonal analysis: two trajectories exist which demonstrate differential contribution to the E8.5 cardiac tube (Meilhac et al, DOI: 10.1016/s1534-5807(04)00133-9).

b) Live imaging: JCF cells contribute to the forming heart (Tyser et al, DOI: 10.1126/science.abb2986; Dominguez et al, DOI: 10.1016/j.cell.2023.01.001).

c) Genetic labelling based lineage tracing: early *Hand1*+ mesodermal cells differentiate and contribute to the cardiac crescent (Zhang et al, DOI: 10.1161/CIRCRESAHA.121.318943).

Molecular events by the initial segregation of the two lineages were not characterized before, which are the main focus of our paper. Our analyses suggest that the JCF lineage segregates earlier from the nascent/mixed mesoderm status, also consistent with the clonal analysis (Meilhac et al, DOI: 10.1016/s1534-5807(04)00133-9).

> *Q4R1: As mentioned in the discussion, the authors identified the MJH and PSH trajectories as nonoverlapping. But, the authors did not discuss major previously published data showing that both FHF and SHF arise from a common transcriptomic progenitor state in the primitive streak (DOI: 10.1126/science.aao4174; DOI: 10.1007/s11886-022-01681-w). The authors should consider and discuss the specifics of why they obtained two completely separate trajectories from the beginning, how these observations conflict with prior published work, and what efforts they have made at validation.*

R4R1: Thank you for the important question. For trajectory analysis, we assigned cells to the trajectory with higher fate probability, resulting in 'non-overlapping' cell sets. However, the statement of 'two non-overlapping trajectories' is inaccurate. We performed analysis of fate divergence between two trajectories (which was not shown in the first version), which suggests, before E7.0, mesodermal cells have similar probabilities to choose either trajectory (Figure S1E). We agree with you and previously published data that the JCF and SHF arise from a common progenitor pool. Correction has been made in the revised manuscript.

> *Q5R1: Figures 1D and E are confusing, as it's unclear why the authors selected only cells at E7.0. Also, panels 1D 'Trajectory' and 'Pseudotime' suggest that the CM trajectory moves from the PSH cells to the MJH. This result is confusing, and the authors should explain this observation.*

R5R1: Thank you for pointing out the confusion. As mentioned in R4R1, trajectory analysis indicates JCFSHF fate segregation by E7.0 and we used Figures 1D and E to characterize the cellular status. By E7.0, JCF progenitors are at EEM or MM status, while SHF progenitors are still at the earlier differentiation stage (NM). This result is consistent with previous clonal analysis (Meilhac et al, DOI: 10.1016/s1534-5807(04)00133-9) which demonstrates an apparent earlier segregation of the first lineage. Our interpretation of the pseudotime analysis is that it represents different levels of differentiation, instead of developmental direction.

> *Q6R1: Regarding the PSH trajectory, it's unclear how the authors can obtain a full cardiac differentiation trajectory from the SHF progenitors as the SHF-derived cardiomyocytes are just starting to invade the heart tube at E8.5 (DOI: 10.7554/eLife.30668).*

R6R1.1: We agree with your opinion. Our trajectory analysis covers E8.5 SHF-derived CM cells and progenitors. Cells that differentiate as CM cells after E8.5 were missed.

> *The above notes some of the discrepancies between the author's trajectory analysis and the historical cardiac development literature. Overall, the discrepancies between the author's trajectory analysis and the historical cardiac development literature are glossed over and not adequately validated.*

R6R1.2: Historical cardiac development related literature provided evidence, using multiple techniques, which support the existence of two cardiac lineages with common progenitors at the beginning and overlapping contribution of the four-chamber heart. Our trajectory analysis is in agreement with this model and provides more detailed molecular insights about lineage segregation by E7.0. Thank you for pointing out our mistakes describing the observations. We have corrected the text and provided additional data (Figure S1D-F and S2), aiming to resolved the confusions.

> *Q7R1: The authors mention analyzing "activated/inhibited genes" from Peng et al. 2019 but didn't specify when Peng's data was collected. Is it temporally relevant to the current study? How can "later stage" pathway enrichment be interpreted in the context of early-stage gene expression?*

R7R1: The gene sets of "activated/inhibited genes" were collected from several published perturbation datasets (Gene Expression Omnibus accession numbers GSE48092, GSE41260, GSE17879, GSE69669, GSE15268 and GSE31544) using mouse ES cells or embryos. For a specific pathway, the gene set is fixed but the gene expression levels, which change over time, reflect the pathway enrichment. This explains the differential pathway enrichment between early and late stages.

> *Q8R1: Motif enrichment: cluster-specific DAEs were analyzed for motifs, but the authors list specific TFs rather than TF families, which is all that motif enrichment can provide. The authors should either list TF families or state clearly that the specific TFs they list were not validated beyond motifs.*

R8R1: Thank you for your comment. For the DAE motif analysis, we firstly inferred the motif and TF families, then tested which specific TFs are expressed in the corresponding cell cluster. We have added this information in the legend of Figure 2D.

> *Q9R1: The core regulatory network is purely predictive. The authors again should refrain from language implying that the TFs in the CRN have any validated role.*

R9R1: Thank you for your kind suggestion. We have revised the manuscript to avoid any misleading implications, as follows:

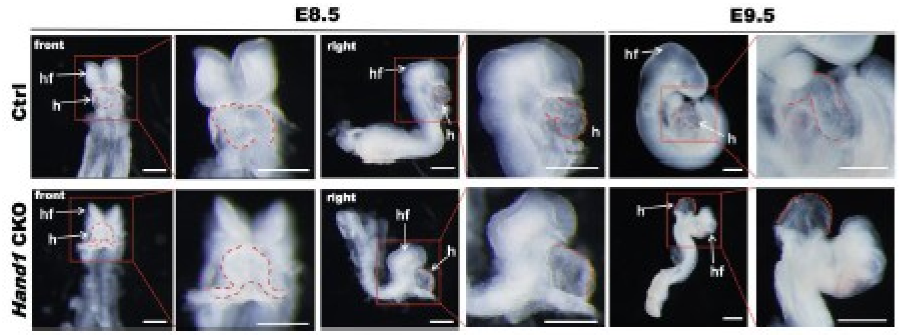
“Through single-cell multi-omics analysis, a predicted core regulatory network (CRN) in JCF is identified, consisting of transcription factors (TFs) GATA4, TEAD4, HAND1 and FOXF1.”

> *Q10R1: Regarding the in vivo analysis of Hand1 CKO embryos, Figures 6 and 7:*
>
> *How can the authors explain the presence of a heart tube in the E9.5 Hand1 CKO embryos (Figure 6B) if, following the authors' model, the FHF/Juxta-cardiac field trajectory is disrupted by Hand1 CKO? A more detailed analysis of the cardiac phenotype of Hand1 CKO embryos would help to assess this question.*

R10R1: Thank you for your valuable suggestion. In the revised manuscript, we have added detailed analysis of the cardiac phenotype of Hand1 CKO embryo (Figure S8C). Data suggest that by E8.5 when heart looping initiate in control group (14/17), the hearts of Hand1 CKO embryos (3/3) still demonstrate a linear tube morphology. By E9.5 when atrium and ventricle become distinct in WT embryos, heart looping of Hand1 CKO embryos is abnormal. The cardiac defects of our MESP1CRE driven Hand1 conditional KO are consistent with those of Hand1-null mutant mice (Doi: 10.1038/ng0398-266; D oi: 10.1038/ng0398-271).

Author response image 1.

The bright field images of E8.5-E9.5 Ctrl and *Hand1* CKO mouse embryos. The arrows indicating the embryonic heart (h) and head folds (hf). Scale bars (E8.5): 200 μm; scale bars (E9.5): 500 μm.



> *Q11R1: The cell proportion differences observed between Ctrl and Hand1 CKO in Figure 6D need to be replicated and an appropriate statistical analysis must be performed to definitely conclude the impact of Hand1 CKO on cell proportions.*

R11R1: We appreciate your valuable suggestion. As Figure 6D is based on scRNA-seq experiment, where replicates were merged as one single sequencing library, statistical analysis is infeasible. To address potential concerns about cell proportions, we added IF staining experiments of EEM marker gene, Vim, in serial embryo sections (Figure S8D). Statistical analysis indicates a significant decrease of VIM+ EEM cell proportion of *Hand1* CKO embryos.

> *Q12R1: The in-vitro cell differentiations are unlikely to recapitulate the complexity of the heart fields invivo, but they are analyzed and interpreted as if they do.*

R12R1: We agree with your opinion. In the revised manuscript, we tuned down the interpretation of the invitro cell differentiation data.

Previous version:

I. “The analysis indicated that HAND1 and FOXF1 could dually regulate MJH specification through directly activating the MJH specific genes and inhibiting the PSH specific genes.”

II. “Together, our data indicated that mutual regulation between HAND1 and FOXF1 could play a key role in MJH cardiac progenitor specification.”

III. “Thus, our data further supported the specific and synergistic roles of HAND1 and FOXF1 in MJH cardiac progenitor specification.”

Revised version:

I. “The analysis indicated that HAND1 and FOXF1 were able to directly activate the JCF specific genes.”

II. “Together, our in vitro experimental data indicated that mutual regulation between HAND1 and FOXF1 could play a key role in activation of JCF specific genes.”

III. “These results suggest that HAND1 and FOXF1 may cooperatively regulate early cardiac lineage specification by promoting JCF-associated gene expression and suppressing alternative mesodermal programs.”

> *Q13R1: The schematic summary of Figure 7F is confusing and should be adjusted based on the following considerations:*

> *(a) the 'Wild-type' side presents 3 main trajectories (SHF, Early HT and JCF), but uses a 2-color code and the authors described only two trajectories everywhere else in the article (aka MJH and PSH). It's unclear how the SHF trajectory (blue line) can contribute to the Early HT, when the Early HT is supposed to be FHF-associated only (DOI: 10.7554/eLife.30668). As mentioned previously in Major comment 3., this model suggests a distinction between FHF and JCF trajectories, which is not investigated in the article.*

R13R1(a): Thank you for your great insights. The paper you mentioned used _Nkx2.5_cre/+; Rosa26tdtomato+/- and _Nkx2.5_eGFP embryos to reconstruct the cardiac morphologies between E7.5 and E8.2. Their 3D models clearly demonstrate the transition from yolk sac to FHF and then SHF (Figure 2A' and A''). The location of yolk sac is defined as JCF in later literature (DOI: 10.1126/science.abb2986). However, as *Nkx2.5* mainly marks cells after the entry of the heart tube, it is unable to reflect the lineage contribution by JCF or SHF. As in R3R1, more and more evidence support the contribution of both lineages to the Early HT, which is discussed in a recent review paper (DOI: 0.1016/j.devcel.2023.01.010).

> *(b) the color code suggests that the MJH (FHF-related) trajectory will give rise to the right ventricle and outflow tract (green line), which is contrary to current knowledge.*

R13R1(b): Thank you for pointing out the confusion. The coloring of outflow tract is not an indication of JCF lineage contribution. We have changed the color of JCF/SHF trajectory in the revised model.

> *Minor comments:*
>
> *Q14R1: How genes were selected to generate Figure 1F? Is this a list of top differentially expressed genes over each pseudotime and/or between pseudotimes?*

R14R1: For each trajectory, we ranked genes by the correlation between expression levels and pseudotime.

Top 1000 genes for each group were selected.

> *Q15R1: Regarding Figure 1G, it's unclear how inhibited signaling can have an increased expression of underlying genes over pseudotimes. Can the authors give more details about this analysis and results?*

R15R1: The increased expression of 'inhibited genes' could be explained as an indication of decreasing signaling levels or compensation effect by other signaling pathways. We appreciate your kind suggestion. Details about this analysis have been added in the Method section.

> *Q16R1: How do the authors explain the visible Hand1 expression in Hand1 CKO in Figure S7C 'EEM markers'? Is this an expected expression in terms of RNA which is not converted into proteins?*

R16R1: Our opinion is that the visible *Hand1* expression caused by the imperfect knock-out efficiency by *Mesp1-Cre* driven system.

> *Q17R1: The authors do not address the potential presence of doublets (merged cells) within their newly generated dataset. While they mention using "SCTransform" for normalization and artifact removal, it's unclear if doublet removal was explicitly performed.*

R17R1: We appreciate your kind reminder. Doublet removal was performed using R package 'DoubletFinder' (DOI: 10.1016/j.cels.2019.03.003). We have added this information in the revised manuscript.

> ***Reviewer #2 (Public review):***
>
> *Summary of goals:*
>
> *The aims of the study were to identify new lineage trajectories for the cardiac lineages of the heart, and to use computational and cell and animal studies to identify and validate new gene regulatory mechanisms involved in these trajectories.*
>
> *Strengths:*
>
> *The study addresses the long-standing yet still not fully answered questions of what drives the earliest specification mechanisms of the heart lineages. The introduction demonstrates a good understanding of the relevant lineage trajectories that have been previously established, and the significance of the work is well described. The study takes advantage of several recently published data sets and attempts to use these in combination to uncover any new mechanisms underlying early mesoderm/cardiac specification mechanisms. A strength of the study is the use of an in vitro model system (mESCs) to assess the functional relevance of the key players identified in the computational analysis, including innovative technology such as CRISPR-guided enhancer modulations. Lastly, the study generates mesoderm-specific Hand1 LOF embryos and assesses the differentiation trajectories in these animals, which represents a strong complementary approach to the in vitro and computational analysis earlier in the paper. The manuscript is clearly written and the methods section is detailed and comprehensive.*
>
> *Comments and Weaknesses:*
>
> *Overall: The computational analysis presented here integrates a large number of published data sets with one new data point (E7.0 single cell ATAC and RNA sequencing). This represents an elegant approach to identifying new information using available data. However, the data presentation at times becomes rather confusing, and relatively strong statements and conclusions are made based on trajectory analysis or other inferred mechanisms while jumping from one data set to another. The cell and in vivo work on Hand1 and Foxf1 is an important part of the study. Some additional experiments in both of these model systems could strongly support the novel aspects that were identified by the computational studies leading into the work.*

We appreciate your positive comments and insightful suggestions. In the revised manuscript, we have incorporated additional analyses and experimental validations to address the concerns raised. Specifically, we added RNA velocity analysis to independently support the identification of the MJH and PSH trajectories, performed immunofluorescence staining of mesodermal and cardiac markers in Hand1 and Foxf1 knockout models, and included Vim staining-based quantification in Hand1 CKO embryos to assess developmental outcomes in *vivo*. Furthermore, we revised potentially overinterpreted conclusions, clarified methodological details of WOT analysis. These revisions have strengthened both the rigor and clarity of the manuscript.

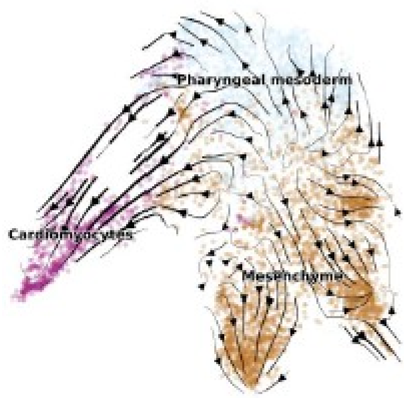
> *Q1R2: Definition of MJH and PSH trajectory:*
>
> *The study uses previously published data sets to identify two main new differentiation trajectories: the MJH and the PSH trajectory (Figure 1). A large majority of subsequent conclusions are based on in-depth analysis of these two trajectories. For this reason, the*

> *method used to identify these trajectories (WTO, which seems a highly biased analysis with many manually chosen set points) should be supported by other commonly used methods such as for example RNA velocity analysis. This would inspire some additional confidence that the MJH and PSH trajectories were chosen as unbiased and rigorous as possible and that any follow-up analysis is biologically relevant.*

R1R2: We appreciate your valuable comments. It is totally agreed that other commonly used methods help strengthen our conclusion about the two main trajectories. To this end, we performed RNA velocity analysis for the cardiac specification. Results support the contribution to CM along the MJH and PSH routes.

Author response image 2.

UMAP layout is colored by cell types. Developmental directions, shown as arrows, are inferred by RNA-velocity analysis.



Actually, several recent studies indicated a convergence cardiac developing model where progenitors reach a myocardial state along two trajectories (DOI: 10.1016/j.devcel.2023.01.010). However, when and how specification between the two routes were unclear. Our data and analysis revealed a clear fate separation by E7.0 from transcriptomic and epigenetic perspectives, where unbiased RNA velocity analysis was performed (Figure 2C).

We would like to clarify how we performed WOT (DOI: 10.1016/j.cell.2019.01.006) analysis: the only manually chosen cell set was the starting set, which was all cardiomyocyte cells by E8.5, of computational reverse lineage tracing. The ancestor cells were predicted in an unbiased manner among all mesodermal cells.

> *Q2R2.1: Identification of MJH and PSH trajectory progenitors:*
>
> *The study defines various mesoderm populations from the published data set (Figure 1A-E), including nascent mesoderm, mixed mesoderm, and extraembryonic mesoderm. It further assigns these mesoderm populations to the newly identified MJH/PSH trajectories. Based on the trajectory definition in Figure 1A it appears that both trajectories include all 3 mesoderm populations, albeit at different proportions and it seems thus challenging to assign these as unique progenitor populations for a distinct trajectory, as is done in the epigenetic study by comparing clusters 8 (MJH) and 2 (PSH)(Figure 2).*

R2R2.1: According to our model, the most significant difference between the two trajectories is their enrichment of EEM and PM cell types (Figure 1B), which represent the middle stages of cardiac development. Both trajectories begin as Mesp1+ Nascent mesoderm cells (Figure

1F), which is supported by Mesp1 lineage tracing (DOI: 10.1161/CIRCRESAHA.121.318943), and ends as cardiomyocytes. Our epigenetic analysis focused on the E7.0 stage when the two trajectories could be clearly separated and when JCF and SHF lineages were at mixed mesoderm and nascent mesoderm states, respectively. However, SHF lineage was predicted to bypass mixed mesoderm state later on.

> *Q2R2.2: Along similar lines, the epigenetic analysis of clusters 2 and 8 did not reveal any distinct differences in H3K4m1, H3K27ac, or H3K4me3 at any of the time points analyzed (Figure 2F). While conceptually very interesting, the data presented do not seem to identify any distinct temporal patterns or differences in clones 2 and 8 (Figure 2H), and thus don't support the conclusion as stated: "the combined transcriptome and chromatin accessibility analysis further supported the early lineage segregation of MJH and the epigenetic priming at gastrulation stage for early cardiac genes".*

R2R2.2: In the epigenetic analysis, we delineated the temporal dynamics of E7.0 cluster-specific DAEs by selecting earlier (E6.5) and later (E7.5) time points. DAEs of C8 and C2 represent regulatory elements for the JCF and SHF lineages, respectively. We also included C1 DAEs as a reference to demonstrate the relative activity of C8 and C2. The overall temporal pattern suggests activation of C8 & C2, as their H3K4me1 and H3K27ac levels surpass C1 over time. Between C8 and C2, the following distinctions could be observed:

a) H3K4me1 levels of C8 are higher by E6.5 and E7.0, with low H3K27ac levels, indicating early priming of C8 DAEs.

b) By E7.5, H3K4me1 levels of C8 are caught up by C2 in E7.5 anterior mesoderm (E7.5_AM, Figure 2F column 3), where cardiac mesoderm is located.

c) H3K4me1 and H3K27ac levels of C8 are similar as C1 in the posterior mesoderm (E7.5_P, Figure 2F column 4) and much higher than C2.

d) From the perspective of chromatin accessibility, hundreds of characteristic DAEs were identified for C2 and C8 (Figure 2D), exemplified by the primed and active enhancers which were predicted to interact with cluster-specific genes (Figure 2H).

Together with the transcriptomic analyses (Figure 2C), these data are consistent with our conclusion about early lineage segregation and epigenetic priming.

> *Q3R2: Function of Hand1 and Foxf1 during early cardiac differentiation:*
>
> *The study incorporated some functional studies by generating Hand1 and Foxf1 KO mESCs and differentiated them into mesoderm cells for RNA sequencing. These lines would present relevant tools to assess the role of Hand1 and Foxf1 in mesoderm formation, and a number of experiments would further support the conclusions, which are made for the most part on transcriptional analysis. For example, the study would benefit from quantification of mesoderm cells and subsequent cardiomyocytes during differentiation (via IF, or more quantitatively, via flow cytometry analysis). These data would help interpret any of the findings in the bulk RNAseq data, and help to assess the function of Hand1 and Foxf1 in generating the cardiac lineages. Conclusions such as "the analysis indicated that HAND1 and FOXF1 could dually regulate MJH specification through directly activating the MJH specific genes and inhibiting PSH specific genes" seem rather strong given the data currently provided.*

R3R2: Thank you for your kind suggestions. We added IF staining of mesodermal (Zic3), JCF (Hand1) and cardiac markers (Tnnt2), followed by cell quantification. Results indicate that Hand1 and Foxf1 knockout leads to reduced commitment to the JCF lineage, evidenced by the loss of Hand1 expression, accumulation of undifferentiated Zic3+ mesoderm, and impaired cardiomyocyte formation (Tnnt2+), consistent with the up-regulation of JCF lineage specific genes and the downregulation of SHF lineage specific genes.

We also revised the conclusion as “These results suggest that HAND1 and FOXF1 may cooperatively regulate early cardiac lineage specification by promoting JCF-associated gene expression and suppressing alternative mesodermal programs.”.

> *(4) Analysis of Hand1 cKO embryos:*
>
> *Adding a mouse model to support the computational analysis is a strong way to conclude the study. Given the availability of these early embryos, some of the findings could be strengthened by performing a similar analysis to Figure 7B&C and by including some of the specific EEM markers found to be differentially regulated to complement the structural analysis of the embryos.*

R4R2: hank you for your positive comments and help. In the revised manuscript, we performed IF staining of EEM marker Vim in a similar fashion as Figure 7B&C (Figure S8D). In comparison with control embryos, the Hand1 CKO embryos demonstrated significant less number of Vim+ cells, further strengthening the conclusion that Hand1 CKO blocked the developmental progression toward JCF direction.

> *Q5R2: Current findings in the context of previous findings:*
>
> *The introduction carefully introduces the concept of lineage specification and different progenitor pools. Given the enormous amount of knowledge already available on Hand1 and Foxf1, and their role in specific lineages of the early heart, some of this information should be added, ideally to the discussion where it can be put into context of what the present findings add to the existing understanding of these transcription factors and their role in early cardiac specification.*

R5R2: We appreciate your positive comments and kind reminder. We have added discussion about how our study could be put into the body of findings on Hand1 and Foxf1. Although these two genes have been validated to be functionally important for heart development, it is unclear when and how they affect this process. Using in-vivo and in-vitro models and single cell multi-omics analyses, we provided evidence to fill the gaps from multiple aspects, including cell state temporal dynamics, regulatory network, and epigenetic regulation underlying the very early cardiac lineage specification.

> ***Reviewer #3 (Public review):***
>
> *Q1R3: In Figure 1A, could the authors justify using E8.5 CMs as the endpoint for the second lineage and better clarify the chamber identities of the E8.5 CMs analysed? Why are the atrial genes in Figure 1C of the PSH trajectory not present in Table S1.1, which lists pseudotime-dependent genes for the MJH/PSH trajectories from Figure 1F?*

R1R3: Thank you for your comments. We used E8.5 CMs as the endpoint of the second (SHF) lineage because this stage represents a critical point where SHF-derived cardiomyocytes have begun distinct differentiation, allowing us to capture terminal lineage states reliably. The chamber identities of E8.5 CMs were determined based on known marker genes (DOI: 10.1186/s13059-025-03633-3). The atrial genes shown in Figure 1C reflect cluster-specific markers that may not meet the strict pseudotime-dependency criteria used to generate Table S1.1, which lists genes dynamically changing along the MJH/PSH trajectories.

> *Q2R3: Could the authors increase the resolution of their trajectory and genomic analyses to distinguish between the FHF (Tbx5+ HCN4+) and the JCF (Mab21l2+/ Hand1+) within*

> *the MJH lineage? Also, clarify if the early extraembryonic mesoderm contributes to the FHF.*

R2R3: Thank you for your great suggestions. To distinguish between the FHF and JCF trajectories, we used early FHF progenitor population (E7.75 *Nkx2-5*+; *Mab21l2*- CM cells) as the starting point and performed WOT lineage inference (Figure S2A). Results suggest that both JCF and SHF progenitors contribute to the FHF, consistent with live imaging-based single cell tracing by Dominguez et al (DOI: 10.1016/j.cell.2023.01.001) and lineage tracing results by Zhang et al (DOI: 10.1161/CIRCRESAHA.121.318943). We also analyzed the expression levels of FHF marker genes (*Tbx5*, *Hcn4*) and observed their activation along both trajectories (Figure S2B).

> *Q3R3: The authors strongly assume that the juxta-cardiac field (JCF), defined by Mab21l2 expression at E7.5 in the extraembryonic mesoderm, contributes to CMs. Could the authors explain the evidence for this? Could the authors identify Mab21l2 expression in the left ventricle (LV) myocardium and septum transversum at E8.5 (see Saito et al., 2013, Biol Open, 2(8): 779-788)? If such a JCF contribution to CMs exists, the extent to which it influences heart development should be clarified or discussed.*

R3R3: Thank you for the important question. For the JCF contribution to the heart tube, several lines of evidence have been published in recent years using micro-dissection of mouse embryonic heart (DOI: 10.1126/science.abb2986), live imaging (DOI: 10.1016/j.cell.2023.01.001) and lineage tracing approaches (DOI: 10.1161/CIRCRESAHA.121.318943). According to Tyser et al (DOI: 10.1126/science.abb2986), *Mab21l2* expression is detected in septum transversum at E8.5 and the *Mab21l2*+ lineage contribute to LV, basically consistent with the literature you mentioned (Saito et al., 2013, Biol Open, 2(8): 779-788). Our lineage inference analyses further support the model and suggest earlier specification by JCF. However, the focus of our work is the transcriptional and epigenetic regulation of underlying the JCF developmental trajectory.

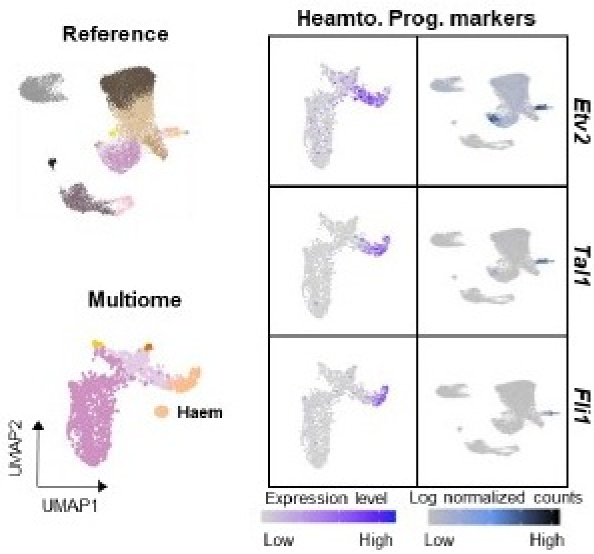
> *Q4R3: Could the authors distinguish the Hand1+ pericardium from JCF progenitors in their single-cell data and explain why they excluded other cell types, such as the endocardium/endothelium and pericardium, or even the endoderm, as endpoints of their trajectory analysis? At the NM and MM mesoderm stages, how did the authors distinguish the earliest cardiac cells from the surrounding developing mesoderm?*

R4R3: We appreciate your insightful question. In our other study (DOI: 10.1186/s13059-025-03633-3), we tried to further divide the CM cells as subclusters and it seems that their difference is mainly driven by the segmentation of the heart tube (e.g. LV, RV, OFT etc.). By the E8.5 stage, we are unable to identify the *Hand1*+ pericardium cluster.

Also, it seems infeasible to distinguish endocardium from other endothelium cells only using singlecell data. High resolution spatial transcriptome data is required. Alternatively, we analyzed the E7.0 mesodermal lineages and determined C5/6 as hematoendothelial progenitors. Marker gene analysis indicate that their lineage segregation has started by this stage (Figure S4C and Author response image 3).

Author response image 3.

UMAP layout, using scRNA-seq (Reference data) and snRNA-seq (Multiome data), is colored by cell types (left). Expression of hematoendothelial progenitor marker genes is shown (right).



We did observe the difference between the earliest cardiac cells from the surrounding developing mesoderm. As in Figure 1D, cells belonging to the JCF lineage (*Hand1* high/*Lefty2* low) were clustered at the EEM/MM end, in contrast to the NM cells.

> *Q5R3: Could the authors contrast their trajectory analysis with those of Lescroart et al. (2018), Zhang et al., Tyser et al., and Krup et al.?*

R5R3: Thank you for the valuable suggestion. We compared our model with the suggested ones and summarized as follows:

(1) Lescroart et al: The JCF and SHF progenitor cells match their DCT2 (Bmp4+) and DCT3 (Foxc2+) clusters, respectively.

(2) Zhang et al: The JCF lineage matches their EEM-DC (developing CM)-CM trajectory. The SHF lineage is consistent with their NM-LPM (lateral plate mesoderm)-DC (developing CM)-CM trajectory. Notably, their EEM-DC-CM also expressed FHF marker (*Tbx5*) at later stages.

(3) Tyser et al: we performed data integration analysis and found the correspondence between JCF progenitors (EEM cells from the cardiac trajectory) and their Me5, as well as SHF progenitors (PM cells from the cardiac trajectory) with Me7. In their model, both Me5 and Me7 contribute to Me4 (representing the FHF), consistent with our results (see Tyser et al., 2021 and Pijuan-Sala et al., 2019).

(4) Krup et al also performed URD lineage inference, providing a model with CM (12) and Cardiac mesoderm (29) as cardiac end points. Their model did not seem to suggest distinct trajectories between JCF and SHF lineages, as both JCF (Hand1) and SHF (Isl1) markers co-expressed in CM.

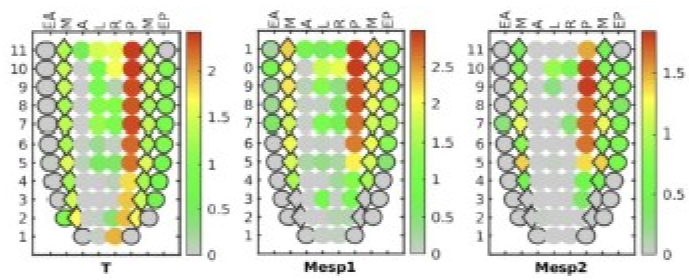
> *Q6R3: Previous studies suggest that Mesp2 expression starts at E8 in the presomitic mesoderm (Saga et al., 1997). Could the authors provide in situ hybridization or HCR staining to confirm the early E7 Mesp2 expression suggested by the pseudo-time analysis of the second lineage.*

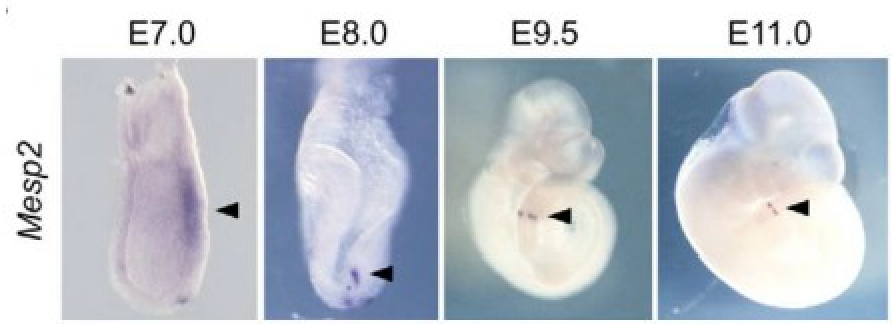
R6R3: We validated the expression of E7 Mesp2 using Geo-seq spatial transcriptome data (Author response image 4, upper). Results suggest the high spatial enrichment of Mesp2 expression in primitive streak (T+) and/or nascent mesoderm (Mesp1+) cells, which correspond to the progenitors of the second lineage.

In situ hybridization data (PMID: 17360776) also supports the early expression of Mesp2 by E7 (Author response image 4, lower).

Author response image 4.

(Upper) E7 Geo-seq data for selected genes: *T*, *Mesp1*, and *Mesp2*. (Lower) *Mesp2* expression during early development; image acquired from Morimoto et al. (PMID: 17360776).



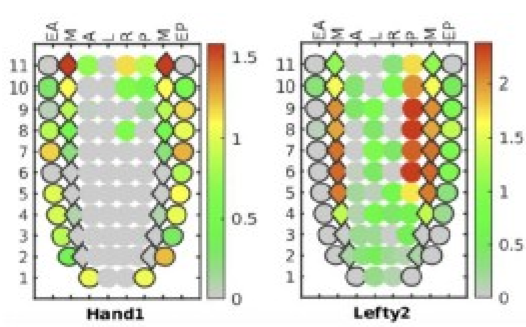


> *Q7R3: Could the authors also confirm the complementary Hand1 and Lefty2 expression patterns at E7 using HCR or in situ hybridization? Hand1 expression in the first lineage is plausible, considering lineage tracing results from Zhang et al.*

R7R3: Thank you for your great suggestion. We observed spatially complementary expression patterns of Hand1 and Lefty2 in the Geo-seq spatial transcriptomic data. In the mesoderm layer, Hand1 is highly expressed in the proximal end. While Lefty2+ cells exhibit preference toward the distal direction.

Author response image 5.

E7 Geo-seq data for selected genes: *Hand1* and *Lefty2*.



> *Q8R3: Could the authors explain why Hand1 and Lefty2+ cells are more likely to be multipotent progenitors, as mentioned in the text?*

R8R3: Thank you for your question. Here, we observed E7.0 Mesp1+ and Lefty2+ nascent mesodermal cells assigned to both the JCF and SHF lineages (Figure 1D), indicating their multipotency. On the other hand, we also found low expressions of JCF markers, *Hand1* and *Msx2*, by the early stage of the SHF trajectory (Figure 1F). Thus, we concluded that both Hand1+ and Lefty2+ E7.0 mesodermal cells are likely to be multipotent.

> *Q9R3: Could the authors comment on the low Mesp1 expression in the mesodermal cells (MM) of the MJH trajectory at E7 (Figure 1D)? Is Mesp1 transiently expressed early in MJH progenitors and then turned off by E7? Have all FHF/JCF/SHF cells expressed Mesp1?*

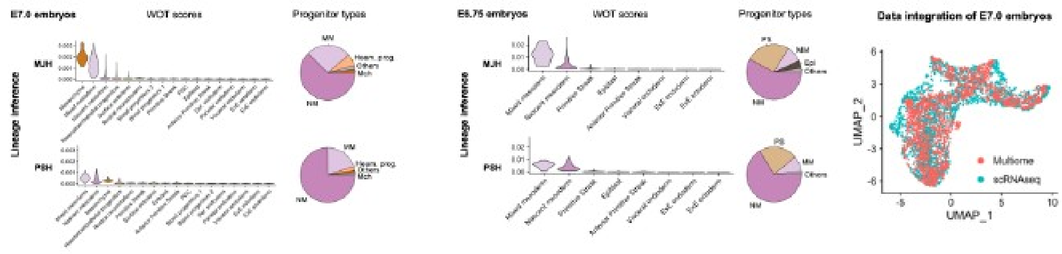
R9R3: Thank you for the insightful questions. Zhang et al. (PMID: 34162224) performed scRNA-seq analysis of *Mesp1* lineage-traced cells, which indicate the contribution of Mesp1+ cells to FHF, JCF, and SHF. This is also supported by Dominguez et al. utilizing live imaging approaches (PMID: 36736300). Our temporal dynamics analysis suggests that along the JCF trajectory, *Mesp1* is turned off as JCF characteristic genes were up regulated (Figure 1F and S1D).

> *Q10R3: Could the authors clarify if their analysis at E7 comprises a mixture of embryonic stages or a precisely defined embryonic stage for both the trajectory and epigenetic analyses? How do the authors know that cells of the second lineage are readily present in the E7 mesoderm they analysed (clusters 0, 1, and 2 for the multiomic analysis)?*

R10R3: Thank you for your questions. Although embryos were collected at E7.0, the developmental stages could be variable. As exemplified by Karl Theiler's book, "The House Mouse: Atlas of Embryonic Development", mesoderm was visible for some E7.0 egg cylinders but not in others. To test whether cells of the second lineage are present in the E7.0 mesoderm, we analyzed the WOT lineage tracing results and the cell type composition by E7.0 (Author response image 6, left panel). Most cells belong to the nascent mesoderm (NM) or mixed mesoderm (MM), while almost no cells were assigned to the primitive streak (PS). To avoid the possibility that the E7.0 embryos represented later stages, we also analyzed the E6.75 cells of the second lineage (Author response image 6, middle panel). Results suggest that NM cells were still the dominant contributors to the second lineage, although ~22.6% cells were assigned to the PS. The abovementioned analyses were performed using the scRNA-seq data. The embryos of the E7.0 single-cell multi-omics represent similar developmental stages as the scRNAseq data, as suggested by the well-aligned UMAPs (Figure S1D, right panel). Thus, we conclude that for the multi-omics data, the cells of the second lineage are also readily present in the mesoderm.

Author response image 6.

(Left and middle) Lineage inference and cell type composition at E7.0 and E6.75. (Right) UMAPs of E7.0 multi-omics and scRNA-seq data.

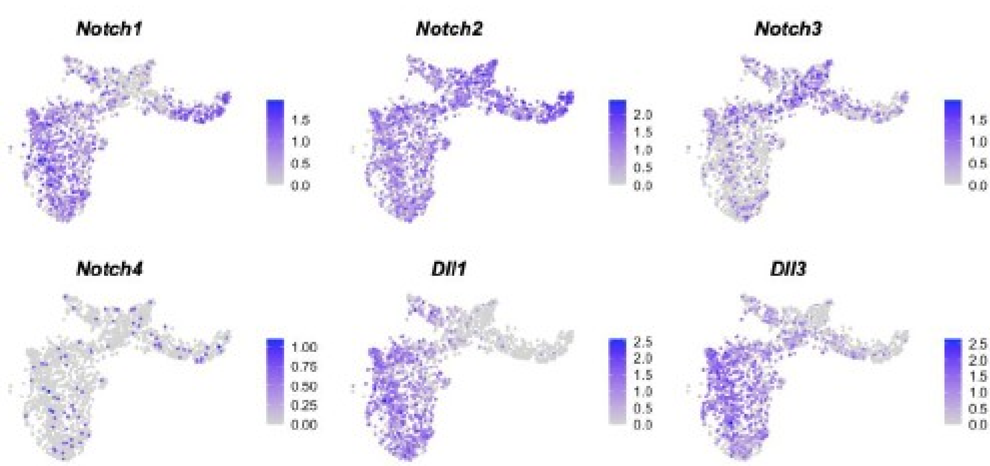


> *Q11R3: Could the authors further comment on the active Notch signaling observed in the first and second lineages, considering that Notch's role in the early steps of endocardial lineage commitment, but not of CMs, during gastrulation has been previously described by Lescroart et al. (2018)?*

R11R3: We appreciate your kind suggestion. As reported by Lescroart et al. (2018), using *Notch1CreERT2/Rosa-tdTomato* mice and tamoxifen administration at E6.5, early expression of *Notch1* mostly marked endocardial cells (ECs, 76.9-83.9%), with minor contribution to the cardiomyocytes (6.0-16.6%) and to the epicardial cells (EPs, 6.0-6.5%). The lineage specificity of *Notch1* is consistent with our E7.0 multi-omics data, where its expression was mainly observed in the NM and hematoendothelial progenitors (Author response image 7). Interestingly, expression of other NOTCH receptor genes (*Notch2* and *Notch3*) and ligand genes (*Dll1* and *Dll3*) in the CM lineages. *Notch3* demonstrate higher expression in the first lineage, while *Dll1* and *Dll3* were highly expressed in the second lineage. The study by Lescroart et al. (2018) emphasized the role of *Notch1* as an EC lineage marker, while our analyses aimed at the activity of the NOTCH pathway.

Author response image 7.

Expression of representative NOTCH genes at E7.0 (multi-omics data).

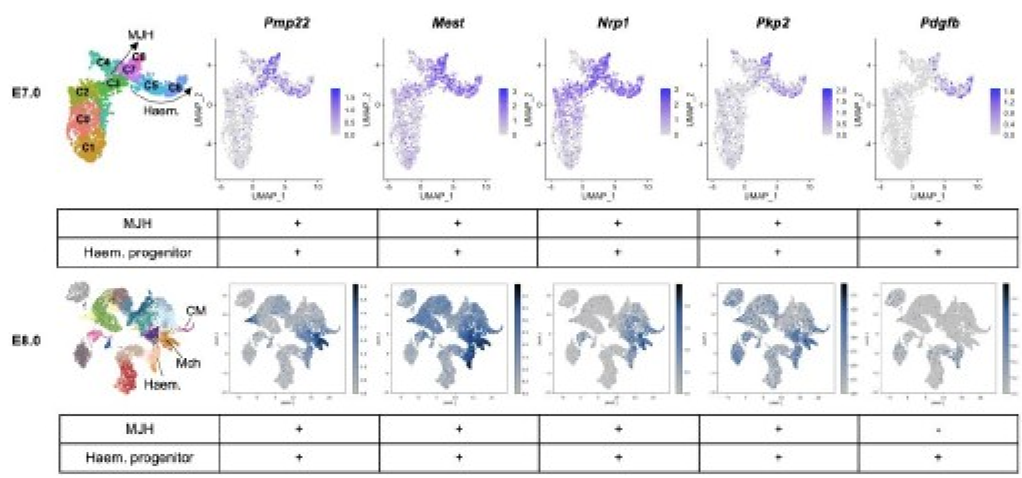


> *Q12R3: In cluster 8, Figure 2D, it seems that levels of accessibility in cluster 8 are relatively high for genes associated with endothelium/endocardium development in addition to MJH genes. Could the authors comment and/or provide further analysis?*

R12R3: Thanks for you for raising this interesting point. To confirm the association of these genes with endothelium (EC) and/or MJH, we analyzed their expression levels by E7.0 (progenitor stage) and E8.0 (differentiated stage) (Author response image 8). Among target genes of MJH-specific DAEs (cluster 3/7/8 in Figure 2D), *Pmp22*, *Mest*, *Npr1*, *Pkp2*, and *Pdgfb* were expressed in the hematoendothelial progenitors. The *Nrp1* gene and PDGF pathway play critical roles in endothelial development by modulating cell migration (PMID: 15920019 and 28167492), which is also important for MJH cells. In addition, we observed common ATAC-seq peaks in both hematoendothelial and MJH clusters (Author response image 9), indicating shared regulatory elements. Interestingly, *Pdgfb* is not expressed by CM in vivo, it is actively expressed in the CM of the in vitro system (Author response image 9). These results indicate regulatory and functional closeness between hematoendothelial and MJH cell groups, at early stages of lineage establishment.

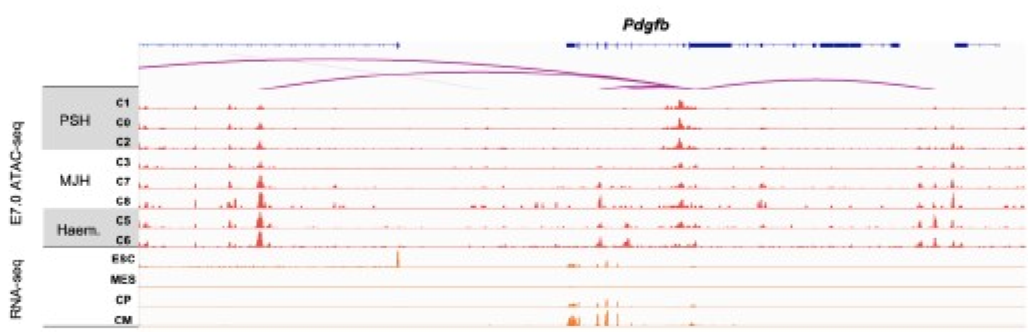
Author response image 8.

Regulatory connection between MJH and endothelial cells (ECs).



Author response image 9.

Representative genome browser snapshots of scATAC-seq (aggregated gene expression and chromatin accessibility for each cluster) and RNA-seq at the *Pdgfb* locus.



> *Q13R3: Can the authors clarify why they state that cluster 8 DAEs are primed before the full activation of their target genes, considering that Bmp4 and Hand1 peak activities seem to coincide with their gene expression in Figure 2G?*

R13R3: Thanks for your great question. The overall analyses indicate low to medium levels of H3K4me1 and H3K27ac by E6.5-7.0 at cluster 8 DAEs, which were fully activated by E7.5 (Figure 2F). Further inspections suggest different epigenetic status of individual DAEs (Figure

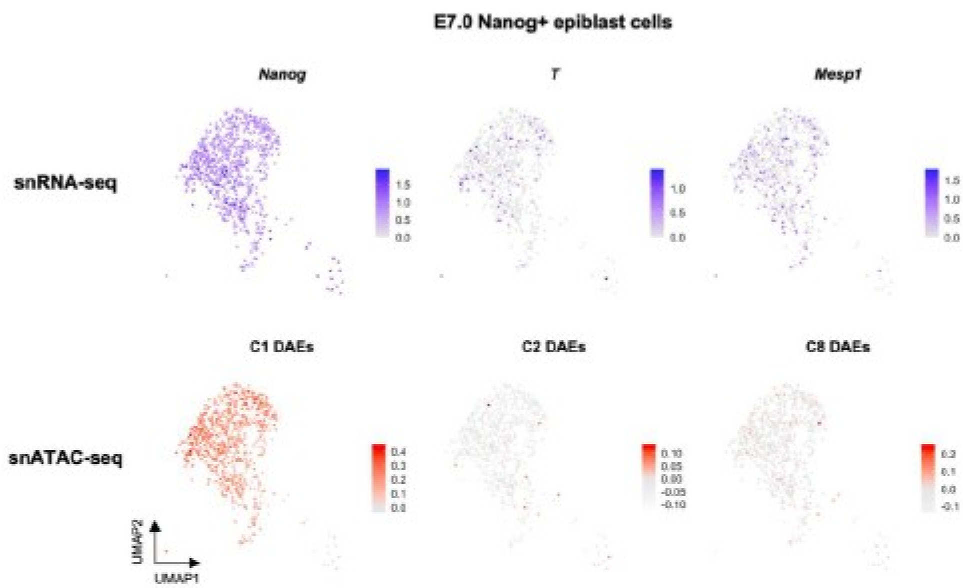
3H), which could be active (K4me1+/K27ac+), primed (K4me1+/K27ac-), or inactive (K4me1-/K27ac-). Thus, we concluded that many DAEs could be primed before full activation. The coincidence of enhancer peak activities and gene expression was observed by aggregating single cell clusters at a single stage E7.0, which does not rule out the possibility that these enhancers are epigenetically primed at earlier stages.

> *Q14R3: Did the authors extend the multiomic analysis to Nanog+ epiblast cells at E7 and investigate if cardiac/mesodermal priming exists before mesodermal induction (defined by T/Mesp1 onset of expression)?*

R14R3: We appreciate your kind suggestion. We observed low levels of *T/Mesp1* expression in the E7.0 Nanog+ epiblast cells (Author response image 10). Interestingly, the T+/Mesp1+ cells were not clustered toward any specific differentiation directions in the UMAP. We also analyzed DAE activities in each single cell by averaging over the C1/C2/C8 DAE sets. The C2 and C8 DAEs were clearly less active than the C1 DAEs. But C2/C8-DAE active cells were observed among the E7.0 Nanog+ epiblast cells. These data indicate the early priming exists in epiblast cells before the commitment to cardiac/mesodermal differentiation.

Author response image 10.

Gene expression and DAE activity levels of E7.0 Nanog+ epiblast cells shown in UMAP layout.



> *Q15R3: In the absence of duplicates, it is impossible to statistically compare the proportions of mesodermal cell populations in Hand1 wild-type and knockout (KO) embryos or to assess for abnormal accumulation of PS, NM, and MM cells. Could the authors analyse the proportions of cells by careful imaging of Hand1 wild-type and KO embryos instead?*

R15R3: Thank you for your important question. To assess the proportions of mesodermal cell populations in E7.25 wild-type and *Hand1*-CKO embryos, we analyzed the serial coronal sections of the extraembryonic portions and performed staining of the *Vim* gene, which marks the extra-embryonic mesodermal (EEM) cells (Figure S8D). We then counted the numbers of mesodermal/Vim+ EEM cells and calculated the relative proportion of Vim+ EEM cells in each section. The proportion of Vim+ EEM cells was statistically lower in the *Hand1*-CKO embryo, consistent with our model that Hand1 deletion led to blocked MJH specification.

> *Q16R3: Could the authors provide high-resolution images for Figure 7 B-C-D as they are currently hard to interpret?*

R16R3: Thank you for your suggestion. We have replaced Figure 7B-C-D with high-resolution images.

> ***Recommendations for the authors:***
>
> ***Reviewing Editor Comments:***
>
> *Discussions among reviewers emphasize the importance of better addressing and validating the trajectory analysis by using more common and alternative bioinformatics and spatial approaches. Further discussion on whether there is a common transcriptional progenitor between the two trajectories is also required to enhance the significance of the study. For functional analysis, further validations are needed as the current data only partially support the claims. Please see public reviews for details.*
>
> ***Reviewer #2 (Recommendations For The Authors):***
>
> *Beyond the suggestions made in the public review, below are some minor aspects for consideration:*
>
> *The manuscript is well written overall but may benefit from a thorough read-through and editing of some minor grammatical errors.*

We have carefully read through the manuscript and corrected minor grammatical errors to improve clarity and readability.

> *Figure 2C: RNA velocity information gets largely lost due to the color choice of EEM and MM (black) on which the direction of arrows can't be appreciated.*

We have updated the color scheme in Figure 2C.

> *Figure 6D: sample information is partially cut off in the graph.*

Sample information is completely shown now.

> *The last paragraph of the discussion has some formatting issues with the references.*

We have corrected the formatting issues with the references.

> *The methods and results section does not comment on if, or how many embryos were pooled for the sequencing analysis performed for this study.*

We have added the numbers of embryos for sequencing analyses in the methods section.

> ***Reviewer #3 (Recommendations For The Authors):***
>
> *Minor:*
>
> *In the discussion, authors could reconsider the sentence: "The process of cardiac lineage segregation is a complex one that may involve TF regulatory networks and signaling pathways," as it is not informative.*

We have re-written the sentence as: “Thus, additional regulation must exist and instructs the process of JCF-SHF lineage segregation.”

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